

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\188-C1-C2-C3-C4- Solid SS Top 2nd stack.pvdb

Date of Analysis : Sep 9,2026 10:04am

PV Elite 28, January 2026



PV Elite 28 Licensee: SL User
 FileName : 188-C1-C2-C3-C4- Solid SS Top 2nd stack Page 2 of 223
 Vessel Design Summary: Step: 37 10:04am Sep 9,2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 560.000 mm ID
 Vessel Design Length, Tangent to Tangent 11445.95 mm
 Specified Datum Line Distance 50.80 mm
 Shell Side Design Temperature 300 °C
 Channel Side Design Temperature 300 °C
 Shell Side Design Pressure 6100.000 KPa.
 Channel Side Design Pressure 6100.000 KPa.
 Shop Shell Side Test Pressure 14618.359 KPa.
 Field Shell Side Test Pressure 8046.557 KPa.
 Shop Channel Side Test Pressure 13142.574 KPa.
 Field Channel Side Test Pressure 9333.928 KPa.
 Wind Design Code ASCE/SEI 7-16
 Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-240 304	...	...	S30400	
Head	SA-240 304	...	...	S30400	
Flange	SA-965 F304	...	...	S30400	
Nozzle	SA-312 TP304	...	...	S30400	
Re-Pad	SA-240 304	...	...	S30400	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-182 F304	...	<= 130	S30400	
Nozzle Flg	SA-105	...	...	K03504	
Tubes	SA-213 TP304L	...	...	S30403	
Tubesheet	SA-965 F304	...	...	S30400	
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Hrz Bolting	SA-193 B7	...	<= 64	G41400	
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (KPa. & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
LHS Dish End	6105.400	103.42	6396.000	1.0000	No	No
LHS Ch Shell	6105.400	103.42	6696.000	1.0000	N/A	No
LHS ch flange	6105.391	103.42	6651.000	1.0000	N/A	No
Main Shell	6105.420	103.42	6997.000	0.0000	N/A	No
Main Shell	6105.420	103.42	10247.000	0.0000	N/A	No
Main Shell	6105.420	103.42	10247.000	0.0000	N/A	No
Main Shell	6105.420	103.42	10247.000	0.0000	N/A	No
Main Shell	6105.420	103.42	10247.000	0.0000	N/A	No
Main Shell	6105.420	103.42	10247.000	0.0000	N/A	No
Main Shell	6105.420	103.42	6997.000	0.0000	N/A	No
RHS Ch Flange	6105.391	103.42	6593.000	1.0000	N/A	No



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RHS Ch Shell	6105.400	103.42	6696.000	1.0000	N/A	No
RHS Dish End	6105.400	103.42	6396.000	1.0000	No	No

Liquid Level: 553.00 mm Dens.: 0.001 kg./cm³ Sp. Gr.: 1.000

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Ellipse	0.0	50.8	28.0	22.0	21.0	2.7	1.00	1.00
Cylinder	800.0	800.0	24.0	24.0	21.9	3.4	1.00	1.00
Body Flg	970.2	167.0	130.0	122.0	116.7	116.7	1.00	1.00
Cylinder	2660.2	1500.0	24.0	24.0	20.8	5.6	1.00	1.00
Cylinder	3660.2	1000.0	36.0	36.0	20.8	5.8	1.00	1.00
Cylinder	5160.2	1500.0	36.0	36.0	20.8	5.8	1.00	1.00
Cylinder	6380.2	1220.0	36.0	36.0	20.8	5.8	1.00	1.00
Cylinder	7880.2	1500.0	36.0	36.0	20.8	5.8	1.00	1.00
Cylinder	8880.2	1000.0	36.0	36.0	20.8	5.8	1.00	1.00
Cylinder	10380.2	1500.0	24.0	24.0	20.8	5.6	1.00	1.00
Body Flg	10744.3	171.0	130.0	122.0	116.4	116.4	1.00	1.00
Cylinder	11344.4	600.0	24.0	24.0	21.9	3.2	1.00	1.00
Ellipse	11395.2	50.8	28.0	22.0	21.0	2.7	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N3-Tube Inlet	WRC-297	0.896	Shell	Passed
N4-Tube Outlet	WRC-297	0.896	Shell	Passed
N1-Shell Inlet	WRC-297	0.861	Shell	Passed
N2-Shell outle	WRC-297	0.861	Shell	Passed

Foundation/Support Summary

Saddle Parameters:

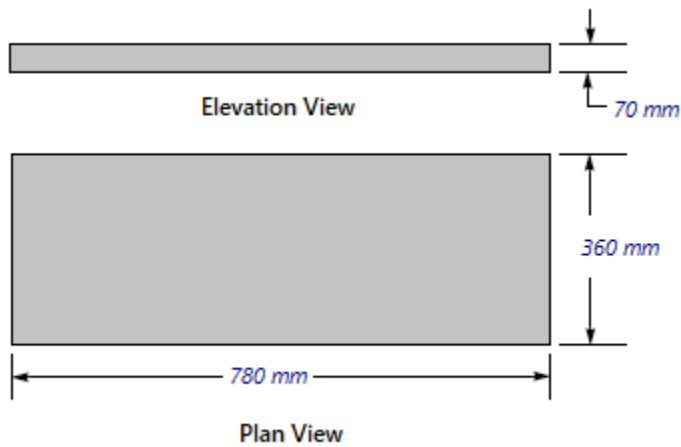
Saddle Width 335.000 mm
 Saddle Bearing Angle 168.000 deg.
 Centerline Dimension 850.000 mm
 Wear Pad Width 530.000 mm
 Wear Pad Thickness 14.000 mm
 Wear Pad Bearing Angle 180.000 deg.
 Distance from Saddle to Tangent 500.000 mm

Baseplate Length 780.000 mm
 Baseplate Thickness 70.000 mm
 Baseplate Width 360.000 mm
 Number of Ribs (including outside ribs) 3
 Rib Thickness 10.000 mm
 Web Thickness 10.000 mm
 Height of Center Web 575.000 mm
 Number of Bolts in Baseplate 4

Baseplate Sketch



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**Summary of Maximum Saddle Loads, Operating Case, Un-Factored:**

Maximum Vertical Saddle Load	30891.16	Kgf
Maximum Transverse Saddle Shear Load	2634.91	Kgf
Maximum Longitudinal Saddle Shear Load	2080.00	Kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	21267.10	Kgf
Maximum Transverse Saddle Shear Load	385.10	Kgf
Maximum Longitudinal Saddle Shear Load	33.64	Kgf

Weights:

Fabricated - Bare w/o removable internals	30052.7	kg.
Shop Test - Fabricated + water (full)	36322.1	kg.
Shipping - Fab. + rem. intls.+ shipping app.	30052.7	kg.
Erected - Fab. + rem. intls.+ insul. (etc)	30052.7	kg.
Empty - Fab. + intls. + details + wghts.	30990.0	kg.
Operating - Empty + operating liquid (no ca)	37298.0	kg.
Field Test - Empty weight + test medium (full)	37259.4	kg.

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 FileName : 188-C1-C2-C3-C4- Solid SS Top 2nd stack Page 5 of 223
 Nozzle Summary: Step: 35 10:04am Sep 9,2026

Nozzle Calculation Summary (KPa. & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N3-Tube Inlet	6575	OK	...	OK	9.10	OK	Passed
N4-Tube Outlet	6570	OK	...	OK	9.10	OK	Passed
N1-Shell Inlet	6181	OK	...	OK	8.10	OK	Passed
N2-Shell outlet	6175	OK	...	OK	8.10	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 6175 Nozzle : N2-Shell outlet [Shellside]
 Minimum MAWP Nozzles : 6570 Nozzle : N4-Tube Outlet [Tubeside]

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
20	N3-Tube Inlet	450.800	0.000	489.750
20	N4-Tube Outlet	450.800	180.000	489.750
40	N1-Shell Inlet	1960.975	0.000	501.650
100	N2-Shell outlet	9680.975	180.000	501.650

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
N3-Tube Inlet	10.000 in	80	WNF	10.750	15.088	453.05	24.00	309.54	900
N4-Tube Outlet	10.000 in	80	WNF	10.750	15.088	453.05	24.00	309.54	900
N1-Shell Inlet	10.000 in	60	WNF	10.750	12.700	426.54	24.00	309.54	900
N2-Shell outlet	10.000 in	60	WNF	10.750	12.700	426.54	24.00	309.54	900

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N3-Tube Inl	SA-312 TP304	14.000	10.000	14.000	24.000	...
N4-Tube Out	SA-312 TP304	14.000	10.000	14.000	24.000	...
N1-Shell In	SA-312 TP304	24.000	10.000	18.000	24.000	...
N2-Shell ou	SA-312 TP304	24.000	10.000	18.000	24.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
N3-Tube Inlet	400.000	0.0	250.00	0.00	LHS Ch Shell
N4-Tube Outlet	400.000	180.0	250.00	0.00	LHS Ch Shell
N1-Shell Inlet	1910.175	0.0	250.00	0.00	Main Shell
N2-Shell outlet	9630.175	180.0	250.00	0.00	Main Shell

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 Bill of Material: Step: 33 10:04am Sep 9,2026

Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 28.0mm THK X 560.0mm ID X 50.8mm	SA-240 304
1	CYLINDER: 24.0mm THK X 560.0mm ID X 800.0mm	SA-240 304
2	WELD NECK FLANGE: 130.0mm THK X 824.0mm OD	SA-965 F304
2	CYLINDER: 24.0mm THK X 560.0mm ID X 1500.0mm	SA-240 304
2	CYLINDER: 36.0mm THK X 560.0mm ID X 1000.0mm	SA-240 304
2	CYLINDER: 36.0mm THK X 560.0mm ID X 1500.0mm	SA-240 304
1	CYLINDER: 36.0mm THK X 560.0mm ID X 1220.0mm	SA-240 304
1	CYLINDER: 24.0mm THK X 560.0mm ID X 600.0mm	SA-240 304
1	INSULATION: 50mm X 100mm THK	Hot Insulation
1	INSULATION: 800mm X 100mm THK	Hot Insulation
2	INSULATION: 180mm X 100mm THK	Hot Insulation
2	TUBESHEET: 824.0mm X 190.0mm THK	SA-965 F304
900	TUBES: 9600.0mm X 12.7mm DIA X 2.1mm THK	SA-213 TP304L
4	INSULATION: 1500mm X 100mm THK	Hot Insulation
2	SADDLE: 335.0mm X 168 DEG	SA-516 70
2	INSULATION: 1000mm X 100mm THK	Hot Insulation
2	CLADDING: 1000.000mm X 3.000mm THK	...
3	CLADDING: 1500.000mm X 3.000mm THK	...
1	INSULATION: 1220mm X 100mm THK	Hot Insulation
1	CLADDING: 1220.000mm X 3.000mm THK	...
1	INSULATION: 600mm X 100mm THK	Hot Insulation
1	INSULATION: 188mm X 100mm THK	Hot Insulation
2	GASKET: 611.0mm OD X 585.0mm ID	...
48	BODY FLANGE BOLTS: 38.1mm DIA	SA-193 B7
48	NUTS FOR BODY FLANGE BOLTS: 38.1mm DIA	...
1	NAMEPLATE	...



PV Elite 28 Licensee: SL User
 FileName : 188-C1-C2-C3-C4- Solid SS Top 2nd stack Page 8 of 223
 MDMT Summary: Step: 36 10:04am Sep 9,2026

Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Notes								
Main Shell [16]		-196						
Main Shell [16]		-196						
Main Shell [16]		-196						
Main Shell [16]		-196						
Main Shell [16]		-196						
Main Shell [16]		-196						
Main Shell [16]		-196						
N1-Shell Inlet[1]	B	-2	-31	-29	0.530	24.000	1.00	No
N2-Shell outle[1]	B	-2	-31	-29	0.530	24.000	1.00	No
Nozzle Flg [5]	A	-18	-48					
Warmest MDMT:		-2	-31					
LHS Dish End [16]		-196						
LHS Ch Shell [16]		-196						
RHS Ch Shell [16]		-196						
RHS Dish End [16]		-196						
N3-Tube Inlet[15]		-196						
N4-Tube Outle[15]		-196						
Bolting [21]		-48						
Bolting [21]		-48						
Warmest MDMT:		-48						
Exchanger Side	Computed MDMT		Required MDMT		Pass/Fail			
Shell	-31.0		-12.0		Pass			
Channel/Tube	-48.0		-12.0		Pass			

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.
 UCS-66(g) was not considered.
 UCS-66(i) was not considered.



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FileName : 188-C1-C2-C3-C4- Solid SS Top 2nd stack Page 9 of 223
MDMT Summary: Step: 36 10:04am Sep 9, 2026

Notes:

Impact test temps were not entered in and not considered in the analysis.
UCS-66(i) applies to impact tested materials not by specification and
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

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FileName : 188-C1-C2-C3-C4- Solid SS Top 2nd stack Page 10 of 223
Warnings and Errors: Step: 0 10:04am Sep 9,2026

Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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 FileName : 188-C1-C2-C3-C4- Solid SS Top 2nd stack Page 11 of 223
 Input Echo: Step: 1 10:04am Sep 9,2026

Units used in this Analysis (MM):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	6.8946	KPa.
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	6100	KPa.
Channel Side Design Pressure	6100	KPa.
Shell Side Design Temperature	300.0	°C
Channel Side Design Temperature	300.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	None	
Service Type, Channel Side	None	
MDMT (CET), Shell Side	-12.0	°C
MDMT (CET), Tube Side	-12.0	°C
User defined MAWP, Shell Side	0	KPa.
User defined MAWP, Channel Side	0	KPa.
User defined MAPnc, Shell Side	0	KPa.
User defined MAPnc, Channel Side	0	KPa.
User defined Test Pres., Shell Side	0	KPa.
User defined Test Pres., Channel Side	0	KPa.

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No



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Field Pressure Test:

Type of Pressure Test UG-99(b) Note [35]
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16	
Wind Load Reduction Scale Factor	1.100	
Basic Wind Speed	92.7	Km/hr
Surface Roughness Category	C: Open Terrain	
Type of Surface	Moderately Smooth	
Base Elevation	1500	mm
Percent Wind for Hydrotest	50.0	
Using User defined Wind Press. Vs Elev.	N	
Height of Hill or Escarpment H or Hh	0	mm
Distance Upwind of Crest Lh	0	mm
Distance from Crest to the Vessel x	0	mm
Type of Terrain (Hill, Escarpment)	Flat	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	
Site Elevation above Sea Level	0	mm

Seismic Design Code	ASCE/SEI 7-16	
Seismic Load Reduction Scale Factor	1.000	
Importance Factor	1.250	
Table Value Fa	1.300	
Table Value Fv	1.500	
Max. Mapped Res. Acceleration [Ss]	0.147	
Max. Eff. Ground Acceleration [S]	0.065	
Force Modification Factor R	3.000	
Site Class	C	
Component Elevation Ratio z/h	0.000	
Amplification Factor Ap	1.000	
Force Factor	0.000	
Consider Vertical Acceleration	Yes	
Minimum Acceleration Multiplier	0.000	
User Value of Sds (used if > 0)	0.000	
User Value of Sd1 (used if > 0)	0.000	
Moment Reduction Factor Tau	1.000	

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y



PV Elite 28 Licensee: SL User
 FileName : 188-C1-C2-C3-C4- Solid SS Top 2nd stack Page 13 of 223
 Input Echo: Step: 1 10:04am Sep 9,2026

Use ASME VIII-1 Appendix 1-9 N
 Perform Blast Load Analysis No
 Material Database Year Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk) Yes
 Do not use Nozzle MDMT Interpretation VIII-1 01-37 No
 Use Table G instead of exact equation for "A" Yes
 Shell Head Joints are Tapered Yes
 Compute "K" in corroded condition Yes
 Use Code Case 2286 No
 Use the MAWP to compute the MDMT Yes
 For thickness ratios <= 0.35, MDMT will be -155F (-104C) Yes
 For PWHT & P1 Materials the MDMT can be < -55F (-48C) No
 Using Metric Material Databases, ASME II D Yes
 Calculate B31.3 type stress for Nozzles with Loads Yes
 Reduce the MDMT due to lower membrane stress Yes
 Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element#	1/13
Element From Node	10
Element To Node	20
Element Type	Elliptical
Description	LHS Dish End
Distance "FROM" to "TO"	50.8 mm
Inside Diameter	560 mm
Element Thickness	22 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	28 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	6100 KPa.
Design Temperature Internal Pressure	300 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	300 °C
Effective Diameter Multiplier	1.2
Material Name	SA-240 304
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	85.9 N./mm ²
Allowable Stress, Hydrotest	184.5 N./mm ²
Material Density	0.008027 kg./cm ³
P Number Thickness	0 mm
Yield Stress, Operating	129 N./mm ²
External Pressure Chart Name	HA-1
UNS Number	S30400
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No
Element From Node	10
Detail Type	Liquid
Detail ID	Liquid: 10
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	550 mm
Liquid Density	0.0009984 kg./cm ³
Element From Node	10
Detail Type	Insulation
Detail ID	Insulation



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Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	50.8	mm
Thickness of Insulation	100	mm
Density	0.000352	kg./cm ³

Element#	2/13
Element From Node	20
Element To Node	30
Element Type	Cylinder
Description	LHS Ch Shell
Distance "FROM" to "TO"	800 mm
Inside Diameter	560 mm
Element Thickness	24 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	24 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	6100 KPa.
Design Temperature Internal Pressure	300 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	300 °C
Effective Diameter Multiplier	1.2
Material Name	SA-240 304
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	20
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	550 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	20
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	800 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

Element From Node	20
Detail Type	Nozzle
Detail ID	N3-Tube Inlet
Dist. from "FROM" Node / Offset dist	400 mm
Nozzle Diameter	10 in.
Nozzle Schedule	80
Nozzle Class	900
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	125.52 Kg
Grade of Attached Flange	GR 2.1
Nozzle Matl	SA-312 TP304

Element From Node	20
Detail Type	Nozzle
Detail ID	N4-Tube Outlet
Dist. from "FROM" Node / Offset dist	400 mm
Nozzle Diameter	10 in.
Nozzle Schedule	80
Nozzle Class	900
Layout Angle	180.0
Blind Flange (Y/N)	N



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Weight of Nozzle (Used if > 0)	125.52	Kgf
Grade of Attached Flange	GR 2.1	
Nozzle Matl	SA-312 TP304	

Element From Node	20	
Detail Type	Weight	
Detail ID	Partition Plate	
Dist. from "FROM" Node / Offset dist	400	mm
Miscellaneous Weight	400	Kgf
Offset from Element Centerline	0	mm

Element#	3/13	
Element From Node	30	
Element To Node	40	
Element Type	Flange	
Description	LHS ch flange	
Distance "FROM" to "TO"	167	mm
Flange Inside Diameter	564	mm
Element Thickness	122	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	130	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	6100	KPa.
Design Temperature Internal Pressure	300	°C
Design External Pressure	103.42	KPa.
Design Temperature External Pressure	300	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-965 F304	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	85.9	N./mm ²
Allowable Stress, Hydrotest	184.5	N./mm ²
Material Density	0.008027	kg./cm ³
P Number Thickness	0	mm
Yield Stress, Operating	129	N./mm ²
External Pressure Chart Name	HA-1	
UNS Number	S30400	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element From Node	30	
Detail Type	Liquid	
Detail ID	Liquid: 30	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	550	mm
Liquid Density	0.0009984	kg./cm ³

Element From Node	30	
Detail Type	Insulation	
Detail ID	Insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	180	mm
Thickness of Insulation	100	mm
Density	0.000352	kg./cm ³

Element#	4/13	
Element From Node	40	
Element To Node	50	
Element Type	Cylinder	
Description	Main Shell	
Distance "FROM" to "TO"	1500	mm



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Inside Diameter	560	mm
Element Thickness	24	mm
Internal Corrosion Allowance	0	mm
Nominal Thickness	24	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	6100	KPa.
Design Temperature Internal Pressure	300	°C
Design External Pressure	103.42	KPa.
Design Temperature External Pressure	300	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-240	304
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	85.9	N./mm ²
Allowable Stress, Hydrotest	184.5	N./mm ²
Material Density	0.008027	kg./cm ³
P Number Thickness	0	mm
Yield Stress, Operating	129	N./mm ²
External Pressure Chart Name	HA-1	
UNS Number	S30400	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	40
Detail Type	Liquid
Detail ID	Liquid: 40
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	553 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	40
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	1500 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

Element From Node	40
Detail Type	Nozzle
Detail ID	N1-Shell Inlet
Dist. from "FROM" Node / Offset dist	750 mm
Nozzle Diameter	10 in.
Nozzle Schedule	60
Nozzle Class	900
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	167.7 Kg
Grade of Attached Flange	GR 2.1
Nozzle Matl	SA-312 TP304

Element#	5/13
Element From Node	50
Element To Node	60
Element Type	Cylinder
Description	Main Shell
Distance "FROM" to "TO"	1000 mm
Inside Diameter	560 mm
Element Thickness	36 mm
Internal Corrosion Allowance	0 mm
Nominal Thickness	36 mm
External Corrosion Allowance	0 mm



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Design Internal Pressure	6100	KPa.
Design Temperature Internal Pressure	300	°C
Design External Pressure	103.42	KPa.
Design Temperature External Pressure	300	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-240	304
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	50	
Detail Type	Saddle	
Detail ID	Left Saddle	
Dist. from "FROM" Node / Offset dist	500	mm
Width of Saddle	335	mm
Height of Saddle at Bottom	850	mm
Saddle Contact Angle	168.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	530	mm
Thickness of Wear Plate	14	mm
Contact Angle, Wear Plate (degrees)	180.0	
Friction coefficient	0.30000001	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm

Element From Node	50	
Detail Type	Liquid	
Detail ID	Liquid: 50	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	553	mm
Liquid Density	0.0009984	kg./cm ³

Element From Node	50	
Detail Type	Insulation	
Detail ID	Insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	1000	mm
Thickness of Insulation	100	mm
Density	0.000352	kg./cm ³

Element From Node	50	
Detail Type	Lining	
Detail ID	Clad	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	1000	mm
Thickness of Lining	3	mm
Density	0.008	kg./cm ³

Element From Node	50	
Detail Type	Weight	
Detail ID	Fireproofing we	
Dist. from "FROM" Node / Offset dist	500	mm
Miscellaneous Weight	300	Kgf
Offset from Element Centerline	950	mm

Element From Node	50	
Detail Type	Weight	
Detail ID	Saddle 1	
Dist. from "FROM" Node / Offset dist	500	mm
Miscellaneous Weight	200	Kgf
Offset from Element Centerline	275	mm



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Element From Node	50
Detail Type	For./Mom.
Detail ID	Stack force
Dist. from "FROM" Node / Offset dist	513 mm
Force in X Direction	1040 Kg-f
Force in Y Direction	0 Kg-f
Force in Z Direction	1040 Kg-f
Moment about X Axis	0 Kg-m.
Moment about Y Axis	0 Kg-m.
Moment about Z Axis	0 Kg-m.
Force/Moment Combination Method	SRSS

Element From Node	50
Detail Type	For./Mom.
Detail ID	Stack force
Dist. from "FROM" Node / Offset dist	513 mm
Force in X Direction	711 Kg-f
Force in Y Direction	0 Kg-f
Force in Z Direction	711 Kg-f
Moment about X Axis	0 Kg-m.
Moment about Y Axis	0 Kg-m.
Moment about Z Axis	0 Kg-m.
Force/Moment Combination Method	SRSS

Element#	6/13
Element From Node	60
Element To Node	70
Element Type	Cylinder
Description	Main Shell
Distance "FROM" to "TO"	1500 mm
Inside Diameter	560 mm
Element Thickness	36 mm
Internal Corrosion Allowance	0 mm
Nominal Thickness	36 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	6100 KPa.
Design Temperature Internal Pressure	300 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	300 °C
Effective Diameter Multiplier	1.2
Material Name	SA-240 304
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	60
Detail Type	Liquid
Detail ID	Liquid: 60
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	553 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	60
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	1500 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

Element From Node	60
Detail Type	Lining
Detail ID	clad



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Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	1500	mm
Thickness of Lining	3	mm
Density	0.008	kg./cm ³

Element#	7/13
Element From Node	70
Element To Node	80
Element Type	Cylinder
Description	Main Shell
Distance "FROM" to "TO"	1220 mm
Inside Diameter	560 mm
Element Thickness	36 mm
Internal Corrosion Allowance	0 mm
Nominal Thickness	36 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	6100 KPa.
Design Temperature Internal Pressure	300 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	300 °C
Effective Diameter Multiplier	1.2
Material Name	SA-240 304
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	70
Detail Type	Liquid
Detail ID	Liquid: 70
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	553 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	70
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	1220 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

Element From Node	70
Detail Type	Lining
Detail ID	Clad
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Lining	1220 mm
Thickness of Lining	3 mm
Density	0.008 kg./cm ³

Element From Node	70
Detail Type	Weight
Detail ID	TOP HE OPERATING
Dist. from "FROM" Node / Offset dist	395.65 mm
Miscellaneous Weight	3500 Kg
Offset from Element Centerline	1700 mm

Element From Node	70
Detail Type	Weight
Detail ID	TOP HE TEST
Dist. from "FROM" Node / Offset dist	395.65 mm
Miscellaneous Weight	3500 Kg
Offset from Element Centerline	1700 mm



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Element From Node	70
Detail Type	Weight
Detail ID	TOP HE EMPTY
Dist. from "FROM" Node / Offset dist	371.64 mm
Miscellaneous Weight	14500 Kg
Offset from Element Centerline	1700 mm

Element#	8/13
Element From Node	80
Element To Node	90
Element Type	Cylinder
Description	Main Shell
Distance "FROM" to "TO"	1500 mm
Inside Diameter	560 mm
Element Thickness	36 mm
Internal Corrosion Allowance	0 mm
Nominal Thickness	36 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	6100 KPa.
Design Temperature Internal Pressure	300 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	300 °C
Effective Diameter Multiplier	1.2
Material Name	SA-240 304
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	80
Detail Type	Liquid
Detail ID	Liquid: 40
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	553 mm
Liquid Density	0.0009984 kg./cm³

Element From Node	80
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	1500 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm³

Element From Node	80
Detail Type	Lining
Detail ID	Clad
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Lining	1500 mm
Thickness of Lining	3 mm
Density	0.008 kg./cm³

Element#	9/13
Element From Node	90
Element To Node	100
Element Type	Cylinder
Description	Main Shell
Distance "FROM" to "TO"	1000 mm
Inside Diameter	560 mm
Element Thickness	36 mm
Internal Corrosion Allowance	0 mm
Nominal Thickness	36 mm



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External Corrosion Allowance	0	mm
Design Internal Pressure	6100	KPa.
Design Temperature Internal Pressure	300	°C
Design External Pressure	103.42	KPa.
Design Temperature External Pressure	300	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-240	304
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	90	
Detail Type	Saddle	
Detail ID	New Saddle	
Dist. from "FROM" Node / Offset dist	500	mm
Width of Saddle	335	mm
Height of Saddle at Bottom	850	mm
Saddle Contact Angle	168.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	530	mm
Thickness of Wear Plate	14	mm
Contact Angle, Wear Plate (degrees)	180.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	720	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm

Element From Node	90	
Detail Type	Liquid	
Detail ID	Liquid: 90	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	553	mm
Liquid Density	0.0009984	kg./cm³

Element From Node	90	
Detail Type	Insulation	
Detail ID	Insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	1000	mm
Thickness of Insulation	100	mm
Density	0.000352	kg./cm³

Element From Node	90	
Detail Type	Lining	
Detail ID	Clad	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	1000	mm
Thickness of Lining	3	mm
Density	0.008	kg./cm³

Element From Node	90	
Detail Type	Weight	
Detail ID	Fireproofing we	
Dist. from "FROM" Node / Offset dist	500	mm
Miscellaneous Weight	300	Kgf
Offset from Element Centerline	950	mm

Element From Node	90	
Detail Type	Weight	
Detail ID	saddle 2	
Dist. from "FROM" Node / Offset dist	500	mm
Miscellaneous Weight	200	Kgf
Offset from Element Centerline	275	mm



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Element From Node	90
Detail Type	For./Mom.
Detail ID	stack force
Dist. from "FROM" Node / Offset dist	513 mm
Force in X Direction	1040 Kgf
Force in Y Direction	0 Kgf
Force in Z Direction	1040 Kgf
Moment about X Axis	0 Kg-m.
Moment about Y Axis	0 Kg-m.
Moment about Z Axis	0 Kg-m.
Force/Moment Combination Method	SRSS

Element From Node	90
Detail Type	For./Mom.
Detail ID	stack force
Dist. from "FROM" Node / Offset dist	513 mm
Force in X Direction	711 Kgf
Force in Y Direction	0 Kgf
Force in Z Direction	711 Kgf
Moment about X Axis	0 Kg-m.
Moment about Y Axis	0 Kg-m.
Moment about Z Axis	0 Kg-m.
Force/Moment Combination Method	SRSS

Element#	10/13
Element From Node	100
Element To Node	110
Element Type	Cylinder
Description	Main Shell
Distance "FROM" to "TO"	1500 mm
Inside Diameter	560 mm
Element Thickness	24 mm
Internal Corrosion Allowance	0 mm
Nominal Thickness	24 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	6100 KPa.
Design Temperature Internal Pressure	300 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	300 °C
Effective Diameter Multiplier	1.2
Material Name	SA-240 304
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	100
Detail Type	Liquid
Detail ID	Liquid: 100
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	553 mm
Liquid Density	0.0009984 kg./cm³

Element From Node	100
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	1500 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm³

Element From Node	100
Detail Type	Lining



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Detail ID	Clad
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Lining	1500 mm
Thickness of Lining	3 mm
Density	0.008 kg./cm ³
Element From Node	100
Detail Type	Nozzle
Detail ID	N2-Shell outlet
Dist. from "FROM" Node / Offset dist	750 mm
Nozzle Diameter	10 in.
Nozzle Schedule	60
Nozzle Class	900
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	167.7 Kg
Grade of Attached Flange	GR 2.1
Nozzle Matl	SA-312 TP304

Element#	11/13
Element From Node	110
Element To Node	120
Element Type	Flange
Description	RHS Ch Flange
Distance "FROM" to "TO"	171 mm
Flange Inside Diameter	560 mm
Element Thickness	122 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	130 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	6100 KPa.
Design Temperature Internal Pressure	300 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	300 °C
Effective Diameter Multiplier	1.2
Material Name	SA-965 F304
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	85.9 N./mm ²
Allowable Stress, Hydrotest	184.5 N./mm ²
Material Density	0.008027 kg./cm ³
P Number Thickness	0 mm
Yield Stress, Operating	129 N./mm ²
External Pressure Chart Name	HA-1
UNS Number	S30400
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element From Node	110
Detail Type	Liquid
Detail ID	Liquid: 50
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	550 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	110
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	180 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³



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Element#	12/13
Element From Node	120
Element To Node	130
Element Type	Cylinder
Description	RHS Ch Shell
Distance "FROM" to "TO"	600 mm
Inside Diameter	560 mm
Element Thickness	24 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	24 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	6100 KPa.
Design Temperature Internal Pressure	300 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	300 °C
Effective Diameter Multiplier	1.2
Material Name	SA-240 304
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	85.9 N./mm ²
Allowable Stress, Hydrotest	184.5 N./mm ²
Material Density	0.008027 kg./cm ³
P Number Thickness	0 mm
Yield Stress, Operating	129 N./mm ²
External Pressure Chart Name	HA-1
UNS Number	S30400
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	120
Detail Type	Liquid
Detail ID	Liquid: 60
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	550 mm
Liquid Density	0.0009984 kg./cm ³
Element From Node	120
Detail Type	Insulation
Detail ID	Hot Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	600 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

Element#	13/13
Element From Node	130
Element To Node	140
Element Type	Elliptical
Description	RHS Dish End
Distance "FROM" to "TO"	50.8 mm
Inside Diameter	560 mm
Element Thickness	22 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	28 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	6100 KPa.
Design Temperature Internal Pressure	300 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	300 °C
Effective Diameter Multiplier	1.2



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Material Name	SA-240 304
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No
Element From Node	130
Detail Type	Liquid
Detail ID	Liquid: 70
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	550 mm
Liquid Density	0.0009984 kg./cm ³
Element From Node	130
Detail Type	Insulation
Detail ID	Hot Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	188.3 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

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XY Coordinate Calculations: Step: 2 10:04am Sep 9,2026

XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
LHS Dish End		50.8	...	50.8	...
LHS Ch Shell		850.8	...	800	...
LHS ch flange		1020.97	...	167	...
Main Shell		2710.98	...	1500	...
Main Shell		3710.98	...	1000	...
Main Shell		5210.97	...	1500	...
Main Shell		6430.97	...	1220	...
Main Shell		7930.97	...	1500	...
Main Shell		8930.97	...	1000	...
Main Shell		10431	...	1500	...
RHS Ch Flange		10795.2	...	171	...
RHS Ch Shell		11395.2	...	600	...
RHS Dish End		11446	...	50.8	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Flange Input Data Values

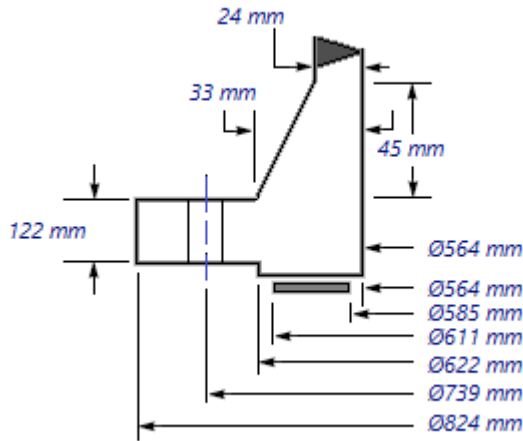
Description: LHS Ch Flang :

LHS ch flange

Flange Type	Integral Weld Neck		
Design Pressure	P	6105.39	KPa.
Design Temperature		300	°C
Internal Corrosion Allowance	ci	1.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	564.000	mm
Flange Outside Diameter	A	824.000	mm
Flange Thickness	t	122.0000	mm
Thickness of Hub at Small End	go	24.0000	mm
Thickness of Hub at Large End	gl	33.0000	mm
Length of Hub	h	45.0000	mm
Flange Material	SA-965 F304		
Flange Material UNS number	S30400		
Flange Allowable Stress At Temperature	Sfo	85.90	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material	SA-193 B7		
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	739.000	mm
Nominal Bolt Diameter	a	38.1000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts	24		
Flange Face Outside Diameter	Fod	622.000	mm
Flange Face Inside Diameter	Fid	564.000	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	611.000	mm
Gasket Inside Diameter	Gi	585.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm
Length of Partition Gasket	lp	550.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	3.0000	
Partition Gasket Design Seating Stress	yPart	68.95	N./mm ²



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ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (6105 \cdot (564/2 + 1)) / (85.9 \cdot 1 - 0.6 \cdot 6105) + Ca$$

$$= 22.0115 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (85.9 \cdot 1 \cdot 23) / (283 + 0.6 \cdot 23)$$

$$= 6656.322 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 24) / (282 + 0.6 \cdot 24)$$

$$= 11173.238 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	121.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	566.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	32.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	23.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	54.500 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	13.000 mm
Basic Gasket Width, $b_o = N/2$	6.500 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	6.425 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	598.151 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 598^2 \cdot 6105$$

$$= 174953.000 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot l_p \cdot b_{Part} \cdot m_{Part} \cdot P$$

$$= 2 \cdot 6.42 \cdot \pi \cdot 598 \cdot 3 \cdot 6105$$

$$+ 2 \cdot 550 \cdot 5 \cdot 3 \cdot 6105$$

$$= 55371.785 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 566^2 \cdot 6105$$

$$= 156650.891 \text{ Kg}$$



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Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 174953 - 156651$$

$$= 18302.121 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(174953 + 55372 + 0, 0)$$

$$= 230324.797 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 6.42 \cdot \pi \cdot 598 + 68.9 \cdot 5 \cdot 550$$

$$= 104215.609 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((230325 + 0 + 4 \cdot 0 / 598) / 172, 104216 / 172)$$

$$= 131.325 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmx]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 38.1 + 6 \cdot 121 / (3 + 0.5)$$

$$= 283.629 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 739 \cdot \sin(\pi / 24)$$

$$= 96.459 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ Bs / (2a + t) }, 1)$$

$$= \max(\sqrt{ 96.5 / (2 \cdot 38.1 + 121) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	131.325	217.548	
Radial Distance between Hub and Bolts:	50.800	54.500	
Radial Distance between Bolts and Edge:	38.100	42.500	
Circ. Spacing between the Bolts:	82.550	96.459	283.629

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 218 \cdot 172 / (68.9 \cdot \pi (611 + 585))$$

$$= 14.443 \text{ mm} [\text{Note: Exceeds actual gasket width, 13.000 }]$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 172 \cdot 218$$

$$= 381546.03 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 230325 - 174953$$

$$= 55371.79 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$



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$$= (739 - 598) / 2$$

$$= 70.4246 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + hg)$$

$$= 0.5 ((739 - 566) / 2 + 70.4)$$

$$= 78.4623 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (739 - 566 - 32) / 2$$

$$= 70.5000 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	156651.	70.5000	1.0000	11044.
Face Pressure, Mt	18302.	78.4623	1.0000	1436.
Gasket Load, Mg	55372.	70.4246	1.0000	3900.
Gasket Seating, Matm	381546.	70.4246	1.0000	26871.

Total Moment for Operation, Mo 16380. Kg-m.
 Total Moment for Gasket seating, Mg 26871. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 114.096 mm
 Hub Ratio, h/h0 = HL / H0 0.394
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.391

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.870
 Factor V per 4.16.5 0.368
 Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 824 / 566$$

$$= 1.456$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169 (1.46^2 \cdot \log_{10}(1.46))) / (1.46^2 - 1)) / (1.46 - 1)$$

$$= 5.340$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.46^2 (1 + 8.55246 \cdot \log_{10}(1.46)) - 1) / ((1.0472 + 1.9448 \cdot 1.46^2) \cdot (1.46 - 1))$$

$$= 1.730$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$$

$$= (1.46^2 (1 + 8.55246 \cdot \log_{10}(1.46)) - 1) / (1.36136 (1.46^2 - 1) \cdot (1.46 - 1))$$

$$= 5.868$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.46^2 + 1) / (1.46^2 - 1)$$

$$= 2.787$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g0^2 \cdot h0 / V$$

$$= 5.87 \cdot 23^2 \cdot 114 / 0.37$$

$$= 962630.125 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:



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$$= F / h_0$$

$$= 0.87/114$$

$$= 0.008 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (121 \cdot 0.0076 + 1) / 1.73 + 121^3/962630$$

$$= 2.952$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g^{1^2})$$

$$= (1 \cdot 16380/566) / (2.95 \cdot 32^2)$$

$$= 93.89 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g^{1^2})$$

$$= (1 \cdot 26871/566) / (2.95 \cdot 32^2)$$

$$= 154.02 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.23 \cdot 16380/566) / (2.95 \cdot 121^2)$$

$$= 14.65 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.23 \cdot 26871/566) / (2.95 \cdot 121^2)$$

$$= 24.03 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (5.34 \cdot 16380 / (121^2 \cdot 566)) - 2.79 \cdot 14.6$$

$$= 62.68 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (5.34 \cdot 26871 / (121^2 \cdot 566)) - 2.79 \cdot 24$$

$$= 102.83 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (93.9 + \max(14.6, 62.7)) / 2$$

$$= 78.28 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (154 + \max(24, 103)) / 2$$

$$= 128.43 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 230325/218$$

$$= 103.83 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (104216/218)$$

$$= 46.98 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

Operating		Gasket Seating	
Actual	Allowed	Actual	Allowed



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Longitudinal Hub	93.89	128.85	154.02	207.00
Radial Flange	14.65	85.90	24.03	138.00
Tangential Flange	62.68	85.90	102.83	138.00
Maximum Average	78.28	85.90	128.43	138.00
Bolting	103.83	172.00	46.98	172.00

Minimum Required Flange Thickness 116.738 mm

Estimated M.A.W.P. (Operating) 6656 KPa.
 Estimated Finished Weight of Flange at given Thk. 296.7 kg.
 Estimated Unfinished Weight of Forging at given Thk 380.0 kg.

Flange Stress results per WRC-538 at computed S_{fmax} (N/mm²):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	265.21	294.86	0.899	265.21	409.99	0.647
Tangential Flange	St	177.07	221.14	0.801	177.07	307.49	0.576
Tangential Flange	Sto	80.17	147.43	0.544	80.17	205.00	0.391
Combined	Sh/f + St	442.29	442.29	1.000	442.29	614.99	0.719
Combined	Sr + St	218.45	442.29	0.494	218.45	614.99	0.355
Combined	Sh + Sto	345.38	442.29	0.781	345.38	614.99	0.562
Radial Flange	Sr	41.38	294.86	0.140	41.38	409.99	0.101
Maximum Stress Ratio				1.000			0.719



Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)
 = max(1; 1.11/1.44)
 = 1.0000

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if h/ho < 0.35, else max (0.25; fn3)
 = max(0.25; 0.44/1.47) if h/ho < 0.35, else max (0.25; 0.23/0.76)
 = 0.3023

In this case, the hub ratio h/ho, 0.3944 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	296.2 N./mm ²
Target assembly gasket stress	SgT	208.0 N./mm ²
Maximum permissible gasket stress	Sgmax	75.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700



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Root Area of a Single Bolt	Abs	9.0645 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	205.0 N./mm ²
Flange Material Operating Yield Stress	Syo	129.0 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.225 deg

Gasket Contact Area [Ag]:
 $= \pi/4 (GO.D.^2 - GI.D.^2) + (PartFactor \cdot Ip \cdot bPart)$
 $= \pi/4 (611^2 - 585^2) + (0.5 \cdot 550 \cdot 5)$
 $= 258.0 \text{ cm}^2$

Target Bolt Stress [Sbsel]: (Eq. O-1)
 $= SgT \cdot Ag / (nb \cdot Ab)$
 $= 208 \cdot 258 / (24 \cdot 9.06)$
 $= 246.7 \text{ N./mm}^2$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)
 $= \min(Sbsel; Sbmax)$
 $= \min(247; 508)$
 $= 246.7 \text{ N./mm}^2$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)
 $= \max(Sbsel; Sbmin)$
 $= \max(247; 290)$
 $= 290.0 \text{ N./mm}^2$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)
 $= \min(Sbsel; Sfmax)$
 $= \min(290; 296)$
 $= 290.0 \text{ N./mm}^2$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)
 $:: Sbsel \geq Sgmin-S(Ag / (nb \cdot Abs))$
 $:: 290 \geq 140 (258 / (24 \cdot 9.06))$
 $:: 290.0 \geq 166.0 \text{ N./mm}^2 \text{ Check Passes}$

Check if Gasket Operating Stress is maintained. (Eq. O-8)
 $:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$
 $:: 290 \geq (97 \cdot 258 + \pi/4 \cdot 6105 \cdot 585^2) / (0.7 \cdot 9.06 \cdot 24)$
 $:: 290.0 \geq 272.1 \text{ N./mm}^2 \text{ Check Passes}$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)
 $:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$
 $:: 290 \leq 75 (258 / (24 \cdot 9.06))$
 $:: 290.0 \leq 88.9 \text{ N./mm}^2 \text{ Check Fails}$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)
 $:: Sbsel \leq Sfmax(thetagmax / thetafmax)$
 $:: 290 \leq 296 (1/0.23)$
 $:: 290.0 \leq 1313.5 \text{ N./mm}^2 \text{ Check Passes}$

Assembly Bolt Torque [Tb]: (Eq. O-2)
 $= Sbsel \cdot K \cdot Abs \cdot \phi b$
 $= 290 \cdot 0.2 \cdot 9.06 \cdot 38.1$
 $= 204.3 \text{ Kg-m.}$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 86 N./mm²
 Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:



Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 26871 / 1 \cdot 9807 \cdot 0.37 / (2.7 \cdot 195128 \cdot 23^2 \cdot 114 \cdot 0.3)$
 $= 0.530$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 16380 / 1 \cdot 9807 \cdot 0.37 / (2.7 \cdot 175604 \cdot 23^2 \cdot 114 \cdot 0.3)$
 $= 0.359$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 26871 / 1 \cdot 9807 \cdot 0.37 / (2.95 \cdot 195128 \cdot 23^2 \cdot 114 \cdot 0.3)$
 $= 0.484$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 16380 / 1 \cdot 9807 \cdot 0.37 / (2.95 \cdot 175604 \cdot 23^2 \cdot 114 \cdot 0.3)$
 $= 0.328$ (should be ≤ 1)

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	92.20	207.00	150.36	207.00
Radial Flange	14.98	138.00	24.42	138.00
Tangential Flange	60.48	138.00	98.63	138.00
Maximum Average	76.34	138.00	124.50	138.00
Bolting	103.83	172.00	46.98	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 114.275 mm

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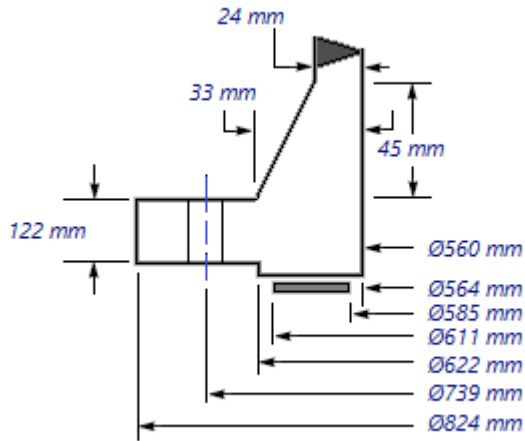
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Flange Input Data Values

Description: New Flange :

RHS Ch Flange

Flange Type	Integral Weld Neck		
Design Pressure	P	6105.39	KPa.
Design Temperature		300	°C
Internal Corrosion Allowance	ci	1.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	560.000	mm
Flange Outside Diameter	A	824.000	mm
Flange Thickness	t	122.0000	mm
Thickness of Hub at Small End	go	24.0000	mm
Thickness of Hub at Large End	gl	33.0000	mm
Length of Hub	h	45.0000	mm
Flange Material	SA-965 F304		
Flange Material UNS number	S30400		
Flange Allowable Stress At Temperature	Sfo	85.90	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material	SA-193 B7		
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	739.000	mm
Nominal Bolt Diameter	a	38.1000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts	24		
Flange Face Outside Diameter	Fod	622.000	mm
Flange Face Inside Diameter	Fid	564.000	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	611.000	mm
Gasket Inside Diameter	Gi	585.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm
Length of Partition Gasket	lp	550.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	3.0000	
Partition Gasket Design Seating Stress	yPart	68.95	N./mm ²



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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + C_a)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (6105 \cdot (560/2 + 1)) / (85.9 \cdot 1 - 0.6 \cdot 6105) + C_a$$

$$= 21.8630 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (85.9 \cdot 1 \cdot 23) / (281 + 0.6 \cdot 23)$$

$$= 6701.480 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 24) / (280 + 0.6 \cdot 24)$$

$$= 11249.145 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	121.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	562.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	32.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	23.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	56.500 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	13.000 mm
Basic Gasket Width, $b_o = N/2$	6.500 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	6.425 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	598.151 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 598^2 \cdot 6105$$

$$= 174953.000 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot l_p \cdot b_{Part} \cdot m_{Part} \cdot P$$

$$= 2 \cdot 6.42 \cdot \pi \cdot 598 \cdot 3 \cdot 6105$$

$$+ 2 \cdot 550 \cdot 5 \cdot 3 \cdot 6105$$

$$= 55371.785 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 562^2 \cdot 6105$$

$$= 154444.547 \text{ Kg}$$



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Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 174953 - 154445$$

$$= 20508.461 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(174953 + 55372 + 0, 0)$$

$$= 230324.797 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 6.42 \cdot \pi \cdot 598 + 68.9 \cdot 5 \cdot 550$$

$$= 104215.609 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((230325 + 0 + 4 \cdot 0 / 598) / 172, 104216 / 172)$$

$$= 131.325 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmx]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 38.1 + 6 \cdot 121 / (3 + 0.5)$$

$$= 283.629 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 739 \cdot \sin(\pi / 24)$$

$$= 96.459 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ Bs / (2a + t) }, 1)$$

$$= \max(\sqrt{ 96.5 / (2 \cdot 38.1 + 121) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	131.325	217.548	
Radial Distance between Hub and Bolts:	50.800	56.500	
Radial Distance between Bolts and Edge:	38.100	42.500	
Circ. Spacing between the Bolts:	82.550	96.459	283.629

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 218 \cdot 172 / (68.9 \cdot \pi (611 + 585))$$

$$= 14.443 \text{ mm} [\text{Note: Exceeds actual gasket width, 13.000 }]$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 172 \cdot 218$$

$$= 381546.03 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 230325 - 174953$$

$$= 55371.79 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$



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$$= (739 - 598) / 2$$

$$= 70.4246 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + hg)$$

$$= 0.5 ((739 - 562) / 2 + 70.4)$$

$$= 79.4623 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (739 - 562 - 32) / 2$$

$$= 72.5000 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	154445.	72.5000	1.0000	11197.
Face Pressure, Mt	20508.	79.4623	1.0000	1630.
Gasket Load, Mg	55372.	70.4246	1.0000	3900.
Gasket Seating, Matm	381546.	70.4246	1.0000	26871.

Total Moment for Operation, Mo 16727. Kg-m.
 Total Moment for Gasket seating, Mg 26871. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 113.693 mm
 Hub Ratio, h/h0 = HL / H0 0.396
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.391

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.870
 Factor V per 4.16.5 0.368
 Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 824 / 562$$

$$= 1.466$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169 (1.47^2 \cdot \log_{10}(1.47))) / (1.47^2 - 1)) / (1.47 - 1)$$

$$= 5.244$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.47^2 (1 + 8.55246 \cdot \log_{10}(1.47)) - 1) / ((1.0472 + 1.9448 \cdot 1.47^2) \cdot (1.47 - 1))$$

$$= 1.725$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$$

$$= (1.47^2 (1 + 8.55246 \cdot \log_{10}(1.47)) - 1) / (1.36136 (1.47^2 - 1) \cdot (1.47 - 1))$$

$$= 5.763$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.47^2 + 1) / (1.47^2 - 1)$$

$$= 2.740$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g0^2 \cdot h0 / V$$

$$= 5.76 \cdot 23^2 \cdot 114 / 0.37$$

$$= 943036.812 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:



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$$= F / h_0$$

$$= 0.87/114$$

$$= 0.008 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (121 \cdot 0.0077 + 1) / 1.73 + 121^3/943037$$

$$= 2.995$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g^{1^2})$$

$$= (1 \cdot 16727/562) / (2.99 \cdot 32^2)$$

$$= 95.17 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g^{1^2})$$

$$= (1 \cdot 26871/562) / (2.99 \cdot 32^2)$$

$$= 152.89 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.23 \cdot 16727/562) / (2.99 \cdot 121^2)$$

$$= 14.88 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.23 \cdot 26871/562) / (2.99 \cdot 121^2)$$

$$= 23.90 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (5.24 \cdot 16727 / (121^2 \cdot 562)) - 2.74 \cdot 14.9$$

$$= 63.80 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (5.24 \cdot 26871 / (121^2 \cdot 562)) - 2.74 \cdot 23.9$$

$$= 102.49 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (95.2 + \max(14.9, 63.8)) / 2$$

$$= 79.49 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (153 + \max(23.9, 102)) / 2$$

$$= 127.69 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 230325/218$$

$$= 103.83 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (104216/218)$$

$$= 46.98 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating
Actual	Allowed	Actual	Allowed



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Longitudinal Hub	95.17	128.85	152.89	207.00
Radial Flange	14.88	85.90	23.90	138.00
Tangential Flange	63.80	85.90	102.49	138.00
Maximum Average	79.49	85.90	127.69	138.00
Bolting	103.83	172.00	46.98	172.00

Minimum Required Flange Thickness 116.357 mm

Estimated M.A.W.P. (Operating) 6598 KPa.
 Estimated Finished Weight of Flange at given Thk. 300.1 kg.
 Estimated Unfinished Weight of Forging at given Thk 384.7 kg.

Flange Stress results per WRC-538 at computed S_{fmax} (N/mm²):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	264.79	294.86	0.898	264.79	409.99	0.646
Tangential Flange	St	177.50	221.14	0.803	177.50	307.49	0.577
Tangential Flange	Sto	80.01	147.43	0.543	80.01	205.00	0.390
Combined	Sh/f + St	442.29	442.29	1.000	442.29	614.99	0.719
Combined	Sr + St	218.88	442.29	0.495	218.88	614.99	0.356
Combined	Sh + Sto	344.80	442.29	0.780	344.80	614.99	0.561
Radial Flange	Sr	41.39	294.86	0.140	41.39	409.99	0.101
Maximum Stress Ratio				1.000			0.719



Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

$$= \max(1; F1_{num} / F1_{den})$$

$$= \max(1; 1.11/1.44)$$

$$= 1.0000$$

Hub Stress Correction Factor [f2]:

$$= \max(0.25; f_{n2}) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; f_{n3})$$

$$= \max(0.25; 0.44/1.47) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; 0.23/0.76)$$

$$= 0.3022$$

In this case, the hub ratio $h/h0$, 0.3958 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, Sf_{max} , until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sb_{max}	507.5 N./mm ²
Minimum permissible bolt stress	Sb_{min}	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sf_{max}	297.9 N./mm ²
Target assembly gasket stress	SgT	208.0 N./mm ²
Maximum permissible gasket stress	Sg_{max}	75.0 N./mm ²
Minimum gasket operating stress	Sg_{min-O}	97.0 N./mm ²
Minimum gasket seating stress	Sg_{min-S}	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	$Phig$	0.700



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Root Area of a Single Bolt	Abs	9.0645 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	205.0 N./mm ²
Flange Material Operating Yield Stress	Syo	129.0 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.224 deg

Gasket Contact Area [Ag]:
 $= \pi/4 (GO.D.^2 - GI.D.^2) + (PartFactor \cdot Ip \cdot bPart)$
 $= \pi/4 (611^2 - 585^2) + (0.5 \cdot 550 \cdot 5)$
 $= 258.0 \text{ cm}^2$

Target Bolt Stress [Sbsel]: (Eq. O-1)
 $= SgT \cdot Ag / (nb \cdot Ab)$
 $= 208 \cdot 258 / (24 \cdot 9.06)$
 $= 246.7 \text{ N./mm}^2$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)
 $= \min(Sbsel; Sbmax)$
 $= \min(247; 508)$
 $= 246.7 \text{ N./mm}^2$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)
 $= \max(Sbsel; Sbmin)$
 $= \max(247; 290)$
 $= 290.0 \text{ N./mm}^2$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)
 $= \min(Sbsel; Sfmax)$
 $= \min(290; 298)$
 $= 290.0 \text{ N./mm}^2$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)
 $:: Sbsel \geq Sgmin-S(Ag / (nb \cdot Abs))$
 $:: 290 \geq 140 (258 / (24 \cdot 9.06))$
 $:: 290.0 \geq 166.0 \text{ N./mm}^2 \text{ Check Passes}$

Check if Gasket Operating Stress is maintained. (Eq. O-8)
 $:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$
 $:: 290 \geq (97 \cdot 258 + \pi/4 \cdot 6105 \cdot 585^2) / (0.7 \cdot 9.06 \cdot 24)$
 $:: 290.0 \geq 272.1 \text{ N./mm}^2 \text{ Check Passes}$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)
 $:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$
 $:: 290 \leq 75 (258 / (24 \cdot 9.06))$
 $:: 290.0 \leq 88.9 \text{ N./mm}^2 \text{ Check Fails}$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)
 $:: Sbsel \leq Sfmax(thetagmax / thetafmax)$
 $:: 290 \leq 298 (1/0.22)$
 $:: 290.0 \leq 1329.1 \text{ N./mm}^2 \text{ Check Passes}$

Assembly Bolt Torque [Tb]: (Eq. O-2)
 $= Sbsel \cdot K \cdot Abs \cdot \phi b$
 $= 290 \cdot 0.2 \cdot 9.06 \cdot 38.1$
 $= 204.3 \text{ Kg-m.}$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 86 N./mm²
 Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:



Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 26871 / 1 \cdot 9807 \cdot 0.37 / (2.72 \cdot 195128 \cdot 23^2 \cdot 114 \cdot 0.3)$
 $= 0.527$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 16727 / 1 \cdot 9807 \cdot 0.37 / (2.72 \cdot 175604 \cdot 23^2 \cdot 114 \cdot 0.3)$
 $= 0.365$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 26871 / 1 \cdot 9807 \cdot 0.37 / (2.99 \cdot 195128 \cdot 23^2 \cdot 114 \cdot 0.3)$
 $= 0.478$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 16727 / 1 \cdot 9807 \cdot 0.37 / (2.99 \cdot 175604 \cdot 23^2 \cdot 114 \cdot 0.3)$
 $= 0.331$ (should be ≤ 1)

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	93.47	207.00	149.30	207.00
Radial Flange	15.21	138.00	24.29	138.00
Tangential Flange	61.56	138.00	98.33	138.00
Maximum Average	77.51	138.00	123.82	138.00
Bolting	103.83	172.00	46.98	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 113.919 mm

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 Internal Pressure Calculations: Step: 5 10:04am Sep 9,2026

Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd KPa.	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) N./mm ²	Element#
LHS Dish End		6105.4	28	1	560	85.9	1
LHS Ch Shell		6105.4	24	1	560	85.9	2
LHS ch flange		6105.4	130	1	564	85.9	3
Main Shell		6105.4	24	...	560	85.9	4
Main Shell		6105.4	36	...	560	85.9	5
Main Shell		6105.4	36	...	560	85.9	6
Main Shell		6105.4	36	...	560	85.9	7
Main Shell		6105.4	36	...	560	85.9	8
Main Shell		6105.4	36	...	560	85.9	9
Main Shell		6105.4	24	...	560	85.9	10
RHS Ch Flange		6105.4	130	1	560	85.9	11
RHS Ch Shell		6105.4	24	1	560	85.9	12
RHS Dish End		6105.4	28	1	560	85.9	13

Element Required Thickness and MAWP :

From	To	Design Pressure KPa.	M.A.W.P. Corroded KPa.	M.A.P. New & Cold KPa.	Minimum Thickness mm	Required Thickness mm	Element#
LHS Dish End		6100	6396	10758	22	21.0214	1
LHS Ch Shell		6100	6696	11249	24	21.8631	2
LHS ch flange		6100	6651	10114	122	116.738	3
Main Shell		6100	6997	11249	24	20.7889	4
Main Shell		6100	10247	16471	36	20.7889	5
Main Shell		6100	10247	16471	36	20.7889	6
Main Shell		6100	10247	16471	36	20.7889	7
Main Shell		6100	10247	16471	36	20.7889	8
Main Shell		6100	10247	16471	36	20.7889	9
Main Shell		6100	6997	11249	24	20.7889	10
RHS Ch Flange		6100	6593	10114	122	116.357	11
RHS Ch Shell		6100	6696	11249	24	21.8631	12
RHS Dish End		6100	6396	10758	22	21.0214	13

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:
 For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 6175 KPa.
 Shell Side MAPnc = 11249 KPa.
 Channel Side MAWP = 6396 KPa.
 Channel Side MAPnc = 10114 KPa.

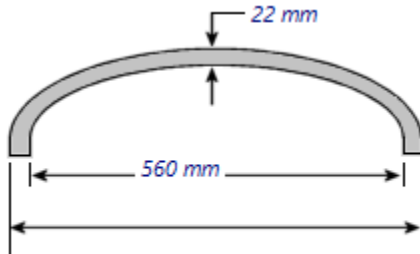
Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: LHS Dish End

Elliptical Head Design Information:			
Design Pressure P	6105.400 KPa.	Int. Design Temperature	300 °C
Material	SA-240 304	Ext. Design Temperature	300 °C
Length	50.800 mm	External Pressure Chart	HA-1
		UNS Number	S30400
Int. Corr. All. c	1.0000 mm	Allowable Stress (ope) S	85.9 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	560.000 mm	Actual Stress	81.9 N./mm ²
Spec. Min. Thk t,e, or sa	22.0000 mm	Spec. Nominal Thk.	28.000 mm
Surface Area	5096.3 cm ²	SG of Contents	1.000
Element Volume	35506.5 cm ³	Weight of Contents	35.5 Kg
Empty Weight	114.1 Kg	Operating Weight	149.6 Kg
		Comp. All. Ope. Stress	75.9 N./mm ²
		Aspect Ratio	2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 7.000 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.500 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (6105 \cdot 562 \cdot 1) / (2 \cdot 85.9 \cdot 1 - 0.2 \cdot 6105)$$

$$= 20.0214 + 1.0000 = 21.0214 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 5.401 KPa.

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 85.9 \cdot 1 \cdot 21) / (1 \cdot 562 + 0.2 \cdot 21)$$

$$= 6402 - 5.4 = 6396 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 22) / (1 \cdot 560 + 0.2 \cdot 22)$$

$$= 10758 \text{ KPa.}$$



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 FileName : 188-C1-C2-C3-C4- Solid SS Top 2nd stack Page 47 of 223
 Element: LHS Dish End Step: 6 10:04am Sep 9,2026

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (6105 \cdot (1 \cdot 562 + 0.2 \cdot 21)) / (2 \cdot 1 \cdot 21)$$

$$= 81.926 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (6105 \cdot 281) / (85.9 \cdot 1 - 0.6 \cdot 6105) + 1$$

$$= 21.863 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

[Less Operating Hydrostatic Head Pressure of 5.401 KPa.](#)

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (85.9 \cdot 1 \cdot 27) / (281 + 0.6 \cdot 27)$$

$$= 7803 - 5.4 = 7798 \text{ KPa.}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6$$

$$= (2 + (562 / (2 \cdot 141)))^2 / 6$$

$$= 0.995280$$

% Elongation per Table UG-79-1 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 19.231 %

Note: Please Check Requirements of Table UHA-44 for Elongation limits.

SA-240 304, Min Metal Temp without impact per UHA-51: -196 °C

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 7.000 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.500 mm

EXTERNAL PRESSURE RESULTS for: LHS Dish End

Elliptical Head From 10 to 20 Ext. Chart: HA-1 at 300 °C

Elastic Modulus from Chart: HA-1 at 300 °C: 170271248 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
21.000	604.00	28.76	0.0048289	59.16

$$MAEP = B / (K_0 \cdot Do / t) = 59.2 / (0.9 \cdot 28.8) = 2285 \text{ KPa.}$$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
1.748	604.00	345.49	0.0004020	32.16

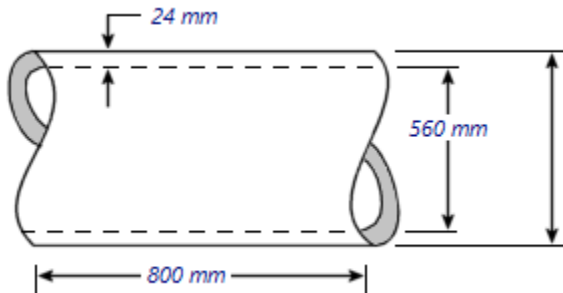
$$MAEP = B / (K_0 \cdot Do / t) = 32.2 / (0.9 \cdot 345) = 103 \text{ KPa.}$$

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Internal Pressure Calculation Results:

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Internal Pressure Results for: LHS Ch Shell

Cylindrical Shell Design Information:			
Design Pressure P	6105.400 KPa.	Int. Design Temperature	300 °C
Material	SA-240 304	Ext. Design Temperature	300 °C
Length	800.000 mm	External Pressure Chart	HA-1
		UNS Number	S30400
Int. Corr. All. c	1.0000 mm	Allowable Stress (ope) S	85.9 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	560.000 mm	Actual Stress	78.3 N./mm ²
Spec. Min. Thk t,e, or sa	24.0000 mm	Spec. Nominal Thk.	24.000 mm
Surface Area	15280.7 cm ²	SG of Contents	1.000
Element Volume	197076.0 cm ³	Weight of Contents	196.2 Kg
Empty Weight	282.8 Kg	Operating Weight	478.9 Kg
		Comp. All. Ope. Stress	76.5 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (6105 \cdot 281) / (85.9 \cdot 1 - 0.6 \cdot 6105) \\
 &= 20.8631 + 1.0000 = 21.8631 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 5.401 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (85.9 \cdot 1 \cdot 23) / (281 + 0.6 \cdot 23) \\
 &= 6701 - 5.4 = 6696 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 24) / (280 + 0.6 \cdot 24) \\
 &= 11249 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (6105 \cdot (281 + 0.6 \cdot 23)) / (1 \cdot 23) \\
 &= 78.260 \text{ N./mm}^2
 \end{aligned}$$



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 FileName : 188-C1-C2-C3-C4- Solid SS Top 2nd stack Page 49 of 223
 Element: LHS Ch Shell Step: 7 10:04am Sep 9,2026

% Elongation per Table UG-79-1 (50*tnom/Rf)*(1-Rf/Ro) 4.110 %
 Note: Please Check Requirements of Table UHA-44 for Elongation limits.

SA-240 304, Min Metal Temp without impact per UHA-51: -196 °C

EXTERNAL PRESSURE RESULTS for: LHS Ch Shell

Cylindrical Shell From 20 to 30 Ext. Chart: HA-1 at 300 °C

Elastic Modulus from Chart: HA-1 at 300 °C: 170271248 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
23.000	608.00	897.47	26.43	1.4761	0.0066127	61.46

$$MAEP = (4*B)/(3*(Do/t)) = (4*61.5)/(3*26.4) = 3100 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
2.449	608.00	897.47	248.27	1.4761	0.0002262	19.26

$$MAEP = (4*B)/(3*(Do/t)) = (4*19.3)/(3*248) = 103 \text{ KPa.}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
23.000	608.00	24134.03	26.43	39.6941	0.0015891	49.01

$$MAEP = (4*B)/(3*(Do/t)) = (4*49)/(3*26.4) = 2472 \text{ KPa.}$$

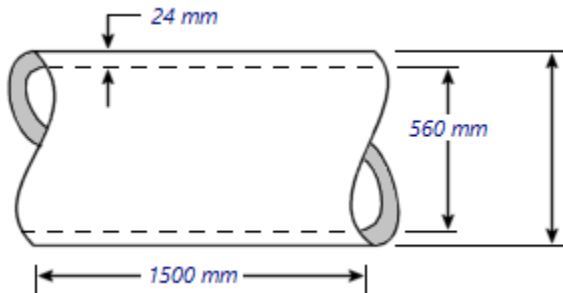
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PV Elite 28 Licensee: SL User
 FileName : 188-C1-C2-C3-C4- Solid SS Top 2nd stack Page 50 of 223
 Element: Main Shell Step: 8 10:04am Sep 9,2026

Internal Pressure Calculation Results:

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Internal Pressure Results for: Main Shell

Cylindrical Shell Design Information:			
Design Pressure P	6105.420 KPa.	Int. Design Temperature	300 °C
Material	SA-240 304	Ext. Design Temperature	300 °C
Length	1500.000 mm	External Pressure Chart	HA-1
		UNS Number	S30400
Int. Corr. All. c	0.0000 mm	Allowable Stress (ope) S	85.9 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	560.000 mm	Actual Stress	74.9 N./mm ²
Spec. Min. Thk t _e , or sa	24.0000 mm	Spec. Nominal Thk.	24.000 mm
Surface Area	28651.3 cm ²	SG of Contents	1.000
Element Volume	198473.1 cm ³	Weight of Contents	198.4 Kg
Empty Weight	530.2 Kg	Operating Weight	728.5 Kg
		Comp. All. Ope. Stress	76.8 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (6105 \cdot 280) / (85.9 \cdot 1 - 0.6 \cdot 6105)$$

$$= 20.7889 + 0.0000 = 20.7889 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 5.421 KPa.

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (85.9 \cdot 1 \cdot 24) / (280 + 0.6 \cdot 24)$$

$$= 7002 - 5.42 = 6997 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 24) / (280 + 0.6 \cdot 24)$$

$$= 11249 \text{ KPa.}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (6105 \cdot (280 + 0.6 \cdot 24)) / (1 \cdot 24)$$

$$= 74.898 \text{ N./mm}^2$$



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 FileName : 188-C1-C2-C3-C4- Solid SS Top 2nd stack Page 51 of 223
 Element: Main Shell Step: 8 10:04am Sep 9,2026

% Elongation per Table UG-79-1 (50*tnom/Rf)*(1-Rf/Ro) 4.110 %
 Note: Please Check Requirements of Table UHA-44 for Elongation limits.

SA-240 304, Min Metal Temp without impact per UHA-51: -196 °C

EXTERNAL PRESSURE RESULTS for: Main Shell

Cylindrical Shell From 40 to 50 Ext. Chart: HA-1 at 300 °C

Elastic Modulus from Chart: HA-1 at 300 °C: 170271248 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
24.000	608.00	9220.00	25.33	15.1645	0.0017432	49.98

$$MAEP = (4*B)/(3*(Do/t)) = (4*50)/(3*25.3) = 2630 \text{ KPa.}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
5.615	608.00	9220.00	108.28	15.1645	0.0000987	8.40

$$MAEP = (4*B)/(3*(Do/t)) = (4*8.4)/(3*108) = 103 \text{ KPa.}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
24.000	608.00	234153.00	25.33	50.0000	0.0017189	49.83

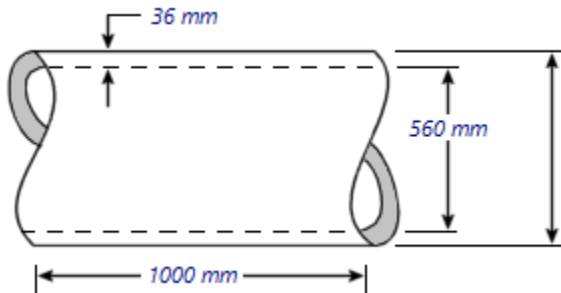
$$MAEP = (4*B)/(3*(Do/t)) = (4*49.8)/(3*25.3) = 2623 \text{ KPa.}$$

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Main Shell

Cylindrical Shell Design Information:			
Design Pressure P	6105.420 KPa.	Int. Design Temperature	300 °C
Material	SA-240 304	Ext. Design Temperature	300 °C
Length	1000.000 mm	External Pressure Chart	HA-1
		UNS Number	S30400
Int. Corr. All. c	0.0000 mm	Allowable Stress (ope) S	85.9 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	560.000 mm	Actual Stress	51.2 N./mm ²
Spec. Min. Thk t,e, or sa	36.0000 mm	Spec. Nominal Thk.	36.000 mm
Surface Area	19854.9 cm ²	SG of Contents	1.000
Element Volume	132315.4 cm ³	Weight of Contents	132.2 Kg
Empty Weight	541.1 Kg	Operating Weight	673.3 Kg
		Comp. All. Ope. Stress	79.2 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (6105 \cdot 280) / (85.9 \cdot 1 - 0.6 \cdot 6105) \\
 &= 20.7889 + 0.0000 = 20.7889 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 5.421 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (85.9 \cdot 1 \cdot 36) / (280 + 0.6 \cdot 36) \\
 &= 10253 - 5.42 = 10247 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 36) / (280 + 0.6 \cdot 36) \\
 &= 16471 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (6105 \cdot (280 + 0.6 \cdot 36)) / (1 \cdot 36) \\
 &= 51.153 \text{ N./mm}^2
 \end{aligned}$$



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 FileName : 188-C1-C2-C3-C4- Solid SS Top 2nd stack Page 53 of 223
 Element: Main Shell Step: 9 10:04am Sep 9,2026

% Elongation per Table UG-79-1 (50*tnom/Rf)*(1-Rf/Ro) 6.040 %
 Note: Please Check Requirements of Table UHA-44 for Elongation limits.

SA-240 304, Min Metal Temp without impact per UHA-51: -196 °C

EXTERNAL PRESSURE RESULTS for: Main Shell

Cylindrical Shell From 50 to 60 Ext. Chart: HA-1 at 300 °C

Elastic Modulus from Chart: HA-1 at 300 °C: 170271248 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
36.000	632.00	9220.00	17.56	14.5886	0.0037299	57.08

$$MAEP = (4*B)/(3*(Do/t)) = (4*57.1)/(3*17.6) = 4335 \text{ KPa.}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
5.831	632.00	9220.00	108.39	14.5886	0.0000988	8.41

$$MAEP = (4*B)/(3*(Do/t)) = (4*8.41)/(3*108) = 103 \text{ KPa.}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
36.000	632.00	386129.59	17.56	50.0000	0.0036870	56.98

$$MAEP = (4*B)/(3*(Do/t)) = (4*57)/(3*17.6) = 4327 \text{ KPa.}$$

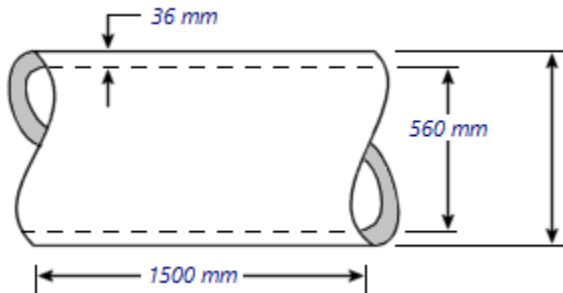
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 FileName : 188-C1-C2-C3-C4- Solid SS Top 2nd stack Page 54 of 223
 Element: Main Shell Step: 10 10:04am Sep 9,2026

Internal Pressure Calculation Results:

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Internal Pressure Results for: Main Shell

Cylindrical Shell Design Information:			
Design Pressure P	6105.420 KPa.	Int. Design Temperature	300 °C
Material	SA-240 304	Ext. Design Temperature	300 °C
Length	1500.000 mm	External Pressure Chart	HA-1
		UNS Number	S30400
Int. Corr. All. c	0.0000 mm	Allowable Stress (ope) S	85.9 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	560.000 mm	Actual Stress	51.2 N./mm ²
Spec. Min. Thk t,e, or sa	36.0000 mm	Spec. Nominal Thk.	36.000 mm
Surface Area	29782.3 cm ²	SG of Contents	1.000
Element Volume	198473.1 cm ³	Weight of Contents	198.4 Kg
Empty Weight	811.6 Kg	Operating Weight	1010.0 Kg
		Comp. All. Ope. Stress	79.2 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (6105 \cdot 280) / (85.9 \cdot 1 - 0.6 \cdot 6105) \\
 &= 20.7889 + 0.0000 = 20.7889 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 5.421 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (85.9 \cdot 1 \cdot 36) / (280 + 0.6 \cdot 36) \\
 &= 10253 - 5.42 = 10247 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 36) / (280 + 0.6 \cdot 36) \\
 &= 16471 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (6105 \cdot (280 + 0.6 \cdot 36)) / (1 \cdot 36) \\
 &= 51.153 \text{ N./mm}^2
 \end{aligned}$$



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 FileName : 188-C1-C2-C3-C4- Solid SS Top 2nd stack Page 55 of 223
 Element: Main Shell Step: 10 10:04am Sep 9,2026

% Elongation per Table UG-79-1 (50*tnom/Rf)*(1-Rf/Ro) 6.040 %
 Note: Please Check Requirements of Table UHA-44 for Elongation limits.

SA-240 304, Min Metal Temp without impact per UHA-51: -196 °C

EXTERNAL PRESSURE RESULTS for: Main Shell

Cylindrical Shell From 60 to 70 Ext. Chart: HA-1 at 300 °C

Elastic Modulus from Chart: HA-1 at 300 °C: 170271248 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
36.000	632.00	9220.00	17.56	14.5886	0.0037299	57.08

$$MAEP = (4*B)/(3*(Do/t)) = (4*57.1)/(3*17.6) = 4335 \text{ KPa.}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
5.831	632.00	9220.00	108.39	14.5886	0.0000988	8.41

$$MAEP = (4*B)/(3*(Do/t)) = (4*8.41)/(3*108) = 103 \text{ KPa.}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
36.000	632.00	386129.59	17.56	50.0000	0.0036870	56.98

$$MAEP = (4*B)/(3*(Do/t)) = (4*57)/(3*17.6) = 4327 \text{ KPa.}$$

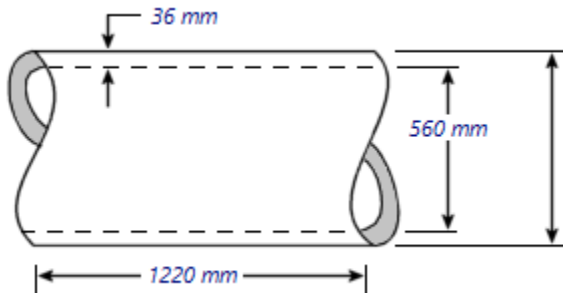
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PV Elite 28 Licensee: SL User
 FileName : 188-C1-C2-C3-C4- Solid SS Top 2nd stack Page 56 of 223
 Element: Main Shell Step: 11 10:04am Sep 9,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Main Shell

Cylindrical Shell Design Information:			
Design Pressure P	6105.420 KPa.	Int. Design Temperature	300 °C
Material	SA-240 304	Ext. Design Temperature	300 °C
Length	1220.000 mm	External Pressure Chart	HA-1
		UNS Number	S30400
Int. Corr. All. c	0.0000 mm	Allowable Stress (ope) S	85.9 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	560.000 mm	Actual Stress	51.2 N./mm ²
Spec. Min. Thk t,e, or sa	36.0000 mm	Spec. Nominal Thk.	36.000 mm
Surface Area	24222.9 cm ²	SG of Contents	1.000
Element Volume	161424.8 cm ³	Weight of Contents	161.3 Kg
Empty Weight	660.1 Kg	Operating Weight	821.5 Kg
		Comp. All. Ope. Stress	79.2 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (6105 \cdot 280) / (85.9 \cdot 1 - 0.6 \cdot 6105) \\
 &= 20.7889 + 0.0000 = 20.7889 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 5.421 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (85.9 \cdot 1 \cdot 36) / (280 + 0.6 \cdot 36) \\
 &= 10253 - 5.42 = 10247 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 36) / (280 + 0.6 \cdot 36) \\
 &= 16471 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (6105 \cdot (280 + 0.6 \cdot 36)) / (1 \cdot 36) \\
 &= 51.153 \text{ N./mm}^2
 \end{aligned}$$



PV Elite 28 Licensee: SL User
 FileName : 188-C1-C2-C3-C4- Solid SS Top 2nd stack Page 57 of 223
 Element: Main Shell Step: 11 10:04am Sep 9,2026

% Elongation per Table UG-79-1 (50*tnom/Rf)*(1-Rf/Ro) 6.040 %
 Note: Please Check Requirements of Table UHA-44 for Elongation limits.

SA-240 304, Min Metal Temp without impact per UHA-51: -196 °C

EXTERNAL PRESSURE RESULTS for: Main Shell

Cylindrical Shell From 70 to 80 Ext. Chart: HA-1 at 300 °C

Elastic Modulus from Chart: HA-1 at 300 °C: 170271248 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
36.000	632.00	9220.00	17.56	14.5886	0.0037299	57.08

$$MAEP = (4*B)/(3*(Do/t)) = (4*57.1)/(3*17.6) = 4335 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
5.831	632.00	9220.00	108.39	14.5886	0.0000988	8.41

$$MAEP = (4*B)/(3*(Do/t)) = (4*8.41)/(3*108) = 103 \text{ KPa.}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
36.000	632.00	386129.59	17.56	50.0000	0.0036870	56.98

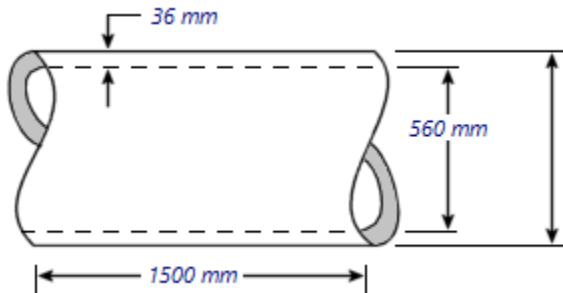
$$MAEP = (4*B)/(3*(Do/t)) = (4*57)/(3*17.6) = 4327 \text{ KPa.}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Main Shell

Cylindrical Shell Design Information:					
Design Pressure	P	6105.420 KPa.	Int. Design Temperature		300 °C
Material		SA-240 304	Ext. Design Temperature		300 °C
Length		1500.000 mm	External Pressure Chart		HA-1
			UNS Number		S30400
Int. Corr. All.	c	0.0000 mm	Allowable Stress (ope)	S	85.9 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	560.000 mm	Actual Stress		51.2 N./mm ²
Spec. Min. Thk	t,e, or sa	36.0000 mm	Spec. Nominal Thk.		36.000 mm
Surface Area		29782.3 cm ²	SG of Contents		1.000
Element Volume		198473.1 cm ³	Weight of Contents		198.4 Kg
Empty Weight		811.6 Kg	Operating Weight		1010.0 Kg
			Comp. All. Ope. Stress		79.2 N./mm ²
Radiography Information:					
User Defined			User Defined		
Circ. Joint Efficiency		100 %	Long. Joint Eff.	E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (6105 \cdot 280) / (85.9 \cdot 1 - 0.6 \cdot 6105) \\
 &= 20.7889 + 0.0000 = 20.7889 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 5.421 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (85.9 \cdot 1 \cdot 36) / (280 + 0.6 \cdot 36) \\
 &= 10253 - 5.42 = 10247 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 36) / (280 + 0.6 \cdot 36) \\
 &= 16471 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (6105 \cdot (280 + 0.6 \cdot 36)) / (1 \cdot 36) \\
 &= 51.153 \text{ N./mm}^2
 \end{aligned}$$



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 FileName : 188-C1-C2-C3-C4- Solid SS Top 2nd stack Page 59 of 223
 Element: Main Shell Step: 12 10:04am Sep 9,2026

% Elongation per Table UG-79-1 (50*tnom/Rf)*(1-Rf/Ro) 6.040 %
 Note: Please Check Requirements of Table UHA-44 for Elongation limits.

SA-240 304, Min Metal Temp without impact per UHA-51: -196 °C

EXTERNAL PRESSURE RESULTS for: Main Shell

Cylindrical Shell From 80 to 90 Ext. Chart: HA-1 at 300 °C

Elastic Modulus from Chart: HA-1 at 300 °C: 170271248 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
36.000	632.00	9220.00	17.56	14.5886	0.0037299	57.08

$$MAEP = (4*B)/(3*(Do/t)) = (4*57.1)/(3*17.6) = 4335 \text{ KPa.}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
5.831	632.00	9220.00	108.39	14.5886	0.0000988	8.41

$$MAEP = (4*B)/(3*(Do/t)) = (4*8.41)/(3*108) = 103 \text{ KPa.}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
36.000	632.00	386129.59	17.56	50.0000	0.0036870	56.98

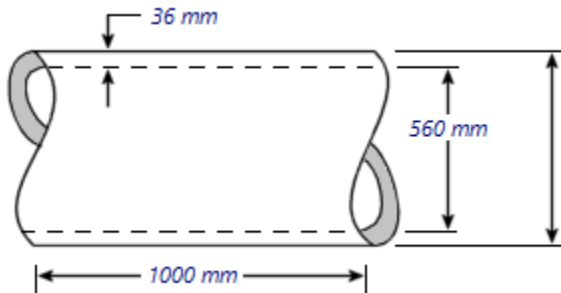
$$MAEP = (4*B)/(3*(Do/t)) = (4*57)/(3*17.6) = 4327 \text{ KPa.}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Main Shell

Cylindrical Shell Design Information:					
Design Pressure	P	6105.420 KPa.	Int. Design Temperature		300 °C
Material		SA-240 304	Ext. Design Temperature		300 °C
Length		1000.000 mm	External Pressure Chart		HA-1
			UNS Number		S30400
Int. Corr. All.	c	0.0000 mm	Allowable Stress (ope)	S	85.9 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	560.000 mm	Actual Stress		51.2 N./mm ²
Spec. Min. Thk t,e, or sa		36.0000 mm	Spec. Nominal Thk.		36.000 mm
Surface Area		19854.9 cm ²	SG of Contents		1.000
Element Volume		132315.4 cm ³	Weight of Contents		132.2 Kg
Empty Weight		541.1 Kg	Operating Weight		673.3 Kg
			Comp. All. Ope. Stress		79.2 N./mm ²
Radiography Information:					
User Defined			User Defined		
Circ. Joint Efficiency		100 %	Long. Joint Eff.	E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (6105 \cdot 280) / (85.9 \cdot 1 - 0.6 \cdot 6105) \\
 &= 20.7889 + 0.0000 = 20.7889 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 5.421 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (85.9 \cdot 1 \cdot 36) / (280 + 0.6 \cdot 36) \\
 &= 10253 - 5.42 = 10247 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 36) / (280 + 0.6 \cdot 36) \\
 &= 16471 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (6105 \cdot (280 + 0.6 \cdot 36)) / (1 \cdot 36) \\
 &= 51.153 \text{ N./mm}^2
 \end{aligned}$$



PV Elite 28 Licensee: SL User
 FileName : 188-C1-C2-C3-C4- Solid SS Top 2nd stack Page 61 of 223
 Element: Main Shell Step: 13 10:04am Sep 9,2026

% Elongation per Table UG-79-1 (50*tnom/Rf)*(1-Rf/Ro) 6.040 %
 Note: Please Check Requirements of Table UHA-44 for Elongation limits.

SA-240 304, Min Metal Temp without impact per UHA-51: -196 °C

EXTERNAL PRESSURE RESULTS for: Main Shell

Cylindrical Shell From 90 to 100 Ext. Chart: HA-1 at 300 °C

Elastic Modulus from Chart: HA-1 at 300 °C: 170271248 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
36.000	632.00	9220.00	17.56	14.5886	0.0037299	57.08

$$MAEP = (4*B)/(3*(Do/t)) = (4*57.1)/(3*17.6) = 4335 \text{ KPa.}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
5.831	632.00	9220.00	108.39	14.5886	0.0000988	8.41

$$MAEP = (4*B)/(3*(Do/t)) = (4*8.41)/(3*108) = 103 \text{ KPa.}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
36.000	632.00	386129.59	17.56	50.0000	0.0036870	56.98

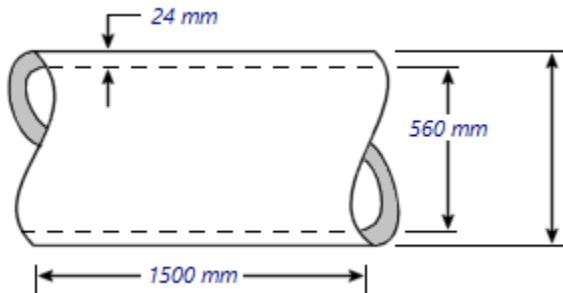
$$MAEP = (4*B)/(3*(Do/t)) = (4*57)/(3*17.6) = 4327 \text{ KPa.}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Main Shell

Cylindrical Shell Design Information:					
Design Pressure	P	6105.420 KPa.	Int. Design Temperature		300 °C
Material		SA-240 304	Ext. Design Temperature		300 °C
Length		1500.000 mm	External Pressure Chart		HA-1
			UNS Number		S30400
Int. Corr. All.	c	0.0000 mm	Allowable Stress (ope)	S	85.9 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	560.000 mm	Actual Stress		74.9 N./mm ²
Spec. Min. Thk t,e, or sa		24.0000 mm	Spec. Nominal Thk.		24.000 mm
Surface Area		28651.3 cm ²	SG of Contents		1.000
Element Volume		198473.1 cm ³	Weight of Contents		198.4 Kg
Empty Weight		530.2 Kg	Operating Weight		728.5 Kg
			Comp. All. Ope. Stress		79.2 N./mm ²
Radiography Information:					
User Defined			User Defined		
Circ. Joint Efficiency		100 %	Long. Joint Eff.	E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (6105 \cdot 280) / (85.9 \cdot 1 - 0.6 \cdot 6105) \\
 &= 20.7889 + 0.0000 = 20.7889 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 5.421 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (85.9 \cdot 1 \cdot 24) / (280 + 0.6 \cdot 24) \\
 &= 7002 - 5.42 = 6997 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 24) / (280 + 0.6 \cdot 24) \\
 &= 11249 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (6105 \cdot (280 + 0.6 \cdot 24)) / (1 \cdot 24) \\
 &= 74.898 \text{ N./mm}^2
 \end{aligned}$$



PV Elite 28 Licensee: SL User
 FileName : 188-C1-C2-C3-C4- Solid SS Top 2nd stack Page 63 of 223
 Element: Main Shell Step: 14 10:04am Sep 9,2026

% Elongation per Table UG-79-1 (50*tnom/Rf)*(1-Rf/Ro) 4.110 %
 Note: Please Check Requirements of Table UHA-44 for Elongation limits.

SA-240 304, Min Metal Temp without impact per UHA-51: -196 °C

EXTERNAL PRESSURE RESULTS for: Main Shell

Cylindrical Shell From 100 to 110 Ext. Chart: HA-1 at 300 °C

Elastic Modulus from Chart: HA-1 at 300 °C: 170271248 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
24.000	608.00	9220.00	25.33	15.1645	0.0017432	49.98

$$MAEP = (4*B)/(3*(Do/t)) = (4*50)/(3*25.3) = 2630 \text{ KPa.}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
5.615	608.00	9220.00	108.28	15.1645	0.0000987	8.40

$$MAEP = (4*B)/(3*(Do/t)) = (4*8.4)/(3*108) = 103 \text{ KPa.}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
24.000	608.00	234153.00	25.33	50.0000	0.0017189	49.83

$$MAEP = (4*B)/(3*(Do/t)) = (4*49.8)/(3*25.3) = 2623 \text{ KPa.}$$

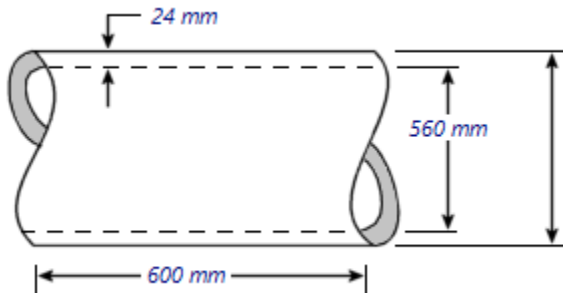
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PV Elite 28 Licensee: SL User
 FileName : 188-C1-C2-C3-C4- Solid SS Top 2nd stack Page 64 of 223
 Element: RHS Ch Shell Step: 15 10:04am Sep 9,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: RHS Ch Shell

Cylindrical Shell Design Information:			
Design Pressure P	6105.400 KPa.	Int. Design Temperature	300 °C
Material	SA-240 304	Ext. Design Temperature	300 °C
Length	600.000 mm	External Pressure Chart	HA-1
		UNS Number	S30400
Int. Corr. All. c	1.0000 mm	Allowable Stress (ope) S	85.9 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	560.000 mm	Actual Stress	78.3 N./mm ²
Spec. Min. Thk t,e, or sa	24.0000 mm	Spec. Nominal Thk.	24.000 mm
Surface Area	11460.5 cm ²	SG of Contents	1.000
Element Volume	147807.0 cm ³	Weight of Contents	147.1 Kg
Empty Weight	212.1 Kg	Operating Weight	359.2 Kg
		Comp. All. Ope. Stress	76.5 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (6105 \cdot 281) / (85.9 \cdot 1 - 0.6 \cdot 6105)$$

$$= 20.8631 + 1.0000 = 21.8631 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 5.401 KPa.

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (85.9 \cdot 1 \cdot 23) / (281 + 0.6 \cdot 23)$$

$$= 6701 - 5.4 = 6696 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 24) / (280 + 0.6 \cdot 24)$$

$$= 11249 \text{ KPa.}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (6105 \cdot (281 + 0.6 \cdot 23)) / (1 \cdot 23)$$

$$= 78.260 \text{ N./mm}^2$$



PV Elite 28 Licensee: SL User
 FileName : 188-C1-C2-C3-C4- Solid SS Top 2nd stack Page 65 of 223
 Element: RHS Ch Shell Step: 15 10:04am Sep 9,2026

% Elongation per Table UG-79-1 (50*tnom/Rf)*(1-Rf/Ro) 4.110 %
 Note: Please Check Requirements of Table UHA-44 for Elongation limits.

SA-240 304, Min Metal Temp without impact per UHA-51: -196 °C

EXTERNAL PRESSURE RESULTS for: RHS Ch Shell

Cylindrical Shell From 120 to 130 Ext. Chart: HA-1 at 300 °C

Elastic Modulus from Chart: HA-1 at 300 °C: 170271248 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
23.000	608.00	697.47	26.43	1.1471	0.0087746	63.32

$$MAEP = (4*B)/(3*(Do/t)) = (4*63.3)/(3*26.4) = 3194 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
2.201	608.00	697.47	276.24	1.1471	0.0002517	21.43

$$MAEP = (4*B)/(3*(Do/t)) = (4*21.4)/(3*276) = 103 \text{ KPa.}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
23.000	608.00	19103.09	26.43	31.4196	0.0015891	49.01

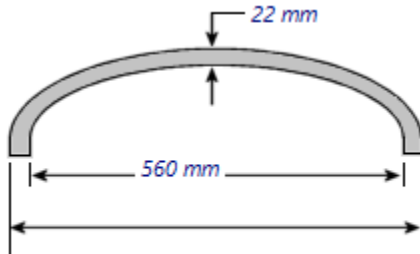
$$MAEP = (4*B)/(3*(Do/t)) = (4*49)/(3*26.4) = 2472 \text{ KPa.}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: RHS Dish End

Elliptical Head Design Information:							
Design Pressure	P	6105.400 KPa.		Int. Design Temperature		300 °C	
Material		SA-240 304		Ext. Design Temperature		300 °C	
Length		50.800 mm		External Pressure Chart		HA-1	
				UNS Number		S30400	
Int. Corr. All.	c	1.0000 mm		Allowable Stress (ope)	S	85.9 N./mm ²	
Ext. Corr. All.	ce	0.0000 mm		Allowable Stress (amb)	Sa	138.0 N./mm ²	
Inside Diameter	Di	560.000 mm		Actual Stress		81.9 N./mm ²	
Spec. Min. Thk t,e, or sa		22.0000 mm		Spec. Nominal Thk.		28.000 mm	
Surface Area		5096.3 cm ²		SG of Contents		1.000	
Element Volume		35506.5 cm ³		Weight of Contents		35.5 Kg	
Empty Weight		114.1 Kg		Operating Weight		149.6 Kg	
				Comp. All. Ope. Stress		76.5 N./mm ²	
				Aspect Ratio		2.00	
Radiography Information:							
User Defined				User Defined			
Circ. Joint Efficiency		100 %		Long. Joint Eff.	E or v	100 %	

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 7.000 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.500 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (6105 \cdot 562 \cdot 1) / (2 \cdot 85.9 \cdot 1 - 0.2 \cdot 6105)$$

$$= 20.0214 + 1.0000 = 21.0214 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 5.401 KPa.

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 85.9 \cdot 1 \cdot 21) / (1 \cdot 562 + 0.2 \cdot 21)$$

$$= 6402 - 5.4 = 6396 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 22) / (1 \cdot 560 + 0.2 \cdot 22)$$

$$= 10758 \text{ KPa.}$$



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 FileName : 188-C1-C2-C3-C4- Solid SS Top 2nd stack Page 67 of 223
 Element: RHS Dish End Step: 16 10:04am Sep 9,2026

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (6105 \cdot (1 \cdot 562 + 0.2 \cdot 21)) / (2 \cdot 1 \cdot 21)$$

$$= 81.926 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (6105 \cdot 281) / (85.9 \cdot 1 - 0.6 \cdot 6105) + 1$$

$$= 21.863 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

[Less Operating Hydrostatic Head Pressure of 5.401 KPa.](#)

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (85.9 \cdot 1 \cdot 27) / (281 + 0.6 \cdot 27)$$

$$= 7803 - 5.4 = 7798 \text{ KPa.}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6$$

$$= (2 + (562 / (2 \cdot 141)))^2 / 6$$

$$= 0.995280$$

% Elongation per Table UG-79-1 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 19.231 %

Note: Please Check Requirements of Table UHA-44 for Elongation limits.

SA-240 304, Min Metal Temp without impact per UHA-51: -196 °C

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 7.000 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.500 mm

EXTERNAL PRESSURE RESULTS for: RHS Dish End

Elliptical Head From 130 to 140 Ext. Chart: HA-1 at 300 °C

Elastic Modulus from Chart: HA-1 at 300 °C: 170271248 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
21.000	604.00	28.76	0.0048289	59.16

$$MAEP = B / (K_0 \cdot Do / t) = 59.2 / (0.9 \cdot 28.8) = 2285 \text{ KPa.}$$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
1.748	604.00	345.49	0.0004020	32.16

$$MAEP = B / (K_0 \cdot Do / t) = 32.2 / (0.9 \cdot 345) = 103 \text{ KPa.}$$



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 Shop Pressure Test Results: Step: 17 10:04am Sep 9,2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	8145.890 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	8046.557 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	14618.359 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	6892.677 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	8722.999 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	8722.999 KPa.
App 27-4	= M.A.W.P.	6175.303 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	9787.172 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	9333.928 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	13142.574 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	8281.454 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	8974.932 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	9333.928 KPa.
App 27-4	= M.A.W.P.	6396.208 KPa.

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
LHS Dish End	168.7	184.5	0.914	13148.06
LHS Ch Shell	161.3	184.5	0.874	13148.06
Main Shell	179.4	184.5	0.972	14623.85
Main Shell	122.5	184.5	0.664	14623.85
Main Shell	122.5	184.5	0.664	14623.85
Main Shell	122.5	184.5	0.664	14623.85
Main Shell	122.5	184.5	0.664	14623.85
Main Shell	122.5	184.5	0.664	14623.85
Main Shell	179.4	184.5	0.972	14623.85
RHS Ch Shell	161.3	184.5	0.874	13148.06
RHS Dish End	168.7	184.5	0.914	13148.06

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N3-Tube Inlet	Nozzle	138.00	116.00	1.190
N3-Tube Inlet	Pad	138.00	85.90	1.606
N4-Tube Outlet	Nozzle	138.00	116.00	1.190
N4-Tube Outlet	Pad	138.00	85.90	1.606
N1-Shell Inlet	Nozzle	138.00	116.00	1.190
N1-Shell Inlet	Pad	138.00	136.00	1.015
N2-Shell outlet	Nozzle	138.00	116.00	1.190
N2-Shell outlet	Pad	138.00	136.00	1.015

Minimum 1.015

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
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LHS Dish End	138.00	85.90	1.606
LHS Ch Shell	138.00	85.90	1.606
LHS ch flange	138.00	85.90	1.606
Main Shell	138.00	85.90	1.606
Main Shell	138.00	85.90	1.606
Main Shell	138.00	85.90	1.606
Main Shell	138.00	85.90	1.606
Main Shell	138.00	85.90	1.606
Main Shell	138.00	85.90	1.606
Main Shell	138.00	85.90	1.606
RHS Ch Flange	138.00	85.90	1.606
RHS Ch Shell	138.00	85.90	1.606
RHS Dish End	138.00	85.90	1.606

Minimum 1.606

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	115.00	97.70	1.177
Tubesheet Material	138.00	85.90	1.606

Minimum 1.177

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N3-Tube Inlet	113.72	184.50	0.616
N4-Tube Outlet	113.72	184.50	0.616
N1-Shell Inlet	173.82	184.50	0.942
N2-Shell outlet	173.82	184.50	0.942

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	8145.890 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	8046.557 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	14618.359 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	6892.677 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	8722.999 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	8722.999 KPa.
App 27-4	= M.A.W.P.	6175.303 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	9787.172 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	9333.928 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	13142.574 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	8281.454 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	8974.932 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	9333.928 KPa.
App 27-4	= M.A.W.P.	6396.208 KPa.

UG-99(b)(35), Test Pressure Calculation [Shell Side]:
 = Test Factor • Design Pressure • Stress Ratio
 = 1.3 • 6100 • 1.01
 = 8046.557 KPa.

UG-99(b) Note 35, Test Pressure Calculation [Channel Side]:
 = Test Factor • Design Pressure • Stress Ratio
 = 1.3 • 6100 • 1.18
 = 9333.928 KPa.

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
LHS Dish End	125.3	184.5	0.679	9339.42
LHS Ch Shell	119.7	184.5	0.649	9339.42
Main Shell	98.8	184.5	0.535	8052.05
Main Shell	67.5	184.5	0.366	8052.05
Main Shell	67.5	184.5	0.366	8052.05
Main Shell	67.5	184.5	0.366	8052.05
Main Shell	67.5	184.5	0.366	8052.05
Main Shell	67.5	184.5	0.366	8052.05
Main Shell	98.8	184.5	0.535	8052.05
RHS Ch Shell	119.7	184.5	0.649	9339.42
RHS Dish End	125.3	184.5	0.679	9339.42

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N3-Tube Inlet	Nozzle	138.00	116.00	1.190
N3-Tube Inlet	Pad	138.00	85.90	1.606
N4-Tube Outlet	Nozzle	138.00	116.00	1.190
N4-Tube Outlet	Pad	138.00	85.90	1.606
N1-Shell Inlet	Nozzle	138.00	116.00	1.190



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N1-Shell Inlet	Pad	138.00	136.00	1.015
N2-Shell outlet	Nozzle	138.00	116.00	1.190
N2-Shell outlet	Pad	138.00	136.00	1.015

Minimum 1.015

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
LHS Dish End	138.00	85.90	1.606
LHS Ch Shell	138.00	85.90	1.606
LHS ch flange	138.00	85.90	1.606
Main Shell	138.00	85.90	1.606
Main Shell	138.00	85.90	1.606
Main Shell	138.00	85.90	1.606
Main Shell	138.00	85.90	1.606
Main Shell	138.00	85.90	1.606
Main Shell	138.00	85.90	1.606
Main Shell	138.00	85.90	1.606
RHS Ch Flange	138.00	85.90	1.606
RHS Ch Shell	138.00	85.90	1.606
RHS Dish End	138.00	85.90	1.606

Minimum 1.606

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	115.00	97.70	1.177
Tubesheet Material	138.00	85.90	1.606

Minimum 1.177

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N3-Tube Inlet	80.78	184.50	0.438
N4-Tube Outlet	80.78	184.50	0.438
N1-Shell Inlet	95.71	184.50	0.519
N2-Shell outlet	95.71	184.50	0.519

Elements Suitable for Test Pressure.

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Input Echo, Tubesheet Number 1, Description:

Shell Data:

Main Shell Description: Main Shell

Shell Test Pressure		8046.56 KPa.
Shell Thickness	ts	24.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	560.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-240 304

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		S30400
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder
Channel Test Pressure		9333.93 KPa.
Channel Thickness	tc	24.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	560.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-240 304

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		S30400
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	900
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	12.7000 mm
Total Straight Tube Length	Lt	9600.00 mm
Straight Tube Length (bet. inner tubsht faces)	L	9220.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-213 TP304L
Tube Material UNS Number		S30403
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		115.00 N./mm ²
Tube Allowable Stress At Ambient		115.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	170.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	15.8750 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	1.6500 mm
Groove Weld Leg	ag	1.6500 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	262.030 mm
Largest Center-to-Center Tube Distance	Ul	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	8.0000 mm
Area of Tube Layout	Ap	3.6 cm ²



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Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf B

Tubesheet Design Metal Temperature	T	21.11 °C
Tubesheet Material		SA-965 F304
Tubesheet Material UNS Number		S30400
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	190.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm
Tubesheet Outside Diameter	A	824.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped?		No
Area of the Untubed Lanes	AL	0.0 cm ²

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	327.000 mm
Tube End condition corresponding to Span (l)	k	1.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Mean Shell Metal Temp. along Shell len.	Tsm	21.11 °C
Mean Tube Metal Temp. along Tube length	Ttm	21.11 °C
Junction Stress Reduction option		Perform Plastic Calculation

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Channel
Flange Outside Diameter	A	824.000 mm
Flange Inside Diameter	B	564.000 mm
Flange Face Outside Diameter	Fod	622.000 mm
Flange Face Inside Diameter	Fid	564.000 mm
Gasket Outside Diameter	Go	611.000 mm
Gasket Inside Diameter	Gi	585.000 mm
Small end Hub thk.	g0	24.0000 mm
Large end Hub thk.	g1	33.0000 mm
Gasket Factor,	m	3.00
Gasket Design Seating Stress	y	68.95 N./mm ²
Flange Facing Sketch		Code Sketch 1a
Gasket Thickness	tg	3.1750 mm
Full face Gasket Flange Option		Program Selects
Length of Partition Gasket	lp	550.000 mm
Width of Partition Gasket	wp	10.0000 mm
Partition Gasket Factor,	mPart	3.0000
Partition Gasket Design Seating Stress	yPart	68.95 N./mm ²
Partition Gasket Facing Sketch		Code Sketch 1a
Partition Gasket Column for Gasket Seating		Code Column II

Bolting Information:

Diameter of Bolt Circle	C	739.000 mm
Nominal Bolt Diameter	dB	38.1000 mm
Thread Series		TEMA Thread Series
Number of Bolts	n	24
Bolt Material		SA-193 B7
Bolt Allowable Stress At Temperature	Sb	172.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00 N./mm ²
Weld between Flange and Shell/Channel		0.0000 mm

Note: The full bolt load Sa*Ab, was used for all load cases.

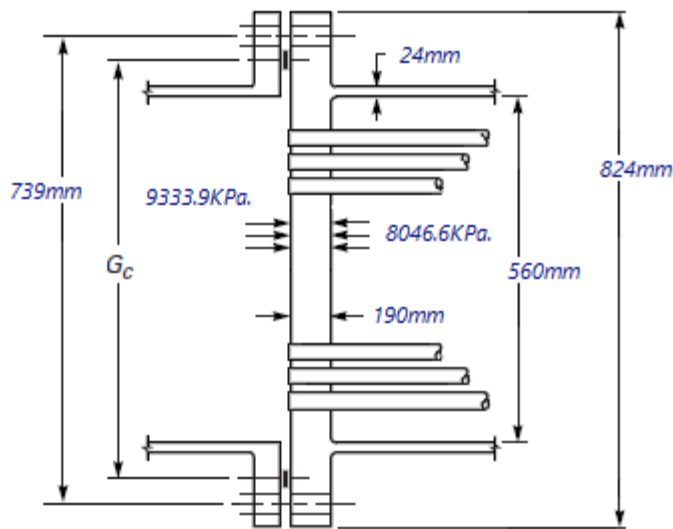


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Tubesheet Integral with Shell
Tubesheet Extended as Flange Yes
Thickness of Extended Portion of Tubesheet Tf 154.0000 mm
Is Bolt Load Transferred to the Tubesheet Yes

Is Exchanger in Creep range (skip EP, Use 3S for Sps) No

Tubesheet Integral w/Shell, Gasketed w/Channel, Extended as a Flange



Configuration b:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Austenitic Group 3 Steels
Shell - TM-1 Material Group G

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000152988 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.19512E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000152988 /°C
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.19512E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.

Channel - TE-1 Austenitic Group 3 Steels
Channel - TM-1 Material Group G

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000152988 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.19512E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.

Tubes - TE-1 Austenitic Group 3 Steels
Tubes - TM-1 Material Group G

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000152988 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.19512E+09 KPa.
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.19512E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.19512E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.



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TubeSheet - TE-1 Austenitic Group 3 Steels
 TubeSheet - TM-1 Material Group G

Elastic Mod. at Design Temperature 21.1 °C 0.19512E+09 KPa.
 Elastic Mod. at Ambient Temperature 21.1 °C 0.19512E+09 KPa.

Note:
 The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (9334 \cdot (12.7/2 - 0)) / (115 \cdot 1 + 0.4 \cdot 9334)$
 $= 0.4992 + 0.0000 = 0.4992 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart HA-3 at 21.11 °C
 Elastic Modulus for Material 193048800.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	12.70	9220.00	6.02	50.0000	0.0303634	101.57

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 28106 \text{ KPa.}$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.9101	12.70	9220.00	13.95	50.0000	0.0056493	84.23

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 84.2) / (3 \cdot 14) = 8048 \text{ KPa.}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.9101 mm
Allowable Internal Pressure at Corroded thickness	44066.75 KPa.
Required Internal Design Pressure	9333.93 KPa.
Allowable External Pressure at Corroded thickness	28106.39 KPa.
Required External Design Pressure	8046.56 KPa.
Required Thickness due to Shell Side pressure	0.9101 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (Goc - Gic) / 2$	13.000 mm
Basic Gasket Width,	$b0 = N / 2$	6.500 mm
Effective Gasket Width,	$b = \text{sqrt}(b0) \cdot 2.5$	6.422 mm
Gasket Reaction Diameter,	$G = Go - 2.0 \cdot b$	598.156 mm

Results for ASME Fixed Tubesheet Calculations for Configuration b:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio $[\rho]$:
 $= ltx / h$ (modified for corrosion if present)
 $= 0/190$
 $= 0.0000$ (must be $0 \leq \rho \leq 1$)



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Effective tube hole diameter [d*]:

$$= \max(dt - 2 \cdot tt(EtT/E) \cdot (StT/S) \cdot (\rho); dt - 2 \cdot tt)$$

$$= \max(12.7 - 2 \cdot 2.11 (195117184/195117184) \cdot (115/138) \cdot (0); 12.7 - 2 \cdot 2.11)$$

$$= 12.7000 \text{ mm}$$

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot ro + dt$$

$$= 2 \cdot 262 + 12.7$$

$$= 536.760 \text{ mm}$$

Determine the basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (15.9 - 12.7) / 15.9$$

$$= 0.200$$

Equivalent outer tube limit radius [ao]:

$$= Do / 2$$

$$= 537/2$$

$$= 268.3800 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{ 1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2) }$$

$$= 15.9 / \sqrt{ 1 - 4 \cdot \min(0 \cdot 100 ; 4 \cdot 537 \cdot 15.9) / (\pi \cdot 537^2) }$$

$$= 15.8750 \text{ mm}$$

Effective ligament efficiency for bending [μ *]:

$$= (p* - d*) / p*$$

$$= (15.9 - 12.7) / 15.9$$

$$= 0.2000$$

Ratio [ps]:

$$= as / ao$$

$$= 280/268$$

$$= 1.0433$$

Ratio [pc]:

$$= ac / ao$$

$$= 299/268$$

$$= 1.1144$$

Compute parameter [xt]:

$$= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot ao))^2$$

$$= 1 - 900 ((12.7 - 2 \cdot 2.11) / (2 \cdot 268))^2$$

$$= 0.7754$$

Determine parameter [xs]:

$$= 1 - Nt(dt / (2 \cdot ao))^2$$

$$= 1 - 900 (12.7 / (2 \cdot 268))^2$$

$$= 0.4962$$

Determine the value [h'g]:

$$= \max((hg - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((8 - 0); 0)$$

$$= 8.000 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$= \pi \cdot ts(Ds + ts) Es / L$$

$$= \pi \cdot 24 (560 + 24) 195117184 / 9220$$

$$= 95024.3828 \text{ Kg/mm}$$

Axial Tube Stiffness [Kt]:



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$$= \pi \cdot t t (D t - t t) E t / L$$

$$= \pi \cdot 2.11 (12.7 - 2.11) 195117184 / 9220$$

$$= 151.4919 \text{ Kg/mm}$$

Stiffness Factor [Ks,t]:

$$= K s / (N t \cdot K t)$$

$$= 95024 / (900 \cdot 151)$$

$$= 0.69695$$

Rigidity Ratio [J]:

$$= 1 / (1 + K s / K j)$$

$$= 1 / (1 + 95024 / 0)$$

$$= 1.0000 (= 1 \text{ if No Exp. Jt.})$$

Shell coefficient [β s]:

$$= ((12(1 - v s^2))^{1/4}) / ((D s + t s) t s)$$

$$= ((12(1 - 0.31^2))^{1/4}) / ((560 + 24) 24)$$

$$= 0.0153 \text{ 1/mm}$$

Shell coefficient [ks]:

$$= \beta s \cdot E s \cdot t s^3 / (6(1 - v s^2))$$

$$= 0.015 \cdot 195117184 \cdot 24^3 / (6(1 - 0.31^2))$$

$$= 7623791104.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λ s]:

$$= 6(D s + t s) \cdot k s / h^3 \cdot (1 + h \cdot \beta s + (h \cdot \beta s)^2 / 2)$$

$$= 6(560 + 24) \cdot 7.62E+09 / 190^3 \cdot (1 + 190 \cdot 0.015 + (190 \cdot 0.015)^2 / 2)$$

$$= 31756838.0000 \text{ KPa}$$

Shell coefficient [δ s]:

$$= D s^2 / (4 \cdot E s \cdot t s) \cdot (1 - D s \cdot v s / (2(D s + t s)))$$

$$= 560^2 / (4 \cdot 195117184 \cdot 24) \cdot (1 - 560 \cdot 0.31 / (2(560 + 24)))$$

$$= 0.14253E-01 \text{ mm/N./mm}^2$$

Intermediate parameters for Tubesheet Gasketed on the Channel Side:
 β_C , k_C , λ_C , δ_C are all = 0

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$$h/p = 11.968504 ; \mu^* = 0.200000 ; E^* = 30311284 \text{ KPa}$$

$$E^*/E = 0.155349 ; \nu^* = 0.458572$$

Note: As h/p (11.969) is > 2, data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor [X_a]:

$$= ((24(1 - \nu^*) N t \cdot E t \cdot t t (d t - t t) a o^2) / (E^* \cdot L \cdot H^3))^{1/4}$$

$$= ((24(1 - 0.46^2) 900 \cdot 195117184 \cdot 2.11$$

$$(12.7 - 2.11) 268^2) / (30311284 \cdot 9220 \cdot 190^3))^{1/4}$$

$$= 1.2929$$

Values from Table 4.18.3

$$Z_d = 0.767414 ; Z_v = 0.166877 ; Z_m = 0.672338$$

$$Z_a = 0.487766E+00 ; Z_w = 0.166877$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$= A / D_o$$

$$= 824 / 537$$

$$= 1.5351$$

Compute coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda s + \lambda c + E \cdot \ln(K))$$

$$= (1 - 0.46) / 30311284 \cdot (31756838 + 0 +$$

$$195117184 \cdot \ln(1.54))$$



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= 2.0611

Compute parameter [φ]:

= (1 + v*)F
 = (1 + 0.46)2.06
 = 3.0062

Compute parameter [Q1]:

= (ps - 1 - φ • Zv)/(1 + φ • Zm)
 = (1.04 - 1 - 3.01 • 0.17)/(1 + 3.01 • 0.67)
 = -0.151719

Compute Parameter [Qz1]:

= (Zd + Q1 • Zw)/2 • Xa⁴
 = (0.77 + -0.15 • 0.17)/2 • 1.29⁴
 = 1.0369

Compute parameter [Qz2]:

= (Zv + Q1 • Zm)/2 • Xa⁴
 = (0.17 + -0.15 • 0.67)/2 • 1.29⁴
 = 0.0906

Compute parameter [U]:

= (Zw + (ps - 1)Zm)Xa⁴ / (1 + φ • Zm)
 = (0.17 + (1.04 - 1)0.67)1.29⁴/(1 + 3.01 • 0.67)
 = 0.1813

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [yb]:

= (Gc - C) / Do (config b)
 = (598 - 739)/537 = -0.26

Compute Parameter [γ]:

= 0.000 mm (For Pressure only cases)

Calculate parameter [ωs]:

= ps • ks • βs • δS(1 + h • βs)
 = 1.04 • 7.62E+09 • 0.015 • 0.014252872 (1 + 190 • 0.015)
 = 6799.4688 mm²

Calculate parameter [ω*s]:

= ao²(ps² - 1) • (ps - 1) / 4 - ωs
 = 268²(1.04² - 1) • (1.04 - 1)/4 - 6799
 = -6730.4946 mm²

Parameter [ωc]:

= pc • kc[δC(CmTheta • βC + CqΔ • h • βC²) -
 ζc(CmTheta + h • CqTheta • h • βC / 2))
 = 1.11 • 0 [0.000 (0 + 0 • 0 • 190) -
 0.000 (0/190 + 0 • 0/2))
 = 0.0000 mm²

Parameter [ω*c]:

= ao²[(pc² + 1) • (pc - 1)/4 - (ps - 1)/2] - ωc
 = 268²[(1.11² + 1) • (1.11 - 1)/4 - (1.04 - 1)/2] - 0
 = 3058.1724 mm²

Pressure [P*S]:

= 0 For Pressure only cases or Configurations d,e,f,A,B,C,D

Pressure [P*C]:

= 0 For Pressure only cases or Configurations b,c,d,B,C,D

ASME Section VIII-2 Part 4.18.8.4 Step 6:



Pressure [P's]:

$$= P_s \{ x_s + 2(1 - x_s) v_t + [2/Kst \cdot (D_s/D_o)^2] v_s - [(p_s^2 - 1)/(J \cdot Kst)] - [(1 - J)/(2J \cdot Kst)] [(D_j^2 - (D_s)^2)/D_o^2] \}$$

$$= 8047 \{ 0.5 + 2(1 - 0.5)0.31 + [2/0.7 \cdot (560/537)^2]0.31 - [(1.04^2 - 1)/(1 \cdot 0.7)] - [(1 - 1)/(2 \cdot 1 \cdot 0.7)] [(0^2 - (560)^2)/537^2] \}$$

$$= 13275.9639 \text{ KPa.}$$

Pressure [P't]:

$$= [x_t + 2(1 - x_t) v_t + 1/(J \cdot Kst)] P_t$$

$$= [0.78 + 2(1 - 0.78)0.31 + 1/(1 \cdot 0.7)] 9334$$

$$= 21929.6680 \text{ KPa.}$$

Pressure [P_y]:

$$= N_t \cdot K_t \cdot \gamma / (\pi \cdot a_o^2)$$

$$= 900 \cdot 151 \cdot 0 / (3.14 \cdot 268^2)$$

$$= 0.0000 \text{ KPa.}$$

Pressure [P_w]:

$$= -\gamma_b \cdot U \cdot W^* / (2 \cdot \pi \cdot a_o^2)$$

$$= -0.26 \cdot 0.18 \cdot 381546 / (2 \cdot \pi \cdot 268^2)$$

$$= 393.2560 \text{ KPa.}$$

Calculate the pressure [Prim]:

$$= -(U/a_o^2) \cdot (\omega_s \cdot P_s - \omega_c \cdot P_t)$$

$$= -(0.18/268^2) \cdot (-6730 \cdot 8047 - 3058 \cdot 9334)$$

$$= 208.1430 \text{ KPa.}$$

Calculate the pressure [P_w]:

$$= (U/a_o^2) \cdot (\omega_s \cdot P_s - \omega_c \cdot P_c)$$

$$= (0.18/268^2) \cdot (6799 \cdot 0 - 0 \cdot 0)$$

$$= 0.0000 \text{ KPa.}$$

Determine the effective pressure [Pe]:

$$= J \cdot Kst / (1 + J \cdot Kst(Qz1 + (p_s - 1)Qz2)) \cdot (P's - P't + P_y + P_w + P_w + Prim)$$

$$= 0.1000E+01 \cdot 0.7 / (1 + 1 \cdot 0.7(1.04 + (1.04 - 1)0.091)) \cdot (13276 - 21930 + 0 + 0 + 393 + 208)$$

$$= -3252.6328 \text{ KPa.}$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$= [((\omega_s \cdot P_s - \omega_c \cdot P_t) - (\omega_s \cdot P_s - \omega_c \cdot P_c)) + W^* \cdot \gamma_b / (2 \cdot \pi)] / (1 + \phi \cdot Z_m)$$

$$= [((-6730 \cdot 8047 - 3058 \cdot 9334) - (6799 \cdot 0 - 0 \cdot 0)) \cdot 0.0001 + 381546 \cdot -0.26 / (2 \cdot \pi)] / (1 + 3.01 \cdot 0.67)$$

$$= -8065.5181 \text{ Kgf}$$

Calculate factor [Q3]:

$$= Q1 + 2 \cdot Q2 / (Pe \cdot a_o^2)$$

$$= -0.15 + 2 \cdot -8066 / (-3253 \cdot 268^2)$$

$$= 0.52348$$

Fm Value from Table 4.18.3 = 0.457139

Tube sheet Bending Stress - original thickness [σ]:

$$= (1.5 \cdot F_m / \mu^*) \cdot (2 \cdot a_o / (H - h'g))^2 \cdot Pe$$

$$= (1.5 \cdot 0.46 / 0.2) \cdot (2 \cdot 268 / (190 - 8))^2 \cdot -3253$$

$$= -97.0035 \text{ N./mm}^2$$

Allowable Tube sheet Bending Stress [σ allowed]:

$$= 1.5 \cdot \beta T \cdot S_y = 1.5 \cdot 0.95 \cdot 205 = 292 \text{ N./mm}^2, 4.18.233$$



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Tubesheet Bending Stress - final thickness [σ_f]:
 $= (1.5 \cdot F_m / \mu \cdot) \cdot (2 \cdot a_o / (h - h'g))^2 \cdot P_e$
 $= (1.5 \cdot 0.37 / 0.2) \cdot (2 \cdot 268 / (72.7 - 8))^2 \cdot -1541$
 $= -292.1194 \text{ N./mm}^2$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:
 $= h + C_{ats} + C_{atc}$
 $= 72.7 + 0 + 0$
 $= 72.7397 \text{ mm}$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ_{limit}]:
 $= (2 \cdot \mu \cdot h / a_o) \cdot (0.8 \cdot \beta T \cdot S_y)$
 $= (2 \cdot 0.2 \cdot 190 / 268) \cdot 156$
 $= 44.1187 \text{ N./mm}^2$

Shear Stress is not required to be computed; [P_e] $\leq$ τ_{limit}

Tubesheet Shear Stress is probably low, use the following req. thk:

Tubesheet thickness (Incl. Corr.) = 3.8100 mm
 Tubesheet Shear Stress = 63.6444 N./mm²

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd]:
 $= \max(H_{reqB}; H_{reqS})$
 $= \max(72.7; 3.81)$
 $= 72.7397 \text{ mm}$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The F_{tmin} and F_{tmax} Coefficients from Table 4.18.4:
 $F_{tmin} = 0.79$, $F_{tmax} = 1.19$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:
 $= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmin}) / (X_t - X_s)$
 $= ((8047 \cdot 0.5 - 9334 \cdot 0.78) - (-3253) \cdot 0.79) / (0.78 - 0.5)$
 $= -2.3821 \text{ N./mm}^2$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:
 $= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmax}) / (X_t - X_s)$
 $= ((8047 \cdot 0.5 - 9334 \cdot 0.78) - (-3253) \cdot 1.19) / (0.78 - 0.5)$
 $= 2.2919 \text{ N./mm}^2$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:
 $= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 2.38 \text{ N./mm}^2$

Allowable Tube Stress [$\sigma_{t,A}$]:
 $= \beta T \cdot S_{y,t} = 161.4971 \text{ N./mm}^2$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:
 $= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w)X_a^4); 1.25)$
 $= \max((3.25 - 0.25(0.77 + 0.52 \cdot 0.17)1.29^4); 1.25)$
 $= 2.0000 \text{ (Should be } \leq 2)$

Determine the factor [rt]:
 $= (\sqrt{dt^2 + (dt - 2 \cdot tt)^2}) / 4$
 $= (\sqrt{12.7^2 + (12.7 - 2 \cdot 2.11)^2}) / 4$
 $= 3.8177 \text{ mm}$



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Determine the factor [Ct]:

$$= \sqrt{(2 \cdot \pi^2 \cdot Et / Syt)}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 195117184 / 170)}$$

$$= 179.1504$$

Determine the factor [Ft]:

$$= k \cdot L / r$$

$$= 1 \cdot 327 / 3.82$$

$$= 85.6531$$

Buckling Allowable Stress [Stb]:

$$= (Sy / t / Fs) \cdot (1 - Ft / (2 \cdot Ct))$$

$$= (170 / 2) \cdot (1 - 85.7 / (2 \cdot 179))$$

$$= 161.497 \text{ N./mm}^2 \text{ (Never greater than Sot)}$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$= \sigma_{t,max} \cdot \text{Tube Area}$$

$$= 2.38 \cdot 0.7$$

$$= 17.05 \text{ Kgf}$$

Tube Metal Area [At]:

$$= \pi (do - t) t$$

$$= \pi (12.7 - 2.11) 2.11$$

$$= 0.702 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (do - t) S$$

$$= \pi \cdot 2.11 (12.7 - 2.11) 161$$

$$= 1156.017 \text{ Kgf}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot ag (do + 0.67 \cdot ag) Sw \text{ (but not } > Ft)$$

$$= 0.85 \cdot \pi \cdot 1.65 (12.7 + 0.67 \cdot 1.65) 161$$

$$= 1001.7075 \text{ Kgf}$$

Weld Strength Factor [fw]:

$$= S / Sw \text{ where } Sw = \min(S \text{ and } St)$$

$$= 161 / 161$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw \text{ (but not } > Ft)$$

$$= 0.55 \cdot \pi \cdot 1.65 (12.7 + 0.67 \cdot 1.65) 161$$

$$= 648.1636 \text{ Kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{(0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) fw \cdot fd} - 0.75 \cdot do \right)$$

$$= 2 \left(\sqrt{(0.75 \cdot 12.7)^2 + 1.07 \cdot 2.11 \cdot (12.7 - 2.11) 1 \cdot 1} - 0.75 \cdot 12.7 \right)$$

$$= 2.364 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.2660 mm
 Minimum Required Groove Weld Leg [agr]: 1.2660 mm

Tube-Tubesheet Jt allowable, 1156 is $\geq$ tube strength 1156 Kgf

Note: This tube-tubesheet joint is a Full Strength joint



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ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:
 $= \beta T \cdot S_{y,s} = 0.95 \cdot 205 = 292 \text{ N./mm}^2$

Axial Membrane Stress in Shell $[\sigma_s, m]$:
 $= a o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a s^2 \cdot P_t / ((D_s + t_s) t_s)$
 $= 268^2 / ((560 + 24) 24) \cdot [-3253 + (1.04^2 - 1)$
 $(8047 - 9334)] + 280^2 \cdot 9334 / ((560 + 24) 24)$
 $= 34.9119 \text{ N./mm}^2$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:
 For a given Shell thickness of 24 mm, the minimum Shell
 length adjacent to the tubesheet should be 208.676 mm

Computations completed for ASME tubesheet configuration: b

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	97.0 <=	292.1 N./mm ²	Pass
Tubesheet Shear Stress	11.5 <=	155.8 N./mm ²	Pass
Maximum Tube Stress	2.4 <=	161.5 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-2.4 <=	-161.5 N./mm ²	Pass
Maximum Force on any one Tube	17.1 <=	1156.0 Kgf	Pass
Axial Membrane Stress in Shell	34.9 <=	292.1 N./mm ²	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	72.740	190.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.266	1.650	Ok
Tube-Tubesheet Groove Weld Leg :	1.266	1.650	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses Bend Allwd Shear Allwd	Case Type	Pass/ Fail
D1uc	111.537 65.559	129 292	31 156	Ps+Pt-Th D1	Ok
D2uc	67.392 ...	43 292	19 156	Ps+Pt-Th D2	Ok
D3uc	72.740 ...	97 292	11 156	Ps+Pt-Th D3	Ok
Max:	111.5371 65.559 mm	0.443	0.197 (Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+)-Th]:
 With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
 With or Without Corrosion Allowance



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[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:
 Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Shell Axial Membrane Stress Summary:

---- Shell Stresses ----					:	---- Shell Band Stress ----					:	Pass
Case#	Ten	Allwd	Cmp	Allwd	:	Ten	Allwd	Cmp	Allwd	:	Fail	
D1uc	3	292	...	...	:	...	...	...	...	:		Ok
D2uc	32	292	...	...	:	...	...	...	...	:		Ok
D3uc	35	292	...	...	:	...	...	...	...	:		Ok
Max RATIO	0.120		...		:	...		...		:		

Tube, Shell and Channel Stress Summary:

---- Tube Stresses ----					Tube Loads		Shell Stress		Channel Stress		Pass
Case#	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	Fail
D1uc	8	161	...	...	55	1156	...	292	...	...	Ok
D2uc	5	161	-5	-161	39	1156	...	292	...	...	Ok
D3uc	2	161	-2	-161	17	1156	...	292	...	...	Ok
Max RATIO	0.048		0.033		0.048		...		...		

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	111.537	190.000	Ok
Tubesheet thickness flanged extension :	65.559	154.000	Ok
Tube thickness :	0.910	2.110	Ok
Tube-tubesheet fillet weld leg :	1.266	1.650	Ok
Tube-tubesheet groove weld leg :	1.266	1.650	Ok

Min Shell length of thk, (24.000) adj. to tubesheet: 208.676 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm
 Due to Pressure Shell Compresses by : -0.343 mm
 Due to Pressure + Thermal Shell Compresses by : -0.343 mm

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0228	...	...	...	...	...
2	-0.3429	...	...	...	...	...
3	-0.3200	...	...	...	...	...
4	...	...	...	...	...	...
5	...	...	...	...	...	...
6	...	...	...	...	...	...
7	...	...	...	...	...	...
8	...	...	...	...	...	...



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External Pressure Calculation Results Summary:

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External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B N./mm ²
10	20	No Calc	604	21	0.0048289	59.1573
20	30	897.467	608	23	0.0066127	61.4609
30	40	No Calc	...	121	No Calc	No Calc
40	50	9220	608	24	0.0017432	49.9778
50	60	9220	632	36	0.0037299	57.0787
60	70	9220	632	36	0.0037299	57.0787
70	80	9220	632	36	0.0037299	57.0787
80	90	9220	632	36	0.0037299	57.0787
90	100	9220	632	36	0.0037299	57.0787
100	110	9220	608	24	0.0017432	49.9778
110	120	No Calc	...	121	No Calc	No Calc
120	130	697.467	608	23	0.0087746	63.3206
130	140	No Calc	604	21	0.0048289	59.1573

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	22	2.74824	103.419	2285
20	30	24	3.44893	103.419	3100
30	40	122	116.738	103.419	No Calc
40	50	24	5.61515	103.419	2630
50	60	36	5.83084	103.419	4335
60	70	36	5.83084	103.419	4335
70	80	36	5.83084	103.419	4335
80	90	36	5.83084	103.419	4335
90	100	36	5.83084	103.419	4335
100	110	24	5.61515	103.419	2630
110	120	122	116.357	103.419	No Calc
120	130	24	3.20096	103.419	3194
130	140	22	2.74824	103.419	2285

Minimum

2285

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	897.467	24134	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	9220	234153	No Calc	No Calc
50	60	9220	386130	No Calc	No Calc
60	70	9220	386130	No Calc	No Calc
70	80	9220	386130	No Calc	No Calc
80	90	9220	386130	No Calc	No Calc
90	100	9220	386130	No Calc	No Calc
100	110	9220	234153	No Calc	No Calc
110	120	No Calc	No Calc	No Calc	No Calc



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External Pressure Calculations: Step: 19 10:04am Sep 9,2026

120	130	697.467	19103.1	No Calc	No Calc
130	140	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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 FileName : 188-C1-C2-C3-C4- Solid SS Top 2nd stack Page 87 of 223
 Element and Detail Weights: Step: 20 10:04am Sep 9,2026

Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	114.101	35506.5	110.026	35843.3	11.4101
20	30	282.77	197076	271.452	198486	28.277
30	40	314.941	43959.2	312.448	44039.1	31.4941
40	50	530.194	198473	530.194	198473	53.0194
50	60	541.089	132315	541.089	132315	54.1089
60	70	811.633	198473	811.633	198473	81.1633
70	80	660.129	161425	660.129	161425	66.0129
80	90	811.633	198473	811.633	198473	81.1633
90	100	541.089	132315	541.089	132315	54.1089
100	110	530.194	198473	530.194	198473	53.0194
110	120	318.497	43339.5	316.021	43418.9	31.8497
120	130	212.078	147807	203.589	148865	21.2078
130	140	114.101	35506.5	110.026	35843.3	11.4101
Total		5782	1723143.00	5749	1726443.62	578



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Liqd	35.4513	-46.6667	5	...	Liquid: 10
10	Insl	27.5421	25.4	...	...	Insulation
20	Liqd	196.162	400	5	...	Liquid: 20
20	Insl	62.7081	400	...	...	Insulation
20	Nozl	138.071	400	401.437	...	N3-Tube Inlet
20	Nozl	138.071	400	-401.437	...	N4-Tube Outlet
20	Wght	400	400	...	...	Partition Plate
30	Liqd	43.6429	83.5	7	...	Liquid: 30
30	Insl	16.5008	90	...	...	Insulation
40	Liqd	198.352	750	3.49999	...	Liquid: 40
40	Insl	117.578	750	...	...	Insulation
40	Nozl	184.473	750	405.412	...	N1-Shell Inlet
50	Sadl	330.015	500	540	...	Left Saddle
50	Liqd	132.235	500	3.49999	...	Liquid: 50
50	Insl	81.0423	500	...	...	Insulation
50	Lini	42.0459	500	...	...	Clad
50	Wght	300	499.995	-950	...	Fireproofing we
50	Wght	200	499.995	275	...	Saddle 1
50	Forc	...	512.995	...	...	Stack force
50	Forc	...	512.995	...	...	Stack force
60	Liqd	198.352	750	3.49999	...	Liquid: 60
60	Insl	121.563	750	...	...	Insulation
60	Lini	63.0689	750	...	...	clad
70	Liqd	161.326	610	3.49999	...	Liquid: 70
70	Insl	98.8716	610	...	...	Insulation
70	Lini	51.2961	610	...	...	Clad
70	Wght	3500	395.645	1700	...	TOP HE OPERATING
70	Wght	3500	395.645	1700	...	TOP HE TEST
70	Wght	14500	371.645	1699.74	-29.9688	TOP HE EMPTY
80	Liqd	198.352	750	3.49999	...	Liquid: 40
80	Insl	121.563	750	...	...	Insulation



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 FileName : 188-C1-C2-C3-C4- Solid SS Top 2nd stack Page 88 of 223
 Element and Detail Weights: Step: 20 10:04am Sep 9,2026

80 Lini	63.0689	750	...	...	Clad
90 Sadl	330.015	500	540	...	New Saddle
90 Liqd	132.235	500	3.49999	...	Liquid: 90
90 Insl	81.0423	500	...	...	Insulation
90 Lini	42.0459	500	...	...	Clad
90 Wght	300	499.996	-950	...	Fireproofing we
90 Wght	200	499.996	275	...	saddle 2
90 Forc	...	512.995	...	...	stack force
90 Forc	...	512.995	...	...	stack force
100 Liqd	198.352	750	3.49999	...	Liquid: 100
100 Insl	117.578	750	...	...	Insulation
100 Lini	63.0689	750	...	...	Clad
100 Nozl	184.473	750	-405.412	...	N2-Shell outlet
110 Liqd	43.1386	85.5	5	...	Liquid: 50
110 Insl	16.711	90	...	...	Insulation
120 Liqd	147.122	300	5	...	Liquid: 60
120 Insl	47.0311	300	...	...	Hot Insulation
130 Liqd	35.4513	97.4667	5	...	Liquid: 70
130 Insl	27.5421	94.15	...	...	Hot Insulation
30 FTsh	703.391	269.175	...	...	
30 Tube	5355.56	4974.17	...	...	Tubes
30 RTsh	703.391	9679.17	...	...	Tubesheet

Total Weight of Each Detail Type:

Saddles	660.0
Liquid	1720.2
Insulation	937.3
Cladding & Overlay	324.6
Nozzles	645.1
Weights	22900.0
Exchanger Components	6762.3
Liquid in Tubes	487.8

Sum of the Detail Weights 34437.3 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	6360.7	6360.7	6360.7	6360.7	6360.7	6360.7
Saddles	660.0	660.0	660.0	660.0	660.0	660.0
Cladding	324.6	324.6	324.6	324.6	324.6	324.6
Nozzles	645.1	645.1	645.1	645.1	645.1	645.1
Wld Weights	15300.0	15300.0	15300.0	15300.0	15300.0	15300.0
Exchanger	6762.3	6762.3	6762.3	6762.3	6762.3	6762.3
Insulation	...	...	...	...	937.3	937.3 *
Test Weights	...	4100.0	...	...	...	...
Ope Weights	...	...	...	...	...	4100.0
Ope. Liquid	...	...	...	...	...	1720.2
Tube Ope Lqd	...	...	...	...	...	487.8
Test Liquid	...	1681.6	...	...	...	...
Tube Tst Lqd	...	487.8	...	...	...	...
Totals	30052.7	36322.1	30052.7	30052.7	30990.0	37298.0

Field Installation Options:

* Insulation installed after lifting.

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.



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 FileName : 188-C1-C2-C3-C4- Solid SS Top 2nd stack Page 89 of 223
 Element and Detail Weights: Step: 20 10:04am Sep 9,2026

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	30052.7 kg.
Shop Test Wt.	- Fabricated weight + water (full)	36322.1 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	30052.7 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	30052.7 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	30990.0 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	37298.0 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	37261.7 kg.
Field Test Wt.	- Empty weight + test medium (full)	37259.4 kg.

Exchanger Tube Data

Volume of Exchanger tubes :	488059.9 cm ³
Weight of Ope Liq in tubes :	487.8 kg.
Weight of Water in tubes :	487.8 kg.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	5096.34	4293.1
20	30	15280.7	14074.3
30	40	7090.26	3100.75
40	50	28651.3	26389.4
50	60	19854.9	17592.9
60	70	29782.3	26389.4
70	80	24222.9	21463.4
80	90	29782.3	26389.4
90	100	19854.9	17592.9
100	110	28651.3	26389.4
110	120	7119.92	3078.76
120	130	11460.5	10555.8
130	140	5096.34	4293.1

Total	231944.0	201602.5 cm ²
Total	249.7	217.0 ft ²

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 FileName : 188-C1-C2-C3-C4- Solid SS Top 2nd stack Page 90 of 223
 Nozzle Flange MAWP: Step: 21 10:04am Sep 9,2026

Nozzle Flange MAWP Results: (KPa. & °C)

Nozzle Description	Flange Rating Ope.	Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
N3-Tube Inlet	9290.1	14910.0	300	900	GR 2.1	3882.5	9290.1	9290.1	9290.1
N4-Tube Outlet	9290.1	14910.0	300	900	GR 2.1	...	...	...	...
N1-Shell Inlet	9290.1	14910.0	300	900	GR 2.1	3882.5	9290.1	5407.5	9290.1
N2-Shell outlet	9290.1	14910.0	300	900	GR 2.1	3882.5	9290.1	5407.5	9290.1

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 9290.075 KPa.
 Lowest Flange Pressure Rating was (Amb)[ShellSide]: 14910.002 KPa.

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 9290.075 KPa.
 Lowest Flange Pressure Rating was (Amb)[TubeSide]: 14910.002 KPa.

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kgf	MAWP PD KPa.	Maximum Pressure Pmax KPa.	Pass Fail
N3-Tube Inlet	293.751	0.50	1802	2294	6396.2	9290.1	Pass
N1-Shell Inlet	293.751	0.50	1802	2294	6175.3	9290.1	Pass
N2-Shell outlet	293.751	0.50	1802	2294	6175.3	9290.1	Pass

Evaluating Flange on Nozzle [N3-Tube Inlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(1802) + 4 \cdot \text{abs}(2294) \cdot 293.751 \leq \pi \cdot 293.751^3 \cdot [(9290.075 - 6396.208) + .500 \cdot 9290.075]$$

$$= 0.31529E+05 \leq 0.61221E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N3-Tube Inlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(9290.075, 0.500 \cdot 9290.075 + 9290.075 - 31528.9 / (\pi \cdot 293.751^3))$$

$$= \min(9290.075, 10052.580)$$

$$= 9290.075 \text{ KPa.}$$

Evaluating Flange on Nozzle [N1-Shell Inlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(1802) + 4 \cdot \text{abs}(2294) \cdot 293.751 \leq \pi \cdot 293.751^3 \cdot [(9290.075 - 6175.303) + .500 \cdot 9290.075]$$

$$= 0.31529E+05 \leq 0.63015E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N1-Shell Inlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(9290.075, 0.500 \cdot 9290.075 + 9290.075 - 31528.9 / (\pi \cdot 293.751^3))$$

$$= \min(9290.075, 10052.580)$$

$$= 9290.075 \text{ KPa.}$$

Evaluating Flange on Nozzle [N2-Shell outlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$



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Nozzle Flange MAWP: Step: 21 10:04am Sep 9, 2026

$$\begin{aligned} &= 16(1802) + 4 \cdot \text{abs}(2294) 293.751 \leq \pi \cdot 293.751^3 [(9290.075 - 6175.303) + .500 \cdot 9290.075] \\ &= 0.31529\text{E}+05 \leq 0.63015\text{E}+05 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N2-Shell outlet [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(9290.075, 0.500 \cdot 9290.075 + 9290.075 - 31528.9 / (\pi \cdot 293.751^3)) \\ &= \min(9290.075, 10052.580) \\ &= 9290.075 \text{ KPa.} \end{aligned}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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 FileName : 188-C1-C2-C3-C4- Solid SS Top 2nd stack Page 92 of 223
 Wind Load Calculation: Step: 22 10:04am Sep 9,2026

Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	92.7 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	1500 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q)/(1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.23 \cdot 0.98)/(1 + 1.7 \cdot 3.4 \cdot 0.23)))$
 $= \min(0.85, 0.91)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.644
Structure Height to Diameter ratio	14.843

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
 Exposure Category: C from Table 26.11-1
 $\alpha = 9.5 : Zg = 274320 \text{ mm}$

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 850 + 1500 = 2350 \text{ mm}$
 $= 7.71 \text{ ft. Imperial Units}$

Velocity Pressure coefficient evaluated at height z $[Kz]$:
 Because $z (7.71 \text{ ft.}) < 15 \text{ ft.}$
 $= 2.01(15 / Zg)^{2/\alpha}$
 $= 2.01(15/900)^{2/9.5}$
 $= 0.85$

Type of Hill: No Hill



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 FileName : 188-C1-C2-C3-C4- Solid SS Top 2nd stack Page 93 of 223
 Wind Load Calculation: Step: 22 10:04am Sep 9,2026

Wind Directionality Factor [Kd]:
 = 1, per Table 26.6-1

As there is No Hill Present: [Kzt]:
 K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:
 = (1 + K1 • K2 • K3)²
 = (1 + 0 • 0 • 0)²
 = 1

Velocity Pressure evaluated at height z, Imperial Units [qz]:
 = max(16, 0.00256 • Kz • Kzt • Kd • Ke • V(mph)²)
 = max(16, 0.00256 • 0.85 • 1 • 1 • 1 • 57.6²)
 = 16 psf [78.1] Kgs/m²

Force on the first element [F]:
 = qz • G • Cf • WindArea
 = 16 • 0.85 • 0.64 • 2.31
 = 20.2 lbs. [9.16] Kgf

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
LHS Dish End	2350.0	0.000	0.000	0.000	0.849	1.000	78.120
LHS Ch Shell	2350.0	0.000	0.000	0.000	0.849	1.000	78.120
LHS ch flange	2350.0	0.000	0.000	0.000	0.849	1.000	78.120
Main Shell	2350.0	0.000	0.000	0.000	0.849	1.000	78.120
Main Shell	2350.0	0.000	0.000	0.000	0.849	1.000	78.120
Main Shell	2350.0	0.000	0.000	0.000	0.849	1.000	78.120
Main Shell	2350.0	0.000	0.000	0.000	0.849	1.000	78.120
Main Shell	2350.0	0.000	0.000	0.000	0.849	1.000	78.120
Main Shell	2350.0	0.000	0.000	0.000	0.849	1.000	78.120
Main Shell	2350.0	0.000	0.000	0.000	0.849	1.000	78.120
RHS Ch Flange	2350.0	0.000	0.000	0.000	0.849	1.000	78.120
RHS Ch Shell	2350.0	0.000	0.000	0.000	0.849	1.000	78.120
RHS Dish End	2350.0	0.000	0.000	0.000	0.849	1.000	78.120

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm	Pressure Kgs/m ²	Force Kgf
Partition Pla	0.00	2350.00	78.12	0.00
Fireproofing	0.00	1500.00	78.12	0.00
Saddle 1	0.00	2625.00	78.12	0.00
TOP HE OPERAT	0.00	4050.00	78.12	0.00
TOP HE TEST	0.00	4050.00	78.12	0.00
TOP HE EMPTY	117913.00	4049.74	78.12	921.14
Fireproofing	0.00	1500.00	78.12	0.00
saddle 2	0.00	2625.00	78.12	0.00

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	2350	964.8	2144.54	78.12	10.0811
20	30	2350	969.6	7756.8	78.12	36.4633
30	40	2350	916.8	1531.06	78.12	7.19721
40	50	2350	969.6	14544	78.12	68.3686
50	60	2350	998.4	9984	78.12	46.9329
60	70	2350	998.4	14976	78.12	70.3994



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Wind Load Calculation: Step: 22 10:04am Sep 9, 2026

70	80	2350	998.4	12180.5	78.12	1070.51
80	90	2350	998.4	14976	78.12	70.3994
90	100	2350	998.4	9984	78.12	46.9329
100	110	2350	969.6	14544	78.12	68.3686
110	120	2350	912	1559.52	78.12	7.33101
120	130	2350	969.6	5817.6	78.12	27.3475
130	140	2350	964.8	2144.54	78.12	10.0811

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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 FileName : 188-C1-C2-C3-C4- Solid SS Top 2nd stack Page 95 of 223
 Earthquake Load Calculation: Step: 23 10:04am Sep 9,2026

Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.300
Table Value Fv	1.500
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	3.000
Site Class	C
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.3 * 0.15 = 0.19 \\ SM1 &= Fv * S1 = 1.5 * 0.065 = 0.097 \\ SDS &= 2/3 * Sms = 2/3 * 0.19 = 0.13 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.097 = 0.065 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_n^x$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(3.710^{0.75})$
 = 0.054 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/33)
 = 0.030

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of ($1.7 \cdot 0.054$, 0.03) per 12.8.2
 = 0.030
 As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 = $0.3 \cdot Sds \cdot W \cdot I$
 = $0.3 \cdot 0.13 \cdot 36638 \cdot 1.25$
 = 1750.377 Kgf

Vertical load per 12.4-4, [Ev]:
 = $0.2 \cdot Sds \cdot W$
 = $0.2 \cdot 0.13 \cdot 36638$
 = 933.5 Kgf

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight Kgf	Element Ope Load Kgf
10	20	280	2442.53	116.636



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Earthquake Load Calculation: Step: 23 10:04am Sep 9, 2026

20	30	280	2442.53	116.636
30	40	282	2442.53	117.469
40	50	280	2442.53	116.636
50	60	280	2442.53	116.636
50	60	280	2442.53	116.636
60	70	280	2442.53	116.636
70	80	280	2442.53	116.636
80	90	280	2442.53	116.636
90	0	280	2442.53	116.636
90	100	280	2442.53	116.636
100	110	280	2442.53	116.636
110	120	280	2442.53	116.636
120	130	280	2442.53	116.636
130	140	280	2442.53	116.636

The Earthquake Weight does not include the weight of the saddles.

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Center of Gravity Calculation: Step: 24 10:04am Sep 9,2026

Shop/Field Installation Options :

Insulation is installed in the Field after being lifted.
Lining is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	5820.975 mm
Center of Gravity of the Liquid	5654.324 mm
Center of Gravity of the Insulation	5723.979 mm
Center of Gravity of the Lining	6570.975 mm
Center of Gravity of the Nozzles	3522.174 mm
Center of Gravity of the Added Weights (Operating)	5493.424 mm
Center of Gravity of the Added Weights (Empty)	5454.686 mm
Center of Gravity of the Tubesheet(s)	5824.975 mm
Center of Gravity of the Tubes	5824.975 mm
Center of Gravity of Bare Shell New and Cold	5751.047 mm
Center of Gravity of Bare Shell Corroded	5753.584 mm
Vessel CG in the Operating Condition	5596.382 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	5579.352 mm
Vessel CG in the Test Condition	5587.021 mm

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 Saddle Calcs: Road Transport: Step: 25 10:04am Sep 9,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Road Transport

Wear Pad Bearing angle is less than saddle angle + saddle/12 degrees. The wear plate will be ignored.

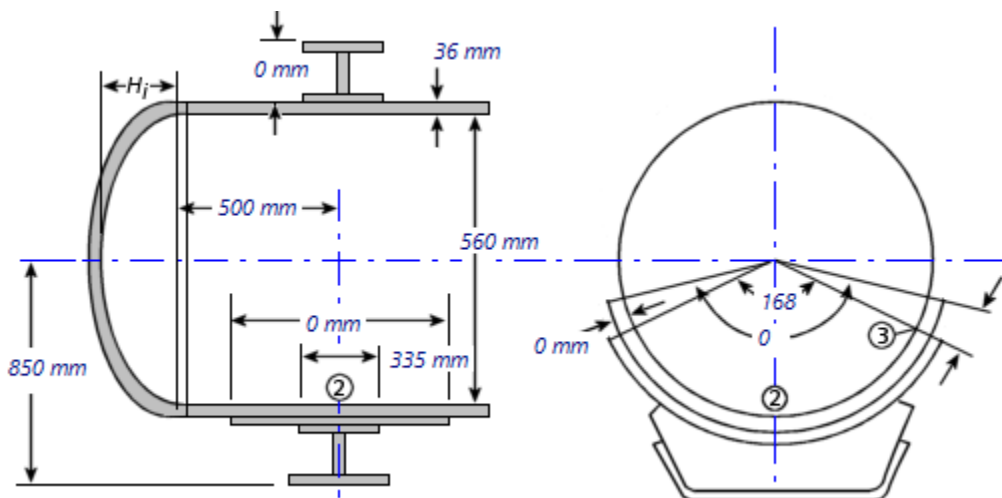
Minimum Wear Plate Angle [θ1]:
 = 180.0 degrees

Note:
 Wear Pad Width (.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 $= \min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a)$
 $= \min(335 + 1.56 \cdot \sqrt{298 \cdot 36}, 2 \cdot 500)$
 $= 496.5786 \text{ mm}$

Input and Calculated Values:

Vessel Mean Radius	Rm	298.00	mm
Shell Thickness used in this Case	t	36.000	mm
Stiffened Vessel Length per 4.15.6	L	9220.00	mm
Distance from Saddle to Vessel tangent	a1 or a	500.00	mm
Saddle Width	b1 or b	335.00	mm
Saddle Bearing Angle	delta or theta	168.00	degrees
Shell Allowable Stress used in Calculation		85.90	N./mm ²
Head Allowable Stress used in Calculation		85.90	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	850.00	mm
Coefficient of Friction	mu	0.30	



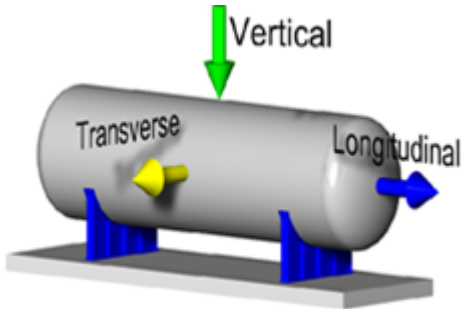
Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		34882.39	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	1.05	g



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Vertical Acceleration Acting with Transverse gw 1.00 g
 Internal Pressure during Transport 0.00 KPa.
 Transportation Wind Load Scalar Factor wlsf 1.00
 Transportation Wind Velocity wv 0.0 Km/hr



Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-61.64	-184.50
Long. Stress at Top of Midspan	61.64	85.90
Long. Stress at Bottom of Midspan	61.64	85.90
Long. Stress at Top of Saddles	3.80	85.90
Long. Stress at Bottom of Saddles	-2.25	-184.50
Long. Stress at Bottom of Saddles	2.25	85.90
Tangential Shear in Shell	18.59	68.72
Circ. Stress at Horn of Saddle	13.65	128.85
Circ. Compressive Stress in Shell	1.23	85.90

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [Ftw]:
 = Transverse Area • 0.00256 • wv² • wlsf
 = 121 • 0.00256 • 0² • 1
 = 0.0 lbf
 = 0.000 Kgf

Longitudinal Force due to Wind during Transport [Flw]:
 = $\pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult}$
 = $\pi \cdot 2.73^2/4 \cdot 0.00256 \cdot 0^2 \cdot 1 \cdot 1.2$
 = 0.0 lbf
 = 0.0 Kgf

Transverse Force due to Lateral Acceleration [Ft]:
 = gz • Shipping Weight • gw + Ftw
 = 0.1 • 29393 • 1 + 0
 = 2939.3 Kgf

Longitudinal Force due to Axial Acceleration [FI]:
 = gx • Shipping Weight • gw + Flw
 = 0.1 • 29393 • 1 + 0
 = 2939.3 Kgf

Saddle Reaction Force due to Transverse Load [Fwt]:
 = Ft/Num of Saddles • B / E
 = 2939/2 • 850/656
 = 1903.1 Kgf

Saddle Reaction Force due to Longitudinal Load [Fwl]:
 = FI • B / Ls
 = 2939 • 850/5220



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= 478.6 Kgf

Saddle Reaction Force due to Vertical Transportation Acceleration [Frv]:

= Saddle Load • gy
 = 16087 • 1.05
 = 16891.809 Kgf

Load Combination Results due to Transport Loading [Q]:

= Saddle Load + Frv + max(Fwl, Fwt)
 = 16087 + 16892 + max(479, 1903)
 = 34882.4 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	35212.40	Kgf
Transverse Shear Load	1469.64	Kgf
Longitudinal Shear Load	2939.27	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= min(2(ShellID/2 + t + WearPadThickness)sin(θ/2) , 2•Rm)
 = min(2(560/2 + 36 + 14)sin(168/2) , 2•298)
 = 656.384 mm

The Computed K values from Table 4.15.1:

K1 = 0.1997	K2 = 0.6539	K3 = 0.3357	K4 = 0.2452
K5 = 0.6404	K6 = 0.0224	K7 = 0.0224	K8 = 0.2720
K9 = 0.1759	K10 = 0.0221	K1* = 0.3382	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]
 = -34882•500[1-(1-500/9220+(298²-0²)/
 (2•500•9220))/(1+(4•0)/(3•9220))]
 = -777.9 Kg-m.

Moment per Equation 4.15.2 [M2]:

= Q•L/4((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L))-4a/L)
 = 34882•9220/4((1+2(298²-0²)/(9220²))/(1+(4•0)/
 (3•9220))-4•500/9220)
 = 63131.9 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ1]:

= P • Rm/(2t) - M2/(π•Rm²t)
 = 0 • 298/(2•36) - 63132/(π•298²•36)
 = -61.64 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ2]:

= P • Rm/(2t) + M2/(π • Rm² • t)
 = 0 • 298/(2 • 36) + 63132/(π • 298² • 36)
 = 61.64 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ*3]:

= P • Rm/(2t) - M1/(K1•π•Rm²t)
 = 0•298/(2•36)--778/(0.2•π•298²•36)
 = 3.80 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ*4]:

= P • Rm/(2t) + M1/(K1* • π • Rm² • t)
 = 0•298/(2•36)+-778/(0.34•π•298²•36)
 = -2.25 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:

= Q(L-2a)/(L+(4•hm/3))



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$$= 34882 (9220 - 2 \cdot 500) / (9220 + (4 \cdot 0/3))$$

$$= 31099.0 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 0.65 \cdot 31099 / (298 \cdot 36)$$

$$= 18.59 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{ (R_m \cdot t) }$$

$$= 0.78 \cdot \sqrt{ (298 \cdot 36) }$$

$$= 80.789 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t (b + X_1 + X_2))$$

$$= -0.64 \cdot 34882 \cdot 0.1 / (36 (335 + 80.8 + 80.8))$$

$$= -1.23 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t (b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$$

$$= -34882 / (4 \cdot 36 (335 + 80.8 + 80.8)) -$$

$$3 \cdot 0.022 \cdot 34882 / (2 \cdot 36^2)$$

$$= -13.65 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{ (R_m \cdot t) }, 2a)$$

$$= \min(335 + 1.56 \cdot \sqrt{ (298 \cdot 36) }, 2 \cdot 500)$$

$$= 496.58 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	780.0000	mm
Baseplate Thickness	Bpthk	70.0000	mm
Baseplate Width	Bpwid	360.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	575.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{ (R \cdot t) } + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{ (280 \cdot 36) } + 530) 36^3) / 12$$

$$= 266.959 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C_1 - Y)^2$$

$$= 267 + 247 (250 - 18)^2$$

$$= 133754.688 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² (2) cm ⁴
Shell	686.6	36.0	18.0	247.2	444931.6	0.134E+06
Wearplate	530.0	14.0	43.0	74.2	319060.0	0.319E+05
Web	10.0	450.0	275.0	45.0	1237500.1	0.787E+04
BasePlate	360.0	70.0	535.0	252.0	13482000.0	0.205E+06
Totals	...	...	...	618.4	15483492.0	0.379E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 15483492/618$$



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$$= 250.386 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 168/2$$

$$= 96.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(96) - 0.5 \cdot \sin(96)^2) / (\pi - 1.68 + \sin(96) \cos(96))$$

$$= 0.2943$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.29 \cdot 34882$$

$$= 10267.4854 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh/As) = 2.7126 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 850 - 280 \cdot \sin(168/2) / 1.47$$

$$= 660.060 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 6777.2944 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 4.3942 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(34882 + 330) \cdot 360) / (4 \cdot 780 \cdot 173)}$$

$$= 26.261 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 168/2) (280 + 36 + 14)$$

$$= 656.384 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nrings} - 1)$$

$$= 656 / (3 - 1)$$

$$= 328.192 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 328 \cdot 360 / 2$$

$$= 590.746 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 35212 / (780 \cdot 360)$$

$$= 12.540 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 591 \cdot 12.5$$

$$= 7407.972 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 32.5 + 16.4$$



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$$= 48.910 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/A_r$$

$$= 7408/48.9$$

$$= 14.854 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	335.0	...	33.5	...	0.313E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 335 - 10$$

$$= 325.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/48.9$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 335/2$$

$$= 167.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(3133/48.9)}$$

$$= 80.035 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 850 - \cos(168/2) (280 + 36 + 14) - 70$$

$$= 745.506 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008/260)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 746/80$$

$$= 9.315$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 2939 / (2 \cdot 780) \cdot 328 \cdot 746/2$$

$$= 230.501 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($9.31 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (9.31)^2 / (2 \cdot 123^2)) 260 /$$

$$(5/3 + 3 \cdot (9.31) / (8 \cdot 123) - (9.31^3) / (8 \cdot 123^3))$$

$$= 153 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 14.9/153 + (231 \cdot 168/31329476) / 173$$

$$= 0.167$$



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Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	325.0	0.0	32.5	0.0	0.313E+04
Web	328.2	10.0	0.0	32.8	0.0	2.73
Totals	...	...	...	65.3	...	0.314E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/65.3$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 335/2$$

$$= 167.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 850 - \cos(330/2)(70 + 0 + 0) - 0$$

$$= 450.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(3136/65.3)}$$

$$= 69.285 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 450/69.3$$

$$= 6.495$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 2939 / (2 \cdot 780)$$

$$= 1.884 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.88 \cdot 328 \cdot 450$$

$$= 278.269 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 278/187200$$

$$= 14.577 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($6.49 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (6.49)^2 / (2 \cdot 123^2)) 260 /$$

$$(5/3 + 3 \cdot (6.49) / (8 \cdot 123) - (6.49^3) / (8 \cdot 123^3))$$

$$= 154 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 22.3/154 + (278 \cdot 168/3136) / 173$$

$$= 0.229$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Wear Pad Bearing angle is less than saddle angle + saddle/12 degrees. The wear plate will be ignored.



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Minimum Wear Plate Angle [01]:
 = 180.0 degrees

Note:
 Wear Pad Width (.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 $= \min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a)$
 $= \min(335 + 1.56 \cdot \sqrt{298 \cdot 36}, 2 \cdot 500)$
 = 496.5786 mm

Input and Calculated Values:

Vessel Mean Radius	Rm	298.00	mm
Shell Thickness used in this Case	t	36.000	mm
Stiffened Vessel Length per 4.15.6	L	9220.00	mm
Distance from Saddle to Vessel tangent	a1 or a	500.00	mm
Saddle Width	b1 or b	335.00	mm
Saddle Bearing Angle	delta or theta	168.00	degrees
Shell Allowable Stress used in Calculation		85.90	N./mm ²
Head Allowable Stress used in Calculation		85.90	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	850.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		29178.95	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	1.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	0.0	Km/hr

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan		-51.56	-184.50
Long. Stress at Top of Midspan		51.56	85.90
Long. Stress at Bottom of Midspan		51.56	85.90
Long. Stress at Top of Saddles		3.18	85.90
Long. Stress at Bottom of Saddles		-1.88	-184.50
Long. Stress at Bottom of Saddles		1.88	85.90
Tangential Shear in Shell		15.55	68.72
Circ. Stress at Horn of Saddle		11.42	128.85
Circ. Compressive Stress in Shell		1.03	85.90

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [Ftw]:
 = Transverse Area • 0.00256 • wv² • wlsf
 = 121 • 0.00256 • 0² • 1
 = 0.0 lbf
 = 0.000 Kgf



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Longitudinal Force due to Wind during Transport [Flw]:
 $= \pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wsf \cdot \text{Wind Dia. Mult}$
 $= \pi \cdot 2.73^2/4 \cdot 0.00256 \cdot 0^2 \cdot 1 \cdot 1.2$
 $= 0.0 \text{ lbf}$
 $= 0.0 \text{ Kgf}$

Transverse Force due to Lateral Acceleration [Ft]:
 $= gz \cdot \text{Shipping Weight} \cdot gw + Ftw$
 $= 0.1 \cdot 29393 \cdot 1 + 0$
 $= 2939.3 \text{ Kgf}$

Longitudinal Force due to Axial Acceleration [FI]:
 $= gx \cdot \text{Shipping Weight} \cdot gw + Flw$
 $= 0.1 \cdot 29393 \cdot 1 + 0$
 $= 2939.3 \text{ Kgf}$

Saddle Reaction Force due to Transverse Load [Fwt]:
 $= Ft/\text{Num of Saddles} \cdot B / E$
 $= 2939/2 \cdot 850/656$
 $= 1903.1 \text{ Kgf}$

Saddle Reaction Force due to Longitudinal Load [Fwl]:
 $= FI \cdot B / Ls$
 $= 2939 \cdot 850/5220$
 $= 478.6 \text{ Kgf}$

Saddle Reaction Force due to Vertical Transportation Acceleration [Frv]:
 $= \text{Saddle Load} \cdot gy$
 $= 13305 \cdot 1.05$
 $= 13970.538 \text{ Kgf}$

Load Combination Results due to Transport Loading [Q]:
 $= \text{Saddle Load} + Frv + \max(Fwl, Fwt)$
 $= 13305 + 13971 + \max(479, 1903)$
 $= 29179.0 \text{ Kgf}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	29508.97	Kgf
Transverse Shear Load	1469.64	Kgf
Longitudinal Shear Load	2939.27	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot Rm)$
 $= \min(2(560/2 + 36 + 14) \sin(168/2), 2 \cdot 298)$
 $= 656.384 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1997	K2 = 0.6539	K3 = 0.3357	K4 = 0.2452
K5 = 0.6404	K6 = 0.0224	K7 = 0.0224	K8 = 0.2720
K9 = 0.1759	K10 = 0.0221	K1* = 0.3382	

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$
 $= -29179 \cdot 500 [1 - (1 - 500/9220 + (298^2 - 0^2)/(2 \cdot 500 \cdot 9220))/(1 + (4 \cdot 0)/(3 \cdot 9220))]$
 $= -650.7 \text{ Kg-m.}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4 [(1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L]$
 $= 29179 \cdot 9220/4 [(1 + 2(298^2 - 0^2)/(9220^2))/(1 + (4 \cdot 0)/(3 \cdot 9220)) - 4 \cdot 500/9220]$



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= 52809.6 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t)$$

$$= 0 \cdot 298 / (2 \cdot 36) - 52810 / (\pi \cdot 298^2 \cdot 36)$$

$$= -51.56 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 298 / (2 \cdot 36) + 52810 / (\pi \cdot 298^2 \cdot 36)$$

$$= 51.56 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t)$$

$$= 0 \cdot 298 / (2 \cdot 36) - 651 / (0.2 \cdot \pi \cdot 298^2 \cdot 36)$$

$$= 3.18 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 298 / (2 \cdot 36) + 651 / (0.34 \cdot \pi \cdot 298^2 \cdot 36)$$

$$= -1.88 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 29179 (9220 - 2 \cdot 500) / (9220 + (4 \cdot 0 / 3))$$

$$= 26014.2 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 0.65 \cdot 26014 / (298 \cdot 36)$$

$$= 15.55 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t}$$

$$= 0.78 \cdot \sqrt{298 \cdot 36}$$

$$= 80.789 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.64 \cdot 29179 \cdot 0.1 / (36(335 + 80.8 + 80.8))$$

$$= -1.03 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$$

$$= -29179 / (4 \cdot 36(335 + 80.8 + 80.8)) -$$

$$3 \cdot 0.022 \cdot 29179 / (2 \cdot 36^2)$$

$$= -11.42 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(335 + 1.56 \cdot \sqrt{298 \cdot 36}, 2 \cdot 500)$$

$$= 496.58 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	780.0000	mm
Baseplate Thickness	Bpthk	70.0000	mm
Baseplate Width	Bpwid	360.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	575.0	mm
Friction Coefficient	mu	0.000	



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Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(280 \cdot 36)} + 530)36^3)/12$$

$$= 266.959 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 267 + 247(250 - 18)^2$$

$$= 133754.688 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² (2) cm ⁴
Shell	686.6	36.0	18.0	247.2	444931.6	0.134E+06
Wearplate	530.0	14.0	43.0	74.2	319060.0	0.319E+05
Web	10.0	450.0	275.0	45.0	1237500.1	0.787E+04
BasePlate	360.0	70.0	535.0	252.0	13482000.0	0.205E+06
Totals	...	...	...	618.4	15483492.0	0.379E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 15483492/618$$

$$= 250.386 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 168/2$$

$$= 96.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(96) - 0.5 \cdot \sin(96)^2) / (\pi - 1.68 + \sin(96) \cos(96))$$

$$= 0.2943$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.29 \cdot 29179$$

$$= 8588.7031 \text{ Kgf}$$

Tension Stress, $St = (Fh/As) = 2.2691 \text{ N./mm}^2$
 Allowed Stress, $Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 850 - 280 \cdot \sin(168/2) / 1.47$$

$$= 660.060 \text{ mm}$$

Bending Moment, $M = Fh \cdot d = 5669.1748 \text{ Kg-m.}$

Bending Stress, $Sb = (M \cdot C1 / I) = 3.6758 \text{ N./mm}^2$
 Allowed Stress, $Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(29179 + 330)360 / (4 \cdot 780 \cdot 173))}$$

$$= 24.040 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 168/2) (280 + 36 + 14)$$



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= 656.384 mm

Distance between Ribs [e]:
 = Web Length / (Nribs - 1)
 = 656 / (3 - 1)
 = 328.192 mm

Baseplate Pressure Area [Ap]:
 = e • Bpwid / 2
 = 328 • 360 / 2
 = 590.746 cm²

Bearing Pressure [Bp]:
 = (Q + SaddleWt) / (BasePlateLength • BasePlateWidth)
 = 29509 / (780 • 360)
 = 10.509 Kg/cm²

Axial Load [P]:
 = Ap • Bp
 = 591 • 10.5
 = 6208.086 Kg

Area of the Rib and Web [Ar]:
 = Rib Area + Web Area
 = 32.5 + 16.4
 = 48.910 cm²

Compressive Stress [Sc]:
 = P / Ar
 = 6208 / 48.9
 = 12.448 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	335.0	...	33.5	...	0.313E+04

Rib dimension [D]:
 = Saddle Width - Web Thickness
 = 335 - 10
 = 325.000 mm

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 0 / 48.9
 = 0.000 mm

Distance to Centroid [C1]:
 = Saddle Width / 2
 = 335 / 2
 = 167.500 mm

Radius of Gyration [r]:
 = $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{3133 / 48.9}$
 = 80.035 mm

Length of Outer Rib [L]:
 = Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 850 - cos(168/2)(280 + 36 + 14) - 70
 = 745.506 mm



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Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008 / 260)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 746 / 80$$

$$= 9.315$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{\text{plen}}) \cdot e \cdot L / 2$$

$$= 2939 / (2 \cdot 780) \cdot 328 \cdot 746 / 2$$

$$= 230.501 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($9.31 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (9.31)^2 / (2 \cdot 123^2)) 260 / (5/3 + 3 \cdot (9.31) / (8 \cdot 123) - (9.31)^3 / (8 \cdot 123^3))$$

$$= 153 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 12.4 / 153 + (231 \cdot 168 / 31329476) / 173$$

$$= 0.151$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	325.0	0.0	32.5	0.0	0.313E+04
Web	328.2	10.0	0.0	32.8	0.0	2.73
Totals	...	...	...	65.3	...	0.314E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 65.3$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 335 / 2$$

$$= 167.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{\text{pthk}}$$

$$= 850 - \cos(330/2) (70 + 0 + 0) - 0$$

$$= 450.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{3136 / 65.3}$$

$$= 69.285 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 450 / 69.3$$

$$= 6.495$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 2939 / (2 \cdot 780)$$

$$= 1.884 \text{ Kg/mm}$$



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Moment at base of inner Rib [Mbase,c]:

$$\begin{aligned}
 &= \text{Unit Force} \cdot e \cdot L \\
 &= 1.88 \cdot 328 \cdot 450 \\
 &= 278.269 \text{ Kg-m.}
 \end{aligned}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$\begin{aligned}
 &= \text{Bending Moment} / \text{Section Modulus} \\
 &= 278/187200 \\
 &= 14.577 \text{ N./mm}^2
 \end{aligned}$$

Compressive Allowable, $KL/r < C_c$ (6.49 < 123) per AISC E2-1 [Sca]:

$$\begin{aligned}
 &= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3)) \\
 &= (1 - (6.49)^2 / (2 \cdot 123^2)) 260 / \\
 &\quad (5/3 + 3 \cdot (6.49) / (8 \cdot 123) - (6.49^3) / (8 \cdot 123^3)) \\
 &= 154 \text{ N./mm}^2
 \end{aligned}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$\begin{aligned}
 &= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba} \\
 &= 18.7/154 + (278 \cdot 168/3136) / 173 \\
 &= 0.205
 \end{aligned}$$

Review notes about nozzle loadings and supports in the Warnings report.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Wear Pad Bearing angle is less than saddle angle + saddle/12 degrees. The wear plate will be ignored.

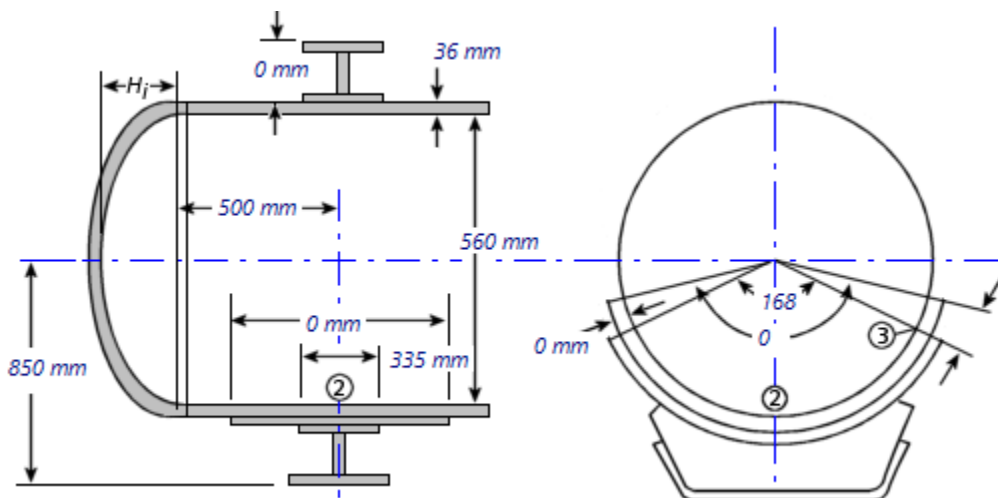
Minimum Wear Plate Angle [θ1]:
 = 180.0 degrees

Note:
 Wear Pad Width (.00) is less than $1.56 \cdot \sqrt{R_m \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 $= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$
 $= \min(335 + 1.56 \cdot \sqrt{298 \cdot 36}, 2 \cdot 500)$
 $= 496.5786 \text{ mm}$

Input and Calculated Values:

Vessel Mean Radius	Rm	298.00	mm
Shell Thickness used in this Case	t	36.000	mm
Stiffened Vessel Length per 4.15.6	L	9220.00	mm
Distance from Saddle to Vessel tangent	a1 or a	500.00	mm
Saddle Width	b1 or b	335.00	mm
Saddle Bearing Angle	delta or theta	168.00	degrees
Shell Allowable Stress used in Calculation		85.90	N./mm ²
Head Allowable Stress used in Calculation		85.90	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	850.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		30561.15	Kgf
Pressure used in Analysis	P	6102.676	KPa.

Horizontal Vessel Analysis Results:	Actual	Allowable
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	N. /mm ²	N. /mm ²
Long. Stress at Top of Midspan	-28.75	-116.10
Long. Stress at Top of Midspan	28.75	85.90
Long. Stress at Bottom of Midspan	79.27	85.90
Long. Stress at Top of Saddles	28.59	85.90
Long. Stress at Bottom of Saddles	23.29	85.90
Tangential Shear in Shell	16.29	68.72
Circ. Stress at Horn of Saddle	11.96	107.38
Circ. Compressive Stress in Shell	1.07	85.90

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr}(F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (1540/2 + 1742) \cdot 850/656$
 $= 9760.8 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(67.3, 2080) \cdot 850/5220$
 $= 231.6 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l,eq}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(1750, 2080) \cdot 850/5220$
 $= 338.7 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= F_{tr}(F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (1750/2 + 1742) \cdot 850/656$
 $= 10168.7 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 20392 + \max(232, 9761, 339, 10169)$
 $= 30561.1 \text{ Kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 766 (0 + \pi/4(0.83 \cdot 1.2)^2)$
 $= 659.734 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	30891.16	Kgf
Transverse Shear Load	2617.47	Kgf
Longitudinal Shear Load	2080.00	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$
 $= \min(2(560/2 + 36 + 14) \sin(168/2), 2 \cdot 298)$
 $= 656.384 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1997	K2 = 0.6539	K3 = 0.3357	K4 = 0.2452
K5 = 0.6404	K6 = 0.0224	K7 = 0.0224	K8 = 0.2720
K9 = 0.1759	K10 = 0.0221	K1* = 0.3382	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:



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$$\begin{aligned}
 &= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)] \\
 &= -30561 \cdot 500 [1 - (1 - 500/9220 + (298^2 - 0^2)/(2 \cdot 500 \cdot 9220))/(1 + (4 \cdot 0)/(3 \cdot 9220))] \\
 &= -681.5 \text{ Kg-m.}
 \end{aligned}$$

Moment per Equation 4.15.2 [M2]:

$$\begin{aligned}
 &= Q \cdot L/4 ((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L) \\
 &= 30561 \cdot 9220/4 ((1 + 2(298^2 - 0^2)/(9220^2))/(1 + (4 \cdot 0)/(3 \cdot 9220)) - 4 \cdot 500/9220) \\
 &= 55311.1 \text{ Kg-m.}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$\begin{aligned}
 &= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t) \\
 &= 6103 \cdot 298/(2 \cdot 36) - 55311/(\pi \cdot 298^2 \cdot 36) \\
 &= -28.75 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$\begin{aligned}
 &= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t) \\
 &= 6103 \cdot 298/(2 \cdot 36) + 55311/(\pi \cdot 298^2 \cdot 36) \\
 &= 79.27 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$\begin{aligned}
 &= P \cdot Rm/(2t) - M1/(K1 \cdot \pi \cdot Rm^2 t) \\
 &= 6103 \cdot 298/(2 \cdot 36) - 681/(0.2 \cdot \pi \cdot 298^2 \cdot 36) \\
 &= 28.59 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$\begin{aligned}
 &= P \cdot Rm/(2t) + M1/(K1 \cdot \pi \cdot Rm^2 \cdot t) \\
 &= 6103 \cdot 298/(2 \cdot 36) + 681/(0.34 \cdot \pi \cdot 298^2 \cdot 36) \\
 &= 23.29 \text{ N./mm}^2
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned}
 &= Q(L - 2a)/(L + (4 \cdot hm/3)) \\
 &= 30561 (9220 - 2 \cdot 500)/(9220 + (4 \cdot 0/3)) \\
 &= 27246.5 \text{ Kgf}
 \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$\begin{aligned}
 &= K2 \cdot T / (Rm \cdot t) \\
 &= 0.65 \cdot 27246/(298 \cdot 36) \\
 &= 16.29 \text{ N./mm}^2
 \end{aligned}$$

Decay Length (4.15.20) [x_1, x_2]:

$$\begin{aligned}
 &= 0.78 \cdot \sqrt{(Rm \cdot t)} \\
 &= 0.78 \cdot \sqrt{(298 \cdot 36)} \\
 &= 80.789 \text{ mm}
 \end{aligned}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$\begin{aligned}
 &= -K5 \cdot Q \cdot k / (t(b + X1 + X2)) \\
 &= -0.64 \cdot 30561 \cdot 0.1/(36(335 + 80.8 + 80.8)) \\
 &= -1.07 \text{ N./mm}^2
 \end{aligned}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.22) [σ_7]:

$$\begin{aligned}
 &= -Q/((4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q/(2 \cdot t^2)) \\
 &= -30561/((4 \cdot 36(335 + 80.8 + 80.8)) - 3 \cdot 0.022 \cdot 30561/(2 \cdot 36^2)) \\
 &= -11.96 \text{ N./mm}^2
 \end{aligned}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$\begin{aligned}
 &= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a) \\
 &= \min(335 + 1.56 \cdot \sqrt{(298 \cdot 36)}, 2 \cdot 500) \\
 &= 496.58 \text{ mm}
 \end{aligned}$$

Distance between Saddle Supports [Ls]:

$$= 5220.0 \text{ mm}$$



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Free Un-Restrained Thermal Expansion between the Saddles [Exp]:
 $= \alpha \cdot Ls(\text{Design Temperature} - \text{Ambient Temperature})$
 $= 0.17718\text{E-}04 \cdot 5220 (300 - 21.1)$
 $= 25.793 \text{ mm}$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	780.0000	mm
Baseplate Thickness	Bpthk	70.0000	mm
Baseplate Width	Bpwid	360.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	204.0	N./mm ²
Height of Web at Center	Hw,c	575.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth}) t^3) / 12$
 $= ((1.56 \cdot \sqrt{(280 \cdot 36)} + 530) 36^3) / 12$
 $= 266.959 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$
 $= 267 + 247 (250 - 18)^2$
 $= 133754.688 \text{ cm}^4$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	686.6	36.0	18.0	247.2	444931.6	0.134E+06
Wearplate	530.0	14.0	43.0	74.2	319060.0	0.319E+05
Web	10.0	450.0	275.0	45.0	1237500.1	0.787E+04
BasePlate	360.0	70.0	535.0	252.0	13482000.0	0.205E+06
Totals	...	...	...	618.4	15483492.0	0.379E+06

Distance to Centroid [C1]:
 $= AY / A$
 $= 15483492/618$
 $= 250.386 \text{ mm}$

Angle [β]:
 $= 180 - \text{Saddle Angle}/2$
 $= 180 - 168/2$
 $= 96.0$

Saddle Splitting Coefficient [K1]:
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$
 $= (1 + \cos(96) - 0.5 \cdot \sin(96)^2) / (\pi - 1.68 + \sin(96) \cos(96))$
 $= 0.2943$

Saddle Splitting Force [Fh]:
 $= K1 \cdot Q$
 $= 0.29 \cdot 30561$
 $= 8995.5469 \text{ Kg}$

Tension Stress, St = (Fh/As) = 2.3766 N./mm²
 Allowed Stress, Sa = 0.6 * Yield Str = 122.4015 N./mm²

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$
 $= 850 - 280 \cdot \sin(168/2) / 1.47$



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= 660.060 mm

Bending Moment, $M = F_h \cdot d = 5937.7212 \text{ Kg-m.}$

Bending Stress, $S_b = (M \cdot C_1 / I) = 3.8499 \text{ N./mm}^2$
 Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 136.0016 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{\frac{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}}{4 \cdot \text{BasePlateLength} \cdot \text{AllStress}}}$
 $= \sqrt{\frac{3(30561 + 330) \cdot 360}{4 \cdot 780 \cdot 136}}$
 = 27.768 mm

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 168/2) (280 + 36 + 14)$
 = 656.384 mm

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 656 / (3 - 1)$
 = 328.192 mm

Baseplate Pressure Area [Ap]:
 $= e \cdot \text{Bpwid} / 2$
 $= 328 \cdot 360 / 2$
 = 590.746 cm²

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$
 $= 30891 / (780 \cdot 360)$
 = 11.001 Kg/cm²

Axial Load [P]:
 $= A_p \cdot B_p$
 $= 591 \cdot 11$
 = 6498.872 Kg

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 32.5 + 16.4$
 = 48.910 cm²

Compressive Stress [Sc]:
 $= P / A_r$
 $= 6499 / 48.9$
 = 13.031 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	335.0	...	33.5	...	0.313E+04

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 335 - 10$
 = 325.000 mm

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 = 0/48.9



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= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
 = 335/2
 = 167.500 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{(3133/48.9)}$
 = 80.035 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 850 - cos(168/2)(280 + 36 + 14) - 70
 = 745.506 mm

Intermediate Term [Cc]:

= $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$
 = $\sqrt{(2 \cdot \pi^2 \cdot 199943008/204)}$
 = 139.095

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 746/80
 = 9.315

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2
 = 2080 / (2 • 780) • 328 • 746/2
 = 163.116 Kg-m.

Compressive Allowable, KL/r < Cc (9.31 < 139) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 = $(1 - (9.31)^2 / (2 \cdot 139^2)) 204 /$
 $(5/3 + 3 \cdot (9.31) / (8 \cdot 139) - (9.31^3) / (8 \cdot 139^3))$
 = 120.3 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I)/Sba
 = 13/120 + (163 • 168/31329476)/136
 = 0.171

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	325.0	0.0	32.5	0.0	0.313E+04
Web	328.2	10.0	0.0	32.8	0.0	2.73
Totals	...	...	...	65.3	...	0.314E+04

Distance to Centroid from Datum [ytot]:

= AY / A
 = 0/65.3
 = 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
 = 335/2
 = 167.500 mm

Length of Inner Rib [L]:

= Saddle Height - Outside Radius - Bpthk
 = 850 - cos(330/2)(70 + 0 + 0) - 0



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= 450.000 mm

Radius of Gyration [r]:
 $= \sqrt{\text{Total Inertia} / \text{Total Area}}$
 $= \sqrt{3136/65.3}$
 = 69.285 mm

Slenderness ratio [KL/r]:
 = KL/r
 = 1 • 450/69.3
 = 6.495

Unit Force [Force,u]:
 $= F_l / (2 \cdot \text{Baseplate Length})$
 $= 2080 / (2 \cdot 780)$
 = 1.333 Kg/mm

Moment at base of inner Rib [Mbase,c]:
 = Unit Force • e • L
 = 1.33 • 328 • 450
 = 196.919 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:
 = Bending Moment / Section Modulus
 = 197/187200
 = 10.316 N./mm²

Compressive Allowable, KL/r < Cc (6.49 < 139) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 $= (1 - (6.49)^2 / (2 \cdot 139^2)) 204 /$
 $(5/3 + 3 \cdot (6.49) / (8 \cdot 139) - (6.49^3) / (8 \cdot 139^3))$
 = 121 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)
 $= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$
 $= 19.5/121 + (197 \cdot 168/3136) / 136$
 = 0.237

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	6.1100 cm ²
BasePlate Allowable Stress	S	136.00 N./mm ²
Area Available in a Single Bolt	BltArea	4.8674 cm ²
Saddle Load Q0 (Weight)	Q0	20722.5 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	338.7 Kg
Maximum Transverse Force	Ft	2617.5 Kg
Maximum Longitudinal Force	F1	2080.0 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:
 $= F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 2080 / (4.87 \cdot 4)$
 = 10.5 N./mm². Must be less than 82.8 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:
 $= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 2617 / (4.87 \cdot 4)$
 = 13.2 N./mm². Must be less than 82.8 N./mm².



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Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = FI / (BoltShearAllowable • Nbolts)
 = 2080 / (82.8 • 4)
 = 0.6159 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = B • Ft + Sum of X Moments
 = 850 • 2617 + 0
 = 2224.89 Kg-m.

Eccentricity (e):
 = Rmom / QO
 = 2225 / 20723
 = 107.36 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = max[Bltarearl, Bltarears, Bltareart]
 = max[0, 0.62, 0]
 = 0.6159 cm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Wear Pad Bearing angle is less than saddle angle + saddle/12 degrees. The wear plate will be ignored.

Minimum Wear Plate Angle [Ø1]:
 = 180.0 degrees

Note:
 Wear Pad Width (.00) is less than 1.56*sqrt(rm*t)
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 = min(b + 1.56*sqrt(Rm • t), 2a)
 = min(335 + 1.56*sqrt(298 • 36), 2 • 500)
 = 496.5786 mm

Input and Calculated Values:

Vessel Mean Radius	Rm	298.00	mm
Shell Thickness used in this Case	t	36.000	mm
Stiffened Vessel Length per 4.15.6	L	9220.00	mm
Distance from Saddle to Vessel tangent	a1 or a	500.00	mm
Saddle Width	b1 or b	335.00	mm
Saddle Bearing Angle	delta or theta	168.00	degrees
Shell Allowable Stress used in Calculation		85.90	N./mm ²
Head Allowable Stress used in Calculation		85.90	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	850.00	mm



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Coefficient of Friction mu 0.00

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case 27415.36 KgF
 Pressure used in Analysis P 6102.676 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-23.19	-116.10
Long. Stress at Top of Midspan	23.19	85.90
Long. Stress at Bottom of Midspan	73.71	85.90
Long. Stress at Top of Saddles	28.25	85.90
Long. Stress at Bottom of Saddles	23.49	85.90
Tangential Shear in Shell	14.61	68.72
Circ. Stress at Horn of Saddle	10.73	107.38
Circ. Compressive Stress in Shell	0.96	85.90

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3 (1540/2 + 1760) \cdot 850/656$
 $= 9828.6\ KgF$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl,wind; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(67.3, 2080) \cdot 850/5220$
 $= 231.6\ KgF$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(Fl,eq; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(1750, 2080) \cdot 850/5220$
 $= 338.7\ KgF$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3 (1750/2 + 1760) \cdot 850/656$
 $= 10236.4\ KgF$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= Saddle\ Load + \max(Fwl, Fwt, Fsl, Fst)$
 $= 17179 + \max(232, 9829, 339, 10236)$
 $= 27415.4\ KgF$

Longitudinal Wind Force [Fl]:
 $= WindScalar \cdot WindPress(Platform\ Area + (\pi/4(OD \cdot WindDiaMult)^2))$
 $= 1.1 \cdot 766 (0 + \pi/4(0.83 \cdot 1.2)^2)$
 $= 659.734\ N$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 27745.37 KgF
 Transverse Shear Load 2634.91 KgF
 Longitudinal Shear Load 2080.00 KgF

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 $= \min(2(ShellID/2 + t + WearPadThickness) \sin(\theta/2) , 2 \cdot Rm)$
 $= \min(2(560/2 + 36 + 14) \sin(168/2) , 2 \cdot 298)$
 $= 656.384\ mm$



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The Computed K values from Table 4.15.1:

K1 = 0.1997 K2 = 0.6539 K3 = 0.3357 K4 = 0.2452
 K5 = 0.6404 K6 = 0.0224 K7 = 0.0224 K8 = 0.2720
 K9 = 0.1759 K10 = 0.0221 K1* = 0.3382

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a \left[1 - \left(1 - \frac{a}{L} + \frac{(Rm^2 - hm^2)}{(2a \cdot L)} \right) / \left(1 + \frac{(4hm)}{3L} \right) \right]$$

$$= -27415 \cdot 500 \left[1 - \left(1 - \frac{500}{9220} + \frac{(298^2 - 0^2)}{(2 \cdot 500 \cdot 9220)} \right) / \left(1 + \frac{(4 \cdot 0)}{(3 \cdot 9220)} \right) \right]$$

$$= -611.4 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \left(\frac{(1 + 2(Rm^2 - hm^2)/(L^2))}{(1 + (4hm^2)/(3L))} - \frac{4a}{L} \right)$$

$$= 27415 \cdot 9220 / 4 \left(\frac{(1 + 2(298^2 - 0^2)/(9220^2))}{(1 + (4 \cdot 0)/(3 \cdot 9220))} - \frac{4 \cdot 500}{9220} \right)$$

$$= 49617.7 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t)$$

$$= 6103 \cdot 298 / (2 \cdot 36) - 49618 / (\pi \cdot 298^2 \cdot 36)$$

$$= -23.19 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 6103 \cdot 298 / (2 \cdot 36) + 49618 / (\pi \cdot 298^2 \cdot 36)$$

$$= 73.71 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 t)$$

$$= 6103 \cdot 298 / (2 \cdot 36) - 611 / (0.2 \cdot \pi \cdot 298^2 \cdot 36)$$

$$= 28.25 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot Rm / (2t) + M1 / (K1^* \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 6103 \cdot 298 / (2 \cdot 36) + 611 / (0.34 \cdot \pi \cdot 298^2 \cdot 36)$$

$$= 23.49 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3))$$

$$= 27415 (9220 - 2 \cdot 500) / (9220 + (4 \cdot 0 / 3))$$

$$= 24441.9 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 0.65 \cdot 24442 / (298 \cdot 36)$$

$$= 14.61 \text{ N./mm}^2$$

Decay Length (4.15.20) [x1, x2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(298 \cdot 36)}$$

$$= 80.789 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.64 \cdot 27415 \cdot 0.1 / (36(335 + 80.8 + 80.8))$$

$$= -0.96 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -27415 / (4 \cdot 36(335 + 80.8 + 80.8)) - 3 \cdot 0.022 \cdot 27415 / (2 \cdot 36^2)$$

$$= -10.73 \text{ N./mm}^2$$



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Effective reinforcing plate width (4.15.24) [B1]:
 $= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$
 $= \min(335 + 1.56 \cdot \sqrt{(298 \cdot 36)}, 2 \cdot 500)$
 $= 496.58 \text{ mm}$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	780.0000	mm
Baseplate Thickness	Bpthk	70.0000	mm
Baseplate Width	Bpwid	360.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	204.0	N./mm ²
Height of Web at Center	Hw,c	575.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$
 $= ((1.56 \cdot \sqrt{(280 \cdot 36)} + 530)36^3)/12$
 $= 266.959 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A\text{Shell}(C1 - Y)^2$
 $= 267 + 247 (250 - 18)^2$
 $= 133754.688 \text{ cm}^4$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	686.6	36.0	18.0	247.2	444931.6	0.134E+06
Wearplate	530.0	14.0	43.0	74.2	319060.0	0.319E+05
Web	10.0	450.0	275.0	45.0	1237500.1	0.787E+04
BasePlate	360.0	70.0	535.0	252.0	13482000.0	0.205E+06
Totals	...	...	...	618.4	15483492.0	0.379E+06

Distance to Centroid [C1]:
 $= AY / A$
 $= 15483492/618$
 $= 250.386 \text{ mm}$

Angle [β]:
 $= 180 - \text{Saddle Angle}/2$
 $= 180 - 168/2$
 $= 96.0$

Saddle Splitting Coefficient [K1]:
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$
 $= (1 + \cos(96) - 0.5 \cdot \sin(96)^2) / (\pi - 1.68 + \sin(96) \cos(96))$
 $= 0.2943$

Saddle Splitting Force [Fh]:
 $= K1 \cdot Q$
 $= 0.29 \cdot 27415$
 $= 8069.5972 \text{ Kgf}$

Tension Stress, St = (Fh/As) = 2.1319 N./mm²
 Allowed Stress, Sa = 0.6 * Yield Str = 122.4015 N./mm²

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$



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$$= 850 - 280 \cdot \sin(168/2)/1.47$$

$$= 660.060 \text{ mm}$$

$$\text{Bending Moment, } M = F_h \cdot d = 5326.5269 \text{ Kg-m.}$$

$$\text{Bending Stress, } S_b = (M \cdot C_1 / I) = 3.4536 \text{ N./mm}^2$$

$$\text{Allowed Stress, } S_a = 2/3 \cdot \text{Yield Str} = 136.0016 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(27415 + 330) \cdot 360) / (4 \cdot 780 \cdot 136)}$$

$$= 26.316 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

$$\text{Web Length Dimension [Web Length]:}$$

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 168/2) (280 + 36 + 14)$$

$$= 656.384 \text{ mm}$$

$$\text{Distance between Ribs [e]:}$$

$$= \text{Web Length} / (\text{Nribs} - 1)$$

$$= 656 / (3 - 1)$$

$$= 328.192 \text{ mm}$$

$$\text{Baseplate Pressure Area [Ap]:}$$

$$= e \cdot \text{Bpwid} / 2$$

$$= 328 \cdot 360 / 2$$

$$= 590.746 \text{ cm}^2$$

$$\text{Bearing Pressure [Bp]:}$$

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 27745 / (780 \cdot 360)$$

$$= 9.881 \text{ Kg/cm}^2$$

$$\text{Axial Load [P]:}$$

$$= A_p \cdot B_p$$

$$= 591 \cdot 9.88$$

$$= 5837.062 \text{ Kg}$$

$$\text{Area of the Rib and Web [Ar]:}$$

$$= \text{Rib Area} + \text{Web Area}$$

$$= 32.5 + 16.4$$

$$= 48.910 \text{ cm}^2$$

$$\text{Compressive Stress [Sc]:}$$

$$= P / A_r$$

$$= 5837 / 48.9$$

$$= 11.704 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	335.0	...	33.5	...	0.313E+04

$$\text{Rib dimension [D]:}$$

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 335 - 10$$

$$= 325.000 \text{ mm}$$

$$\text{Distance to Centroid from Datum [ytot]:}$$

$$= AY / A$$



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= 0/48.9
 = 0.000 mm

Distance to Centroid [C1]:
 = Saddle Width / 2
 = 335/2
 = 167.500 mm

Radius of Gyration [r]:
 $= \sqrt{\text{Total Inertia} / \text{Total Area}}$
 $= \sqrt{3133/48.9}$
 = 80.035 mm

Length of Outer Rib [L]:
 = Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 850 - cos(168/2)(280 + 36 + 14) - 70
 = 745.506 mm

Intermediate Term [Cc]:
 $= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 $= \sqrt{2 \cdot \pi^2 \cdot 199943008/204}$
 = 139.095

Slenderness ratio [KL/r]:
 = KL/r
 = 1 • 746/80
 = 9.315

Bending Moment [Rm]:
 $= FI / (2 \cdot B \cdot \text{plen}) \cdot e \cdot L / 2$
 $= 2080 / (2 \cdot 780) \cdot 328 \cdot 746/2$
 = 163.116 Kg-m.

Compressive Allowable, $KL/r < Cc$ (9.31 < 139) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 $= (1 - (9.31)^2 / (2 \cdot 139^2)) 204 / (5/3 + 3 \cdot (9.31) / (8 \cdot 139) - (9.31^3) / (8 \cdot 139^3))$
 = 120.3 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)
 $= Sc/Sca + (Rm \cdot C1 / I) / Sba$
 $= 11.7/120 + (163 \cdot 168/31329476) / 136$
 = 0.160

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	325.0	0.0	32.5	0.0	0.313E+04
Web	328.2	10.0	0.0	32.8	0.0	2.73
Totals	...	...	...	65.3	...	0.314E+04

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 0/65.3
 = 0.000 mm

Distance to Centroid [C1]:
 = Saddle Width / 2
 = 335/2
 = 167.500 mm

Length of Inner Rib [L]:
 = Saddle Height - Outside Radius - Bpthk



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$$= 850 - \cos(330/2)(70 + 0 + 0) - 0$$

$$= 450.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{3136/65.3}$$

$$= 69.285 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 450/69.3$$

$$= 6.495$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 2080 / (2 \cdot 780)$$

$$= 1.333 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.33 \cdot 328 \cdot 450$$

$$= 196.919 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 197/187200$$

$$= 10.316 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($6.49 < 139$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (6.49)^2 / (2 \cdot 139^2)) 204 / (5/3 + 3 \cdot (6.49) / (8 \cdot 139) - (6.49)^3 / (8 \cdot 139^3))$$

$$= 121 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 17.5/121 + (197 \cdot 168/3136)/136$$

$$= 0.221$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	6.1100 cm ²
BasePlate Allowable Stress	S	136.00 N./mm ²
Area Available in a Single Bolt	BltArea	4.8674 cm ²
Saddle Load Q0 (Weight)	Q0	17509.0 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	338.7 Kg
Maximum Transverse Force	Ft	2634.9 Kg
Maximum Longitudinal Force	F1	2080.0 Kg
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:

$$= 6216.8 \text{ Kg}$$

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= \max(F_f, F_1) / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 6217 / (4.87 \cdot 4)$$

$$= 31.3 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$



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Shear Stress in a Single Bolt, Transverse Direction [tb,t]:
= $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $2635 / (4.87 \cdot 4)$
= 13.3 N./mm². Must be less than 82.8 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= $F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$
= $2080 / (82.8 \cdot 4)$
= 0.6159 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot F_t + \text{Sum of X Moments}$
= $850 \cdot 2635 + 0$
= 2239.72 Kg-m.

Eccentricity (e):
= R_{mom} / QO
= $2240 / 17509$
= 127.92 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= $\max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$
= $\max[0, 0.62, 0]$
= 0.6159 cm²

Review notes about nozzle loadings and supports in the Warnings report.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Wear Pad Bearing angle is less than saddle angle + saddle/12 degrees. The wear plate will be ignored.

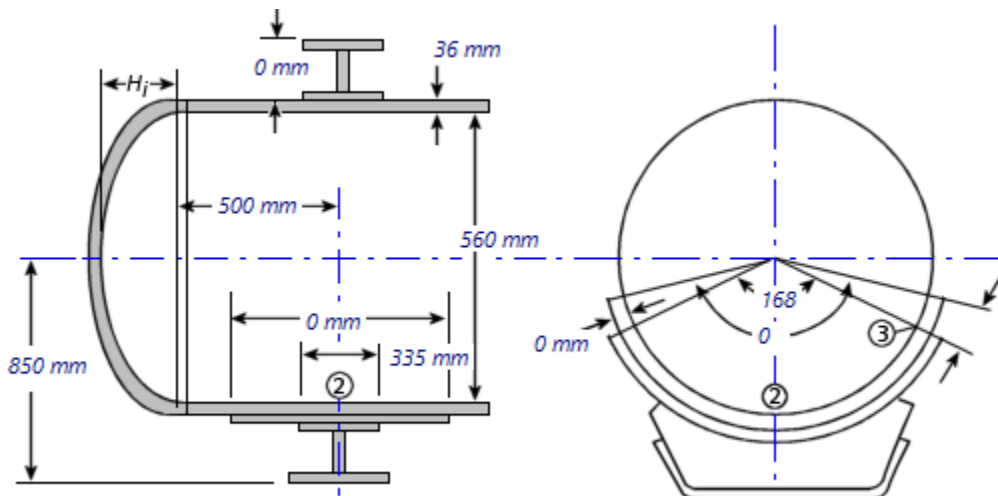
Minimum Wear Plate Angle [θ1]:
 = 180.0 degrees

Note:
 Wear Pad Width (.00) is less than $1.56 \cdot \sqrt{R_m \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 $= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$
 $= \min(335 + 1.56 \cdot \sqrt{298 \cdot 36}, 2 \cdot 500)$
 $= 496.5786 \text{ mm}$

Input and Calculated Values:

Vessel Mean Radius	Rm	298.00	mm
Shell Thickness used in this Case	t	36.000	mm
Stiffened Vessel Length per 4.15.6	L	9220.00	mm
Distance from Saddle to Vessel tangent	a1 or a	500.00	mm
Saddle Width	b1 or b	335.00	mm
Saddle Bearing Angle	delta or theta	168.00	degrees
Shell Allowable Stress used in Calculation		184.50	N./mm ²
Head Allowable Stress used in Calculation		184.50	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	850.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces	20937.08	Kgf
Pressure used in Analysis	P	8049.478 KPa.

Horizontal Vessel Analysis Results:	Actual	Allowable
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	N. /mm ²	N. /mm ²
Long. Stress at Top of Midspan	-3.68	-184.50
Long. Stress at Top of Midspan	3.68	184.50
Long. Stress at Bottom of Midspan	70.32	184.50
Long. Stress at Top of Saddles	35.60	184.50
Long. Stress at Bottom of Saddles	31.97	184.50
Tangential Shear in Shell	11.16	109.26
Circ. Stress at Horn of Saddle	8.20	276.74
Circ. Compressive Stress in Shell	0.74	184.50

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (770/2 + 0) \cdot 850/656$
 $= 1496.1 \text{ Kg}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(33.6, 0) \cdot 850/5220$
 $= 5.5 \text{ Kg}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 19441 + \max(5.48, 1496, 0, 0)$
 $= 20937.1 \text{ Kg}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 766 (0 + \pi/4 (0.83 \cdot 1.2)^2)$
 $= 659.734 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	21267.10	Kgf
Transverse Shear Load	385.10	Kgf
Longitudinal Shear Load	33.64	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$
 $= \min(2(560/2 + 36 + 14) \sin(168/2), 2 \cdot 298)$
 $= 656.384 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1997	K2 = 0.6539	K3 = 0.3357	K4 = 0.2452
K5 = 0.6404	K6 = 0.0224	K7 = 0.0224	K8 = 0.2720
K9 = 0.1759	K10 = 0.0221	K1* = 0.3382	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$
 $= -20937 \cdot 500 [1 - (1 - 500/9220 + (298^2 - 0^2)/(2 \cdot 500 \cdot 9220)) / (1 + (4 \cdot 0)/(3 \cdot 9220))]$
 $= -466.9 \text{ Kg-m}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m)/(3L)) - 4a/L)$
 $= 20937 \cdot 9220/4 ((1 + 2(298^2 - 0^2)/(9220^2)) / (1 + (4 \cdot 0)/(3 \cdot 9220)) - 4 \cdot 500/9220)$
 $= 37893.0 \text{ Kg-m}$



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Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t)$$

$$= 8049 \cdot 298 / (2 \cdot 36) - 37893 / (\pi \cdot 298^2 \cdot 36)$$

$$= -3.68 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 8049 \cdot 298 / (2 \cdot 36) + 37893 / (\pi \cdot 298^2 \cdot 36)$$

$$= 70.32 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t)$$

$$= 8049 \cdot 298 / (2 \cdot 36) - 467 / (0.2 \cdot \pi \cdot 298^2 \cdot 36)$$

$$= 35.60 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 8049 \cdot 298 / (2 \cdot 36) + 467 / (0.34 \cdot \pi \cdot 298^2 \cdot 36)$$

$$= 31.97 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 20937 (9220 - 2 \cdot 500) / (9220 + (4 \cdot 0 / 3))$$

$$= 18666.2 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 0.65 \cdot 18666 / (298 \cdot 36)$$

$$= 11.16 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(298 \cdot 36)}$$

$$= 80.789 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.64 \cdot 20937 \cdot 0.1 / (36(335 + 80.8 + 80.8))$$

$$= -0.74 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$$

$$= -20937 / (4 \cdot 36(335 + 80.8 + 80.8)) - 3 \cdot 0.022 \cdot 20937 / (2 \cdot 36^2)$$

$$= -8.20 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(335 + 1.56 \cdot \sqrt{(298 \cdot 36)}, 2 \cdot 500)$$

$$= 496.58 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	780.0000	mm
Baseplate Thickness	Bpthk	70.0000	mm
Baseplate Width	Bpwid	360.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	575.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)



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Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$
 $= ((1.56 \cdot \sqrt{(280 \cdot 36)} + 530)^3) / 12$
 $= 266.959 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$
 $= 267 + 247 (250 - 18)^2$
 $= 133754.688 \text{ cm}^4$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	686.6	36.0	18.0	247.2	444931.6	0.134E+06
Wearplate	530.0	14.0	43.0	74.2	319060.0	0.319E+05
Web	10.0	450.0	275.0	45.0	1237500.1	0.787E+04
BasePlate	360.0	70.0	535.0	252.0	13482000.0	0.205E+06
Totals	...	...	...	618.4	15483492.0	0.379E+06

Distance to Centroid [C1]:
 $= AY / A$
 $= 15483492 / 618$
 $= 250.386 \text{ mm}$

Angle [β]:
 $= 180 - \text{Saddle Angle} / 2$
 $= 180 - 168 / 2$
 $= 96.0$

Saddle Splitting Coefficient [K1]:
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$
 $= (1 + \cos(96) - 0.5 \cdot \sin(96)^2) / (\pi - 1.68 + \sin(96) \cos(96))$
 $= 0.2943$

Saddle Splitting Force [Fh]:
 $= K1 \cdot Q$
 $= 0.29 \cdot 20937$
 $= 6162.7441 \text{ Kgf}$

Tension Stress, $St = (Fh / As) = 1.6282 \text{ N./mm}^2$
 Allowed Stress, $Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\text{theta} / 2) / (\text{theta} / 2 \text{ in radians})$
 $= 850 - 280 \cdot \sin(168 / 2) / 1.47$
 $= 660.060 \text{ mm}$

Bending Moment, $M = Fh \cdot d = 4067.8635 \text{ Kg-m.}$

Bending Stress, $Sb = (M \cdot C1 / I) = 2.6375 \text{ N./mm}^2$
 Allowed Stress, $Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:
 $= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$
 $= \sqrt{(3(20937 + 330) \cdot 360) / (4 \cdot 780 \cdot 173)}$
 $= 20.409 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 168 / 2) (280 + 36 + 14)$
 $= 656.384 \text{ mm}$



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Distance between Ribs [e]:
 = Web Length / (Nribs - 1)
 = 656/(3 - 1)
 = 328.192 mm

Baseplate Pressure Area [Ap]:
 = e • Bpwid / 2
 = 328 • 360/2
 = 590.746 cm²

Bearing Pressure [Bp]:
 = (Q + SaddleWt) / (BasePlateLength • BasePlateWidth)
 = 21267/(780 • 360)
 = 7.574 Kgf/cm²

Axial Load [P]:
 = Ap • Bp
 = 591 • 7.57
 = 4474.165 Kgf

Area of the Rib and Web [Ar]:
 = Rib Area + Web Area
 = 32.5 + 16.4
 = 48.910 cm²

Compressive Stress [Sc]:
 = P/Ar
 = 4474/48.9
 = 8.971 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	335.0	...	33.5	...	0.313E+04

Rib dimension [D]:
 = Saddle Width - Web Thickness
 = 335 - 10
 = 325.000 mm

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 0/48.9
 = 0.000 mm

Distance to Centroid [C1]:
 = Saddle Width / 2
 = 335/2
 = 167.500 mm

Radius of Gyration [r]:
 = $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{3133/48.9}$
 = 80.035 mm

Length of Outer Rib [L]:
 = Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 850 - cos(168/2)(280 + 36 + 14) - 70
 = 745.506 mm

Intermediate Term [Cc]:
 = $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$



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$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008 / 260)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 746 / 80$$

$$= 9.315$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{pln}) \cdot e \cdot L / 2$$

$$= 33.6 / (2 \cdot 780) \cdot 328 \cdot 746 / 2$$

$$= 2.638 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($9.31 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (9.31)^2 / (2 \cdot 123^2)) 260 / (5/3 + 3 \cdot (9.31) / (8 \cdot 123) - (9.31^3) / (8 \cdot 123^3))$$

$$= 153 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 8.97 / 153 + (2.64 \cdot 168 / 31329476) / 173$$

$$= 0.059$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	325.0	0.0	32.5	0.0	0.313E+04
Web	328.2	10.0	0.0	32.8	0.0	2.73
Totals	...	...	...	65.3	...	0.314E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 65.3$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 335 / 2$$

$$= 167.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 850 - \cos(330/2) (70 + 0 + 0) - 0$$

$$= 450.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(3136 / 65.3)}$$

$$= 69.285 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 450 / 69.3$$

$$= 6.495$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 33.6 / (2 \cdot 780)$$

$$= 0.022 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$



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$$= 0.022 \cdot 328 \cdot 450$$

$$= 3.185 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 3.18/187200$$

$$= 0.167 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($6.49 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (Kl/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (Kl/r) / (8 \cdot C_c) - (Kl/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (6.49)^2 / (2 \cdot 123^2)) 260 / (5/3 + 3 \cdot (6.49) / (8 \cdot 123) - (6.49^3) / (8 \cdot 123^3))$$

$$= 154 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 13.4/154 + (3.18 \cdot 168/3136) / 173$$

$$= 0.088$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	6.1100 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	4.8674 cm ²
Saddle Load Q0 (Weight)	Q0	19771.0 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	5.5 Kg
Maximum Transverse Force	Ft	385.1 Kg
Maximum Longitudinal Force	Fl	33.6 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 33.6 / (4.87 \cdot 4)$$

$$= 0.2 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 385 / (4.87 \cdot 4)$$

$$= 1.9 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} Q_0 > Q_L \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 33.6 / (103 \cdot 4)$$

$$= 0.0080 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot F_t + \text{Sum of X Moments}$$

$$= 850 \cdot 385 + 0$$

$$= 327.34 \text{ Kg-m.}$$

Eccentricity (e):

$$= R_{mom} / Q_0$$



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= 327/19771
 = 16.56 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = max[Bltarearl, Bltarears, Bltareart]
 = max[0, 0.008, 0]
 = 0.0080 cm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Wear Pad Bearing angle is less than saddle angle + saddle/12 degrees. The wear plate will be ignored.

Minimum Wear Plate Angle [Ø1]:
 = 180.0 degrees

Note:
 Wear Pad Width (.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 = min(b + $1.56 \cdot \sqrt{Rm \cdot t}$), 2a)
 = min(335 + $1.56 \cdot \sqrt{298 \cdot 36}$), 2 • 500)
 = 496.5786 mm

Input and Calculated Values:

Vessel Mean Radius	Rm	298.00	mm
Shell Thickness used in this Case	t	36.000	mm
Stiffened Vessel Length per 4.15.6	L	9220.00	mm
Distance from Saddle to Vessel tangent	a1 or a	500.00	mm
Saddle Width	b1 or b	335.00	mm
Saddle Bearing Angle	delta or theta	168.00	degrees
Shell Allowable Stress used in Calculation		184.50	N./mm²
Head Allowable Stress used in Calculation		184.50	N./mm²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	850.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		17757.67	Kgf
Pressure used in Analysis	P	8049.478	KPa.

Horizontal Vessel Analysis Results:		Actual N./mm²	Allowable N./mm²
Long. Stress at Top of Midspan		1.94	184.50
Long. Stress at Bottom of Midspan		64.70	184.50
Long. Stress at Top of Saddles		35.25	184.50
Long. Stress at Bottom of Saddles		32.17	184.50
Tangential Shear in Shell		9.46	109.26
Circ. Stress at Horn of Saddle		6.95	276.74
Circ. Compressive Stress in Shell		0.62	184.50



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Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (770/2 + 0) \cdot 850/656$
 $= 1496.1 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l, \text{wind}}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(33.6, 0) \cdot 850/5220$
 $= 5.5 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 16262 + \max(5.48, 1496, 0, 0)$
 $= 17757.7 \text{ Kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 766 (0 + \pi/4 (0.83 \cdot 1.2)^2)$
 $= 659.734 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	18087.69	Kgf
Transverse Shear Load	385.10	Kgf
Longitudinal Shear Load	33.64	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$
 $= \min(2(560/2 + 36 + 14) \sin(168/2), 2 \cdot 298)$
 $= 656.384 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1997	K2 = 0.6539	K3 = 0.3357	K4 = 0.2452
K5 = 0.6404	K6 = 0.0224	K7 = 0.0224	K8 = 0.2720
K9 = 0.1759	K10 = 0.0221	K1* = 0.3382	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$
 $= -17758 \cdot 500 [1 - (1 - 500/9220 + (298^2 - 0^2)/(2 \cdot 500 \cdot 9220)) / (1 + (4 \cdot 0)/(3 \cdot 9220))]$
 $= -396.0 \text{ Kg-m.}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$
 $= 17758 \cdot 9220/4 ((1 + 2(298^2 - 0^2)/(9220^2)) / (1 + (4 \cdot 0)/(3 \cdot 9220)) - 4 \cdot 500/9220)$
 $= 32138.8 \text{ Kg-m.}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 t)$
 $= 8049 \cdot 298/(2 \cdot 36) - 32139/(\pi \cdot 298^2 \cdot 36)$
 $= 1.94 \text{ N./mm}^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 $= P \cdot R_m/(2t) + M_2/(\pi \cdot R_m^2 \cdot t)$
 $= 8049 \cdot 298/(2 \cdot 36) + 32139/(\pi \cdot 298^2 \cdot 36)$
 $= 64.70 \text{ N./mm}^2$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:
 $= P \cdot R_m/(2t) - M_1/(K_1 \cdot \pi \cdot R_m^2 t)$



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$$= 8049 \cdot 298 / (2 \cdot 36) - 396 / (0.2 \cdot \pi \cdot 298^2 \cdot 36)$$

$$= 35.25 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 8049 \cdot 298 / (2 \cdot 36) + 396 / (0.34 \cdot \pi \cdot 298^2 \cdot 36)$$

$$= 32.17 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 17758 (9220 - 2 \cdot 500) / (9220 + (4 \cdot 0 / 3))$$

$$= 15831.7 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 0.65 \cdot 15832 / (298 \cdot 36)$$

$$= 9.46 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(298 \cdot 36)}$$

$$= 80.789 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.64 \cdot 17758 \cdot 0.1 / (36(335 + 80.8 + 80.8))$$

$$= -0.62 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -17758 / (4 \cdot 36(335 + 80.8 + 80.8)) - 3 \cdot 0.022 \cdot 17758 / (2 \cdot 36^2)$$

$$= -6.95 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(335 + 1.56 \cdot \sqrt{(298 \cdot 36)}, 2 \cdot 500)$$

$$= 496.58 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	780.0000	mm
Baseplate Thickness	Bpthk	70.0000	mm
Baseplate Width	Bpwid	360.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	575.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(280 \cdot 36)} + 530)36^3) / 12$$

$$= 266.959 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 267 + 247(250 - 18)^2$$

$$= 133754.688 \text{ cm}^4$$

B

D

Y

A

AY

I + AD⁽²⁾



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	mm	mm	mm	cm ²	mm ³	cm ⁴
Shell	686.6	36.0	18.0	247.2	444931.6	0.134E+06
Wearplate	530.0	14.0	43.0	74.2	319060.0	0.319E+05
Web	10.0	450.0	275.0	45.0	1237500.1	0.787E+04
BasePlate	360.0	70.0	535.0	252.0	13482000.0	0.205E+06
Totals	...	...	...	618.4	15483492.0	0.379E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 15483492/618$$

$$= 250.386 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 168/2$$

$$= 96.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(96) - 0.5 \cdot \sin(96)^2) / (\pi - 1.68 + \sin(96) \cos(96))$$

$$= 0.2943$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.29 \cdot 17758$$

$$= 5226.8979 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 1.3809 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 850 - 280 \cdot \sin(168/2) / 1.47$$

$$= 660.060 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 3450.1362 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 2.2370 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{ (3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})) }$$

$$= \sqrt{ (3(17758 + 330) 360 / (4 \cdot 780 \cdot 173)) }$$

$$= 18.821 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 168/2) (280 + 36 + 14)$$

$$= 656.384 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 656 / (3 - 1)$$

$$= 328.192 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 328 \cdot 360 / 2$$

$$= 590.746 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$



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$$= 18088 / (780 \cdot 360)$$

$$= 6.441 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 591 \cdot 6.44$$

$$= 3805.282 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 32.5 + 16.4$$

$$= 48.910 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 3805 / 48.9$$

$$= 7.630 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	335.0	...	33.5	...	0.313E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 335 - 10$$

$$= 325.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 48.9$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 335 / 2$$

$$= 167.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(3133 / 48.9)}$$

$$= 80.035 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 850 - \cos(168/2) (280 + 36 + 14) - 70$$

$$= 745.506 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008 / 260)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 746 / 80$$

$$= 9.315$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 33.6 / (2 \cdot 780) \cdot 328 \cdot 746 / 2$$

$$= 2.638 \text{ Kg-m.}$$



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Compressive Allowable, $KL/r < C_c$ ($9.31 < 123$) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$
 $= (1 - (9.31)^2 / (2 \cdot 123^2)) 260 /$
 $(5/3 + 3 \cdot (9.31) / (8 \cdot 123) - (9.31^3) / (8 \cdot 123^3))$
 $= 153 \text{ N./mm}^2$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$
 $= 7.63 / 153 + (2.64 \cdot 168 / 31329476) / 173$
 $= 0.051$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	325.0	0.0	32.5	0.0	0.313E+04
Web	328.2	10.0	0.0	32.8	0.0	2.73
Totals	...	...	...	65.3	...	0.314E+04

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0 / 65.3$
 $= 0.000 \text{ mm}$

Distance to Centroid [C1]:
 $= \text{Saddle Width} / 2$
 $= 335 / 2$
 $= 167.500 \text{ mm}$

Length of Inner Rib [L]:
 $= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$
 $= 850 - \cos(330/2) (70 + 0 + 0) - 0$
 $= 450.000 \text{ mm}$

Radius of Gyration [r]:
 $= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$
 $= \sqrt{ (3136 / 65.3) }$
 $= 69.285 \text{ mm}$

Slenderness ratio [KL/r]:
 $= KL/r$
 $= 1 \cdot 450 / 69.3$
 $= 6.495$

Unit Force [Force,u]:
 $= F_l / (2 \cdot \text{Baseplate Length})$
 $= 33.6 / (2 \cdot 780)$
 $= 0.022 \text{ Kg/mm}$

Moment at base of inner Rib [Mbase,c]:
 $= \text{Unit Force} \cdot e \cdot L$
 $= 0.022 \cdot 328 \cdot 450$
 $= 3.185 \text{ Kg-m.}$

Bending Stress due to Transverse Force and Weight Load [$\sigma_{B,base,c}$]:
 $= \text{Bending Moment} / \text{Section Modulus}$
 $= 3.18 / 187200$
 $= 0.167 \text{ N./mm}^2$

Compressive Allowable, $KL/r < C_c$ ($6.49 < 123$) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$
 $= (1 - (6.49)^2 / (2 \cdot 123^2)) 260 /$
 $(5/3 + 3 \cdot (6.49) / (8 \cdot 123) - (6.49^3) / (8 \cdot 123^3))$



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= 154 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I)/Sba
 = 11.4/154 + (3.18 • 168/3136)/173
 = 0.075

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	6.1100 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	4.8674 cm ²
Saddle Load Q0 (Weight)	Q0	16591.6 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	5.5 Kg
Maximum Transverse Force	Ft	385.1 Kg
Maximum Longitudinal Force	Fl	33.6 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_{b,l}]:

= Fl / (Bolt Area • Number of Bolts)
 = 33.6 / (4.87 • 4)
 = 0.2 N./mm². Must be less than 103.2 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [τ_{b,t}]:

= Ft / (Bolt Area • Number of Bolts)
 = 385 / (4.87 • 4)
 = 1.9 N./mm². Must be less than 103.2 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (Q0 > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:

= Fl / (BoltShearAllowable • Nbolts)
 = 33.6 / (103 • 4)
 = 0.0080 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= B • Ft + Sum of X Moments
 = 850 • 385 + 0
 = 327.34 Kg-m.

Eccentricity (e):

= Rmom / Q0
 = 327/16592
 = 19.73 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= max[Bltarearl, Bltarears, Bltareart]
 = max[0, 0.008, 0]
 = 0.0080 cm²



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Review notes about nozzle loadings and supports in the Warnings report.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : External Pressure

Wear Pad Bearing angle is less than saddle angle + saddle/12 degrees. The wear plate will be ignored.

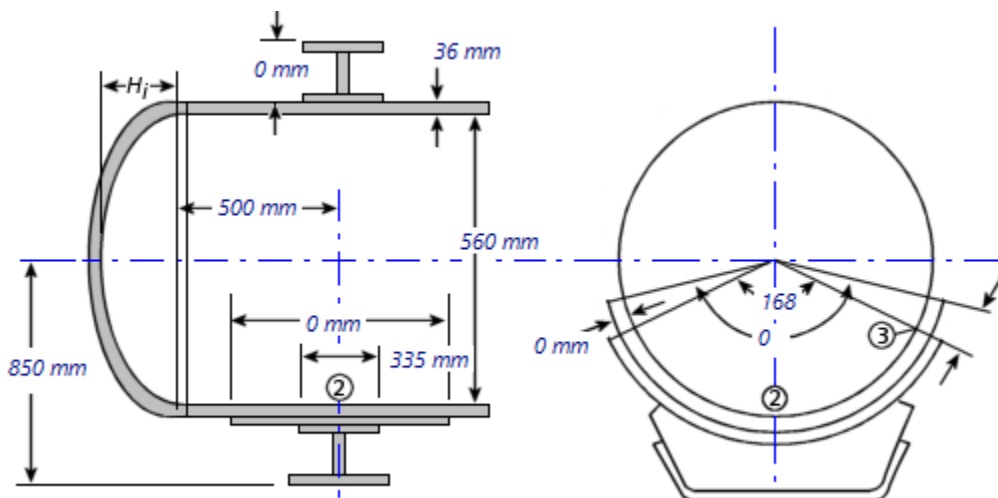
Minimum Wear Plate Angle [θ1]:
 = 180.0 degrees

Note:
 Wear Pad Width (.00) is less than $1.56 \cdot \sqrt{R_m \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 $= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$
 $= \min(335 + 1.56 \cdot \sqrt{298 \cdot 36}, 2 \cdot 500)$
 $= 496.5786 \text{ mm}$

Input and Calculated Values:

Vessel Mean Radius	Rm	298.00	mm
Shell Thickness used in this Case	t	36.000	mm
Stiffened Vessel Length per 4.15.6	L	9220.00	mm
Distance from Saddle to Vessel tangent	a1 or a	500.00	mm
Saddle Width	b1 or b	335.00	mm
Saddle Bearing Angle	delta or theta	168.00	degrees
Shell Allowable Stress used in Calculation		85.90	N./mm ²
Head Allowable Stress used in Calculation		85.90	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	850.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case		30561.15	Kgf
Pressure used in Analysis	P	-103.419	KPa.

Horizontal Vessel Analysis Results:	Actual	Allowable
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	N. /mm ²	N. /mm ²
Long. Stress at Top of Midspan	-54.43	-116.10
Long. Stress at Top of Midspan	54.43	85.90
Long. Stress at Bottom of Midspan	53.58	85.90
Long. Stress at Top of Saddles	2.90	85.90
Long. Stress at Bottom of Saddles	-2.40	-116.10
Long. Stress at Bottom of Saddles	2.40	85.90
Tangential Shear in Shell	16.29	68.72
Circ. Stress at Horn of Saddle	11.96	107.38
Circ. Compressive Stress in Shell	1.07	85.90

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num \text{ of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (1540/2 + 1742) \cdot 850/656$
 $= 9760.8 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl,wind; \text{Sum of X Forces}) \cdot B / Ls$
 $= \max(67.3, 2080) \cdot 850/5220$
 $= 231.6 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(Fl,eq; \text{Sum of X Forces}) \cdot B / Ls$
 $= \max(1750, 2080) \cdot 850/5220$
 $= 338.7 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= Ftr(Ft/Num \text{ of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (1750/2 + 1742) \cdot 850/656$
 $= 10168.7 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(Fwl, Fwt, Fsl, Fst)$
 $= 20392 + \max(232, 9761, 339, 10169)$
 $= 30561.1 \text{ Kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 766 (0 + \pi/4 (0.83 \cdot 1.2)^2)$
 $= 659.734 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	30891.16	Kgf
Transverse Shear Load	2617.47	Kgf
Longitudinal Shear Load	2080.00	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot Rm)$
 $= \min(2(560/2 + 36 + 14) \sin(168/2), 2 \cdot 298)$
 $= 656.384 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1997	K2 = 0.6539	K3 = 0.3357	K4 = 0.2452
K5 = 0.6404	K6 = 0.0224	K7 = 0.0224	K8 = 0.2720
K9 = 0.1759	K10 = 0.0221	K1* = 0.3382	

Note: Dimension a is greater than or equal to $Rm/2$.



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Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$$

$$= -30561 \cdot 500 [1 - (1 - 500/9220 + (298^2 - 0^2)/(2 \cdot 500 \cdot 9220)) / (1 + (4 \cdot 0)/(3 \cdot 9220))]$$

$$= -681.5 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 [(1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L]$$

$$= 30561 \cdot 9220/4 [(1 + 2(298^2 - 0^2)/(9220^2)) / (1 + (4 \cdot 0)/(3 \cdot 9220)) - 4 \cdot 500/9220]$$

$$= 55311.1 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t)$$

$$= -103 \cdot 298/(2 \cdot 36) - 55311/(\pi \cdot 298^2 \cdot 36)$$

$$= -54.43 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$$

$$= -103 \cdot 298/(2 \cdot 36) + 55311/(\pi \cdot 298^2 \cdot 36)$$

$$= 53.58 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:

$$= P \cdot Rm/(2t) - M1/(K1 \cdot \pi \cdot Rm^2 t)$$

$$= -103 \cdot 298/(2 \cdot 36) - 681/(0.2 \cdot \pi \cdot 298^2 \cdot 36)$$

$$= 2.90 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

$$= P \cdot Rm/(2t) + M1/(K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= -103 \cdot 298/(2 \cdot 36) + 681/(0.34 \cdot \pi \cdot 298^2 \cdot 36)$$

$$= -2.40 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot hm/3))$$

$$= 30561 (9220 - 2 \cdot 500)/(9220 + (4 \cdot 0/3))$$

$$= 27246.5 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 0.65 \cdot 27246/(298 \cdot 36)$$

$$= 16.29 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(298 \cdot 36)}$$

$$= 80.789 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.64 \cdot 30561 \cdot 0.1/(36(335 + 80.8 + 80.8))$$

$$= -1.07 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.22) [σ_7]:

$$= -Q/(4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q/(2 \cdot t^2)$$

$$= -30561/(4 \cdot 36(335 + 80.8 + 80.8)) - 3 \cdot 0.022 \cdot 30561/(2 \cdot 36^2)$$

$$= -11.96 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(335 + 1.56 \cdot \sqrt{(298 \cdot 36)}, 2 \cdot 500)$$

$$= 496.58 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length Bplen 780.0000 mm



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Baseplate Thickness	Bpthk	70.0000	mm
Baseplate Width	Bpwid	360.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	204.0	N./mm ²
Height of Web at Center	Hw,c	575.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$
 $= ((1.56 \cdot \sqrt{(280 \cdot 36)} + 530)^3) / 12$
 $= 266.959 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$
 $= 267 + 247 (250 - 18)^2$
 $= 133754.688 \text{ cm}^4$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	686.6	36.0	18.0	247.2	444931.6	0.134E+06
Wearplate	530.0	14.0	43.0	74.2	319060.0	0.319E+05
Web	10.0	450.0	275.0	45.0	1237500.1	0.787E+04
BasePlate	360.0	70.0	535.0	252.0	13482000.0	0.205E+06
Totals	...	...	...	618.4	15483492.0	0.379E+06

Distance to Centroid [C1]:
 $= AY / A$
 $= 15483492 / 618$
 $= 250.386 \text{ mm}$

Angle [β]:
 $= 180 - \text{Saddle Angle} / 2$
 $= 180 - 168 / 2$
 $= 96.0$

Saddle Splitting Coefficient [K1]:
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$
 $= (1 + \cos(96) - 0.5 \cdot \sin(96)^2) / (\pi - 1.68 + \sin(96) \cos(96))$
 $= 0.2943$

Saddle Splitting Force [Fh]:
 $= K1 \cdot Q$
 $= 0.29 \cdot 30561$
 $= 8995.5469 \text{ Kgf}$

Tension Stress, St = (Fh/As) = 2.3766 N./mm²
 Allowed Stress, Sa = 0.6 * Yield Str = 122.4015 N./mm²

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$
 $= 850 - 280 \cdot \sin(168/2) / 1.47$
 $= 660.060 \text{ mm}$

Bending Moment, M = Fh * d = 5937.7212 Kg-m.

Bending Stress, Sb = (M * C1 / I) = 3.8499 N./mm²
 Allowed Stress, Sa = 2/3 * Yield Str = 136.0016 N./mm²



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Minimum Thickness of Baseplate per Moss:

$$= \sqrt{\frac{3(Q + \text{SaddleWt}) \text{BasePlateWidth}}{4 \cdot \text{BasePlateLength} \cdot \text{AllStress}}}$$

$$= \sqrt{\frac{3(30561 + 330)360}{4 \cdot 780 \cdot 136}}$$

$$= 27.768 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 168/2) (280 + 36 + 14)$$

$$= 656.384 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 656 / (3 - 1)$$

$$= 328.192 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 328 \cdot 360 / 2$$

$$= 590.746 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 30891 / (780 \cdot 360)$$

$$= 11.001 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 591 \cdot 11$$

$$= 6498.872 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 32.5 + 16.4$$

$$= 48.910 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 6499 / 48.9$$

$$= 13.031 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	335.0	...	33.5	...	0.313E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 335 - 10$$

$$= 325.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 0 / 48.9$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 335 / 2$$

$$= 167.500 \text{ mm}$$



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Radius of Gyration [r]:
 $= \sqrt{(\text{Total Inertia} / \text{Total Area})}$
 $= \sqrt{(3133/48.9)}$
 $= 80.035 \text{ mm}$

Length of Outer Rib [L]:
 $= \text{Saddle Height} - \cos(\text{theta}/2)(\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$
 $= 850 - \cos(168/2)(280 + 36 + 14) - 70$
 $= 745.506 \text{ mm}$

Intermediate Term [Cc]:
 $= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$
 $= \sqrt{(2 \cdot \pi^2 \cdot 199943008/204)}$
 $= 139.095$

Slenderness ratio [KL/r]:
 $= KL/r$
 $= 1 \cdot 746/80$
 $= 9.315$

Bending Moment [Rm]:
 $= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$
 $= 2080 / (2 \cdot 780) \cdot 328 \cdot 746/2$
 $= 163.116 \text{ Kg-m.}$

Compressive Allowable, $KL/r < Cc$ ($9.31 < 139$) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 $= (1 - (9.31)^2 / (2 \cdot 139^2)) 204 /$
 $(5/3 + 3 \cdot (9.31) / (8 \cdot 139) - (9.31^3) / (8 \cdot 139^3))$
 $= 120.3 \text{ N./mm}^2$

AISC Unity Check of Outside Ribs (must be ≤ 1)
 $= Sc/Sca + (Rm \cdot C1 / I) / Sba$
 $= 13/120 + (163 \cdot 168/31329476) / 136$
 $= 0.171$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	325.0	0.0	32.5	0.0	0.313E+04
Web	328.2	10.0	0.0	32.8	0.0	2.73
Totals	...	...	...	65.3	...	0.314E+04

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0/65.3$
 $= 0.000 \text{ mm}$

Distance to Centroid [C1]:
 $= \text{Saddle Width} / 2$
 $= 335/2$
 $= 167.500 \text{ mm}$

Length of Inner Rib [L]:
 $= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$
 $= 850 - \cos(330/2)(70 + 0 + 0) - 0$
 $= 450.000 \text{ mm}$

Radius of Gyration [r]:
 $= \sqrt{(\text{Total Inertia} / \text{Total Area})}$
 $= \sqrt{(3136/65.3)}$
 $= 69.285 \text{ mm}$



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Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 450/69.3$$

$$= 6.495$$

Unit Force [Force,u]:

$$= F_l / (2 \cdot \text{Baseplate Length})$$

$$= 2080 / (2 \cdot 780)$$

$$= 1.333 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.33 \cdot 328 \cdot 450$$

$$= 196.919 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 197/187200$$

$$= 10.316 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($6.49 < 139$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (6.49)^2 / (2 \cdot 139^2)) 204 /$$

$$(5/3 + 3 \cdot (6.49) / (8 \cdot 139) - (6.49)^3 / (8 \cdot 139^3))$$

$$= 121 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 19.5/121 + (197 \cdot 168/3136) / 136$$

$$= 0.237$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Wear Pad Bearing angle is less than saddle angle + saddle/12 degrees. The wear plate will be ignored.

Minimum Wear Plate Angle [θ_1]:

$$= 180.0 \text{ degrees}$$

Note:

Wear Pad Width (.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$ and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(335 + 1.56 \cdot \sqrt{298 \cdot 36}, 2 \cdot 500)$$

$$= 496.5786 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	298.00	mm
Shell Thickness used in this Case	t	36.000	mm
Stiffened Vessel Length per 4.15.6	L	9220.00	mm
Distance from Saddle to Vessel tangent	a1 or a	500.00	mm
Saddle Width	b1 or b	335.00	mm
Saddle Bearing Angle	delta or theta	168.00	degrees
Shell Allowable Stress used in Calculation		85.90	N./mm ²
Head Allowable Stress used in Calculation		85.90	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	850.00	mm
Coefficient of Friction	mu	0.00	



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Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case 27415.36 Kgf
 Pressure used in Analysis P -103.419 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-48.88	-116.10
Long. Stress at Top of Midspan	48.88	85.90
Long. Stress at Bottom of Midspan	48.02	85.90
Long. Stress at Top of Saddles	2.56	85.90
Long. Stress at Bottom of Saddles	-2.19	-116.10
Long. Stress at Bottom of Saddles	2.19	85.90
Tangential Shear in Shell	14.61	68.72
Circ. Stress at Horn of Saddle	10.73	107.38
Circ. Compressive Stress in Shell	0.96	85.90

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3 (1540/2 + 1760) \cdot 850/656$
 $= 9828.6\ Kgf$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl,wind; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(67.3, 2080) \cdot 850/5220$
 $= 231.6\ Kgf$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(Fl,eq; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(1750, 2080) \cdot 850/5220$
 $= 338.7\ Kgf$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3 (1750/2 + 1760) \cdot 850/656$
 $= 10236.4\ Kgf$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= Saddle\ Load + \max(Fwl, Fwt, Fsl, Fst)$
 $= 17179 + \max(232, 9829, 339, 10236)$
 $= 27415.4\ Kgf$

Longitudinal Wind Force [Fl]:
 $= WindScalar \cdot WindPress(Platform\ Area + (\pi/4(OD \cdot WindDiaMult)^2))$
 $= 1.1 \cdot 766 (0 + \pi/4(0.83 \cdot 1.2)^2)$
 $= 659.734\ N$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 27745.37 Kgf
 Transverse Shear Load 2634.91 Kgf
 Longitudinal Shear Load 2080.00 Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 $= \min(2(ShellID/2 + t + WearPadThickness) \sin(\theta/2), 2 \cdot Rm)$
 $= \min(2(560/2 + 36 + 14) \sin(168/2), 2 \cdot 298)$
 $= 656.384\ mm$



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The Computed K values from Table 4.15.1:

K1 = 0.1997 K2 = 0.6539 K3 = 0.3357 K4 = 0.2452
 K5 = 0.6404 K6 = 0.0224 K7 = 0.0224 K8 = 0.2720
 K9 = 0.1759 K10 = 0.0221 K1* = 0.3382

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a \left[1 - \left(1 - \frac{a}{L} + \frac{(Rm^2 - hm^2)}{(2a \cdot L)} \right) / \left(1 + \frac{(4hm)}{3L} \right) \right]$$

$$= -27415 \cdot 500 \left[1 - \left(1 - \frac{500}{9220} + \frac{(298^2 - 0^2)}{(2 \cdot 500 \cdot 9220)} \right) / \left(1 + \frac{(4 \cdot 0)}{(3 \cdot 9220)} \right) \right]$$

$$= -611.4 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \left(\frac{(1 + 2(Rm^2 - hm^2)/(L^2))}{(1 + (4hm^2)/(3L))} - \frac{4a}{L} \right)$$

$$= 27415 \cdot 9220 / 4 \left(\frac{(1 + 2(298^2 - 0^2)/(9220^2))}{(1 + (4 \cdot 0)/(3 \cdot 9220))} - \frac{4 \cdot 500}{9220} \right)$$

$$= 49617.7 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t)$$

$$= -103 \cdot 298 / (2 \cdot 36) - 49618 / (\pi \cdot 298^2 \cdot 36)$$

$$= -48.88 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= -103 \cdot 298 / (2 \cdot 36) + 49618 / (\pi \cdot 298^2 \cdot 36)$$

$$= 48.02 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 t)$$

$$= -103 \cdot 298 / (2 \cdot 36) - 611 / (0.2 \cdot \pi \cdot 298^2 \cdot 36)$$

$$= 2.56 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

$$= P \cdot Rm / (2t) + M1 / (K1* \cdot \pi \cdot Rm^2 \cdot t)$$

$$= -103 \cdot 298 / (2 \cdot 36) + 611 / (0.34 \cdot \pi \cdot 298^2 \cdot 36)$$

$$= -2.19 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3))$$

$$= 27415 (9220 - 2 \cdot 500) / (9220 + (4 \cdot 0 / 3))$$

$$= 24441.9 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 0.65 \cdot 24442 / (298 \cdot 36)$$

$$= 14.61 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(298 \cdot 36)}$$

$$= 80.789 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.64 \cdot 27415 \cdot 0.1 / (36(335 + 80.8 + 80.8))$$

$$= -0.96 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -27415 / (4 \cdot 36(335 + 80.8 + 80.8)) - 3 \cdot 0.022 \cdot 27415 / (2 \cdot 36^2)$$

$$= -10.73 \text{ N./mm}^2$$



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Effective reinforcing plate width (4.15.24) [B1]:
 $= \min(b + 1.56 \cdot \sqrt{ Rm \cdot t }, 2a)$
 $= \min(335 + 1.56 \cdot \sqrt{ 298 \cdot 36 }, 2 \cdot 500)$
 $= 496.58 \text{ mm}$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	780.0000	mm
Baseplate Thickness	Bpthk	70.0000	mm
Baseplate Width	Bpwid	360.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	204.0	N./mm ²
Height of Web at Center	Hw,c	575.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{ R \cdot t } + \text{WearPlateWidth})t^3)/12$
 $= ((1.56 \cdot \sqrt{ 280 \cdot 36 } + 530)36^3)/12$
 $= 266.959 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$
 $= 267 + 247 (250 - 18)^2$
 $= 133754.688 \text{ cm}^4$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	686.6	36.0	18.0	247.2	444931.6	0.134E+06
Wearplate	530.0	14.0	43.0	74.2	319060.0	0.319E+05
Web	10.0	450.0	275.0	45.0	1237500.1	0.787E+04
BasePlate	360.0	70.0	535.0	252.0	13482000.0	0.205E+06
Totals	...	...	...	618.4	15483492.0	0.379E+06

Distance to Centroid [C1]:
 $= AY / A$
 $= 15483492/618$
 $= 250.386 \text{ mm}$

Angle [β]:
 $= 180 - \text{Saddle Angle}/2$
 $= 180 - 168/2$
 $= 96.0$

Saddle Splitting Coefficient [K1]:
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$
 $= (1 + \cos(96) - 0.5 \cdot \sin(96)^2) / (\pi - 1.68 + \sin(96) \cos(96))$
 $= 0.2943$

Saddle Splitting Force [Fh]:
 $= K1 \cdot Q$
 $= 0.29 \cdot 27415$
 $= 8069.5972 \text{ Kgf}$

Tension Stress, St = (Fh/As) = 2.1319 N./mm²
 Allowed Stress, Sa = 0.6 * Yield Str = 122.4015 N./mm²

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$



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$$= 850 - 280 \cdot \sin(168/2)/1.47$$

$$= 660.060 \text{ mm}$$

$$\text{Bending Moment, } M = F_h \cdot d = 5326.5269 \text{ Kg-m.}$$

$$\text{Bending Stress, } S_b = (M \cdot C_1 / I) = 3.4536 \text{ N./mm}^2$$

$$\text{Allowed Stress, } S_a = 2/3 \cdot \text{Yield Str} = 136.0016 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{\frac{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}}{4 \cdot \text{BasePlateLength} \cdot \text{AllStress}}}$$

$$= \sqrt{\frac{3(27415 + 330) \cdot 360}{4 \cdot 780 \cdot 136}}$$

$$= 26.316 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

$$\text{Web Length Dimension [Web Length]:}$$

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 168/2) (280 + 36 + 14)$$

$$= 656.384 \text{ mm}$$

$$\text{Distance between Ribs [e]:}$$

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 656 / (3 - 1)$$

$$= 328.192 \text{ mm}$$

$$\text{Baseplate Pressure Area [Ap]:}$$

$$= e \cdot \text{Bpwid} / 2$$

$$= 328 \cdot 360 / 2$$

$$= 590.746 \text{ cm}^2$$

$$\text{Bearing Pressure [Bp]:}$$

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 27745 / (780 \cdot 360)$$

$$= 9.881 \text{ Kg/cm}^2$$

$$\text{Axial Load [P]:}$$

$$= A_p \cdot B_p$$

$$= 591 \cdot 9.88$$

$$= 5837.062 \text{ Kg}$$

$$\text{Area of the Rib and Web [Ar]:}$$

$$= \text{Rib Area} + \text{Web Area}$$

$$= 32.5 + 16.4$$

$$= 48.910 \text{ cm}^2$$

$$\text{Compressive Stress [Sc]:}$$

$$= P / A_r$$

$$= 5837 / 48.9$$

$$= 11.704 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	335.0	...	33.5	...	0.313E+04

$$\text{Rib dimension [D]:}$$

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 335 - 10$$

$$= 325.000 \text{ mm}$$

$$\text{Distance to Centroid from Datum [ytot]:}$$

$$= AY / A$$



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= 0/48.9
 = 0.000 mm

Distance to Centroid [C1]:
 = Saddle Width / 2
 = 335/2
 = 167.500 mm

Radius of Gyration [r]:
 $= \sqrt{\text{Total Inertia} / \text{Total Area}}$
 $= \sqrt{3133/48.9}$
 = 80.035 mm

Length of Outer Rib [L]:
 = Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 850 - cos(168/2)(280 + 36 + 14) - 70
 = 745.506 mm

Intermediate Term [Cc]:
 $= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 $= \sqrt{2 \cdot \pi^2 \cdot 199943008/204}$
 = 139.095

Slenderness ratio [KL/r]:
 = KL/r
 = 1 • 746/80
 = 9.315

Bending Moment [Rm]:
 $= FI / (2 \cdot B \cdot \text{plen}) \cdot e \cdot L / 2$
 $= 2080 / (2 \cdot 780) \cdot 328 \cdot 746/2$
 = 163.116 Kg-m.

Compressive Allowable, $KL/r < Cc$ (9.31 < 139) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 $= (1 - (9.31)^2 / (2 \cdot 139^2)) 204 / (5/3 + 3 \cdot (9.31) / (8 \cdot 139) - (9.31^3) / (8 \cdot 139^3))$
 = 120.3 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)
 $= Sc/Sca + (Rm \cdot C1 / I) / Sba$
 $= 11.7/120 + (163 \cdot 168/31329476) / 136$
 = 0.160

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	325.0	0.0	32.5	0.0	0.313E+04
Web	328.2	10.0	0.0	32.8	0.0	2.73
Totals	...	...	...	65.3	...	0.314E+04

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 0/65.3
 = 0.000 mm

Distance to Centroid [C1]:
 = Saddle Width / 2
 = 335/2
 = 167.500 mm

Length of Inner Rib [L]:
 = Saddle Height - Outside Radius - Bpthk



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$$= 850 - \cos(330/2)(70 + 0 + 0) - 0$$

$$= 450.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{3136/65.3}$$

$$= 69.285 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 450/69.3$$

$$= 6.495$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 2080 / (2 \cdot 780)$$

$$= 1.333 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.33 \cdot 328 \cdot 450$$

$$= 196.919 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 197/187200$$

$$= 10.316 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($6.49 < 139$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (6.49)^2 / (2 \cdot 139^2)) 204 / (5/3 + 3 \cdot (6.49) / (8 \cdot 139) - (6.49^3) / (8 \cdot 139^3))$$

$$= 121 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / Sba$$

$$= 17.5/121 + (197 \cdot 168/3136) / 136$$

$$= 0.221$$

Review notes about nozzle loadings and supports in the Warnings report.

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 Nozzle Calcs.: N3-Tube Inlet Nozl: 9 10:04am Sep 9,2026

Input, Nozzle Desc: N3-Tube Inlet

From: 20

Pressure for Reinforcement Calculations	P	6100.000	KPa.
Temperature for Internal Pressure	Temp	300	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	300	°C
Parent Material		SA-240 304	
Parent Allowable Stress at Temperature	Sv	85.90	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	560.00	mm
Design Length of Section	L	897.4667	mm
Shell Finished (Minimum) Thickness	t	24.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		450.80	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

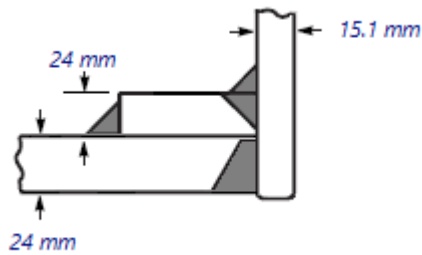
Type of Element Connected to the Parent : Nozzle

Material		SA-312 TP304	
Material UNS Number		S30400	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	116.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		10.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-182 F304	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-240 304	
Pad Allowable Stress at Temperature	Sp	85.90	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	453.0500	mm
Thickness of Pad	te	24.0000	mm
Weld leg size between Pad and Shell	Wp	14.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	24.0000	mm
Reinforcing Pad Width		90.0000	mm
Flange Class		900	
Flange Grade		GR 2.1	

The Pressure Design option was Design Pressure + static head.



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 Nozzle Calcs.: N3-Tube Inlet Nozl: 9 10:04am Sep 9,2026



Reinforcement CALCULATION, Description: N3-Tube Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 9.562 in.
 Actual Thickness Used in Calculation 0.594 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6100 \cdot 281 / (85.9 \cdot 1 - 0.6 \cdot 6100)$
 $= 20.8438 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6100 \cdot 122 / (116 \cdot 1 - 0.6 \cdot 6100)$
 $= 6.6486 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.9026 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	489.7496	mm
Parallel to Vessel Wall, opening length	d	244.8748	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		57.5000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 116 / 85.9)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 116 / 85.9)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 85.9 / 85.9)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

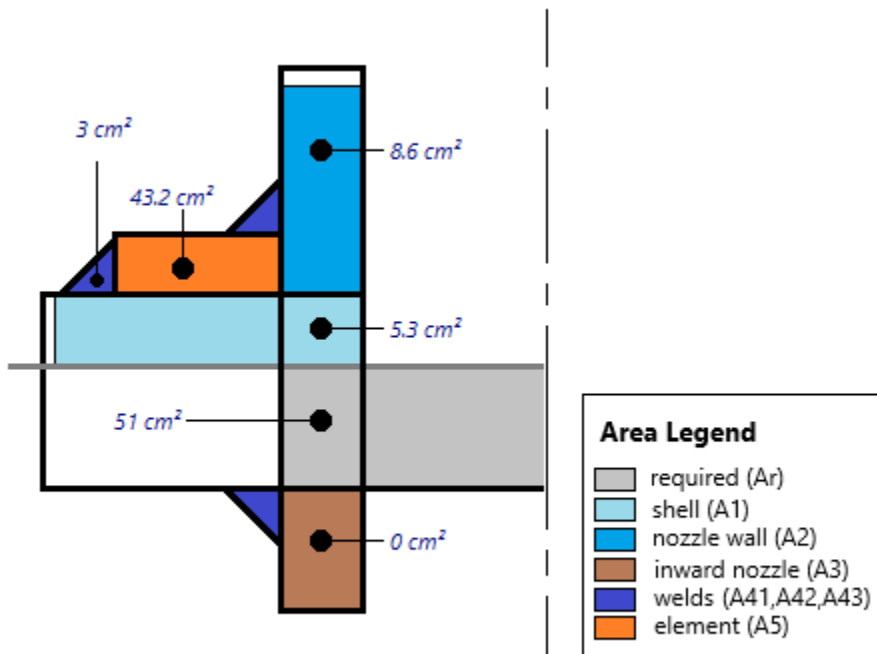


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	51.041	2.998	NA
Area in Shell	A1	5.280	50.324	NA
Area in Nozzle Wall	A2	8.555	15.163	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	2.960	2.960	NA
Area in Element	A5	43.200	43.200	NA
TOTAL AREA AVAILABLE	Atot	59.995	111.647	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 415.7433 24.0000 mm
 Based on given Pad Diameter: 453.0500 19.0258 mm
 Based on the Estimated Diameter Limit: 488.1621 15.9203 mm

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (245 \cdot 20.8 \cdot 1 + 2 \cdot 14.1 \cdot 20.8 \cdot 1 \cdot (1 - 1))$$

$$= 51.041 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 245(1 \cdot 23 - 1 \cdot 20.8) - 2 \cdot 14.1$$

$$(1 \cdot 23 - 1 \cdot 20.8) \cdot (1 - 1)$$



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$$= 5.280 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 57.5) \cdot (14.1 - 6.65) \cdot 1$$

$$= 8.555 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= Wo^2 \cdot fr3 + (Wi-can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1 + 14^2 \cdot 1$$

$$= 2.960 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$$

$$= (453 - 273) \cdot 24 \cdot 1$$

$$= 43.200 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 7.6486 mm
Wall Thickness per UG16(b),	tr16b = 2.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 21.8438 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 21.8438 mm
Wall Thickness, shell/head, external pressure	trb2 = 1.3386 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 2.5000 mm
Wall Thickness per table UG-45	tb3 = 9.1026 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[9.1, \max(21.8, 2.5)]$$

$$= 9.1026 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(7.65, 9.1)$$

$$= 9.1026 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 15.088 = 13.202 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 48.1,	Allowable	: 116.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 269.4 N./mm ²	Passed
Occasional	: 25.1,	Allowable	: 154.3 N./mm ²	Passed
Shear	: 15.1,	Allowable	: 81.2 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

SA-240 304, Min Metal Temp without impact per UHA-51: -196 °C

SA-312 TP304, Min Metal Temp without impact per UHA-51: -196 °C

Weld Size Calculations, Description: N3-Tube Inlet

Intermediate Calc. for nozzle/shell Welds	Tmin	14.0876 mm
Intermediate Calc. for pad/shell Welds	TminPad	19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot Wo \text{ mm}$
Pad Weld	$9.5000 = 0.5 \cdot TminPad$	$9.8980 = 0.7 \cdot Wp \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)



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Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$
 $= \max(0, (51 - 5.28 + 2 \cdot 14.1 \cdot 1 \cdot (1 \cdot 23 - 20.8)) \cdot 85.9)$
 $= 40615.54 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (8.55 + 43.2 + 2.96 - 0 \cdot 1) \cdot 85.9$
 $= 47926.09 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (8.55 + 0 + 1 + (6.48)) \cdot 85.9$
 $= 14045.56 \text{ Kg}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (8.55 + 0 + 2.96 + 43.2 + (6.48)) \cdot 85.9$
 $= 53602.35 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.14/2.0) \cdot 273 \cdot 10 \cdot 0.49 \cdot 85.9$
 $= 18409. \text{ Kg}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.14/2.0) \cdot 453 \cdot 14 \cdot 0.49 \cdot 85.9$
 $= 42762. \text{ Kg}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.14 \cdot 129) \cdot (15.1 - 1) \cdot 0.7 \cdot 116$
 $= 47449. \text{ Kg}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
 $= (3.14/2) \cdot 273 \cdot 24 \cdot 0.74 \cdot 116$
 $= 90103. \text{ Kg}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$
 $= (3.14/2.0) \cdot 273 \cdot (14 - 0) \cdot 0.74 \cdot 116$
 $= 52560. \text{ Kg}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SPEW + SNW) = (42762 + 47449) = 90211 \text{ Kg}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (18409 + 90103 + 52560 + 0) = 161071 \text{ Kg}$

Failure Path [Path 3-3]:
 $= (Spew + Tngw + Sinw)$
 $= (42762 + 52560 + 0) = 95322 \text{ Kg}$



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Summary of Failure Path Calculations:

Path 1-1 = 90210 Kgf, must exceed W = 40615 Kgf or W1 = 47926 Kgf
 Path 2-2 = 161071 Kgf, must exceed W = 40615 Kgf or W2 = 14045 Kgf
 Path 3-3 = 95321 Kgf, must exceed W = 40615 Kgf or W3 = 53602 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6575 KPa.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 35.5395 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 309.5395 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 244.87 mm

Input Echo, WRC297 Item 1, Description: N3-Tube Inlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	560.000	mm
Shell Thickness	24.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-240 304	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 85.90	N./mm ²
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.0000	mm
Nozzle Diameter	242.875	mm
Nozzle Thickness	15.0876	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-312 TP304	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 116.00	N./mm ²
Diameter of Reinforcing Pad	453.050	mm
Thickness of Reinforcing Pad	24.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	6100.00	KPa.
Radial Load	P	-2294.50	Kgf
Circumferential Shear	Vc	1657.15	Kgf
Longitudinal Shear	Vl	1657.15	Kgf
Circumferential Moment	Mc	1274.24	Kg-m.
Longitudinal Moment	Ml	1274.24	Kg-m.
Torsional Moment	Mt	1465.38	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 273.05 + 2 \cdot 1.65 \cdot \sqrt{(304.5(24.0 - 1.0))}$$

$$= 549.217 \text{ mm}$$



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WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 609.000 mm
 Nozzle Outside Diameter (d) = 273.050 mm
 Vessel Thickness used (T) = 47.000 mm
 Nozzle Thickness used (t) = 14.088 mm
 T / t = 3.336
 d / t = 19.382

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.614$$

Nr/P = 0.086
 Mr/P = 0.079
 M0/P = 0.098
 N0/P = 0.220
 MrD/Mc = 0.157
 NrDL/Mc = 0.131
 M0d/Mc = 0.204
 N0DL/Mc = 0.140
 MrD/Ml = 0.121
 NrDL/Ml = 0.102
 M0D/Ml = 0.163
 N0DL/Ml = 0.278

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-5	-5	5	5
Outplane Bending (ML)	-20	20	20	-20
Normal Pressure Stress	33	39	33	39
Outplane Stress Summary	15	50	67	21

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	4	-4	4	-4
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-15	15	15	-15
Inplane Pressure Stress	16	16	16	16
Inplane Stress Summary	5	25	39	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1



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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	16	50	67	21

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	-2	-2	2	2
Outplane Bending (Mc)	-25	25	25	-25
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	16	16	16	16
Outplane Stress Summary	-3	35	53	-9

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	4	-4	4	-4
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-19	19	19	-19
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	33	39	33	39
Inplane Stress Summary	17	52	61	19

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2
Shear Stress Summary	1	1	3	3

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	20	52	62	29

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-5	-5	5	5



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Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	60	67	60	67
Hoop Stress Summary	56	63	68	75

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	24	-24	24	-24
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-17	-17	17	17
Axial Bending (ML)	-96	96	96	-96
Axial Pressure Stress	30	30	30	30
Axial Stress Summary	-58	87	170	-71

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	8	8	8	8
Shear Stress Summary	11	11	6	6

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	117	91	171	147

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-2	-2	2	2
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	60	67	60	67
Hoop Stress Summary	59	66	65	72

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	24	-24	24	-24
Axial Membrane (Mc)	-17	-17	17	17
Axial Bending (Mc)	-126	126	126	-126
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	30	30	30	30
Axial Stress Summary	-87	116	200	-100



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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	8	8	8	8
Shear Stress Summary	6	6	11	11

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	147	117	201	174

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$
 $= 273.05 + 2 \cdot 1.65 \cdot \sqrt{292.5(24.0 - 1.0)}$
 $= 543.720 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	585.000 mm
Nozzle Outside Diameter	(d) =	453.050 mm
Vessel Thickness used	(T) =	23.000 mm
Nozzle Thickness used	(t) =	104.088 mm
T / t	=	0.221
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.906$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.050
Mr/P	=	0.084
M0/P	=	0.026
N0/P	=	0.056
MrD/Mc	=	0.256
NrDL/Mc	=	0.090
M0d/Mc	=	0.077
N0DL/Mc	=	0.085
MrD/Ml	=	0.069
NrDL/Ml	=	0.049
M0D/Ml	=	0.021
N0DL/Ml	=	0.059

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-3	-3	3	3
Outplane Bending (Ml)	-6	6	6	-6
Normal Pressure Stress	71	77	71	77
Outplane Stress Summary	71	76	90	69



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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stresses parallel to longitudinal plane)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	21	-21	21	-21
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-21	21	21	-21
Inplane Pressure Stress	35	35	35	35
Inplane Stress Summary	34	35	83	-2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress normal to longitudinal plane)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	0	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	71	76	90	72

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	-4	-4	4	4
Outplane Bending (Mc)	-23	23	23	-23
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	35	35	35	35
Outplane Stress Summary	16	51	73	11

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	21	-21	21	-21
Inplane Membrane (Mc)	-4	-4	4	4
Inplane Bending (Mc)	-80	80	80	-80
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	71	77	71	77
Inplane Stress Summary	10	133	179	-16

Vessel Stresses (N./mm²):



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CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			0	0	0	0
Torsional Shear (Mt)			1	1	1	1
Shear Stress Summary			0	0	2	2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress			16	133	179	29

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	33	39	33	39	33	39	33	39
Circ. Pl (SUS)	-3	-3	7	7	-2	-2	2	2
Circ. Q (SUS)	-15	15	25	-25	-15	15	23	-23
Long. Pm (SUS)	16	16	16	16	16	16	16	16
Long. Pl (SUS)	-2	-2	2	2	0	0	4	4
Long. Q (SUS)	-11	11	19	-19	-20	20	30	-30
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	33.0	39.0	33.0	39.0	33.0	39.0	33.0	39.0
Pm+Pl (SUS)	30.0	36.0	40.0	46.0	31.0	37.0	35.0	41.0
Pm+Pl+Q (Total)	15.3	51.2	65.1	22.4	20.4	52.2	58.5	28.3

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	39.00	85.90	Passed
Pm+Pl (SUS)	46.00	128.85	Passed
Pm+Pl+Q (TOTAL)	65.14	257.70	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	71	77	71	77	71	77	71	77
Circ. Pl (SUS)	-1	-1	5	5	-2	-2	6	6



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Circ. Q (SUS)	0	0	12	-12	-59	59	101	-101
Long. Pm (SUS)	35	35	35	35	35	35	35	35
Long. P1 (SUS)	0	0	4	4	-2	-2	6	6
Long. Q (SUS)	0	0	42	-42	-17	17	29	-29
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	71.0	77.0	71.0	77.0	71.0	77.0	71.0	77.0
Pm+P1 (SUS)	70.0	76.0	76.0	82.0	69.0	75.0	77.0	83.0
Pm+P1+Q (Total)	70.0	76.0	88.1	73.0	16.2	134.0	178.0	30.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	77.00	85.90	Passed
Pm+P1 (SUS)	83.00	128.85	Passed
Pm+P1+Q (TOTAL)	178.01	257.70	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	60	67	60	67	60	67	60	67
Circ. P1 (SUS)	-3	-3	7	7	0	0	4	4
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	30	30	30	30	30	30	30	30
Long. P1 (SUS)	-16	-16	18	18	-16	-16	18	18
Long. Q (SUS)	-72	72	120	-120	-102	102	150	-150
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	8	8	8	8	8	8	8	8
Pm (SUS)	60.0	67.0	60.0	67.0	60.0	67.0	60.0	67.0
Pm+P1 (SUS)	57.1	64.1	67.2	74.2	60.1	67.1	64.2	71.2
Pm+P1+Q (Total)	116.7	89.9	168.4	146.5	148.5	116.7	198.7	174.2

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	67.00	116.00	Passed
Pm+P1 (SUS)	74.15	174.00	Passed
Pm+P1+Q (TOTAL)	198.74	348.00	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.



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 Nozzle Calcs.: N4-Tube Outlet Nozl: 10 10:04am Sep 9,2026

Input, Nozzle Desc: N4-Tube Outlet

From: 20

Pressure for Reinforcement Calculations	P	6105.391	KPa.
Temperature for Internal Pressure	Temp	300	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	300	°C
Parent Material		SA-240 304	
Parent Allowable Stress at Temperature	Sv	85.90	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	560.00	mm
Design Length of Section	L	897.4667	mm
Shell Finished (Minimum) Thickness	t	24.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		450.80	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

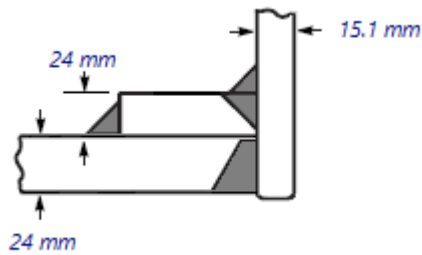
Type of Element Connected to the Parent : Nozzle

Material		SA-312 TP304	
Material UNS Number		S30400	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	116.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		10.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-182 F304	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-240 304	
Pad Allowable Stress at Temperature	Sp	85.90	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	453.0500	mm
Thickness of Pad	te	24.0000	mm
Weld leg size between Pad and Shell	Wp	14.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	24.0000	mm
Reinforcing Pad Width		90.0000	mm
Flange Class		900	
Flange Grade		GR 2.1	

The Pressure Design option was Design Pressure + static head.



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Reinforcement CALCULATION, Description: N4-Tube Outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 9.562 in.
 Actual Thickness Used in Calculation 0.594 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6105 \cdot 281 / (85.9 \cdot 1 - 0.6 \cdot 6105)$
 $= 20.8630 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6105 \cdot 122 / (116 \cdot 1 - 0.6 \cdot 6105)$
 $= 6.6547 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.9026 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	489.7496	mm
Parallel to Vessel Wall, opening length	d	244.8748	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		57.5000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 116 / 85.9)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 116 / 85.9)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 85.9 / 85.9)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

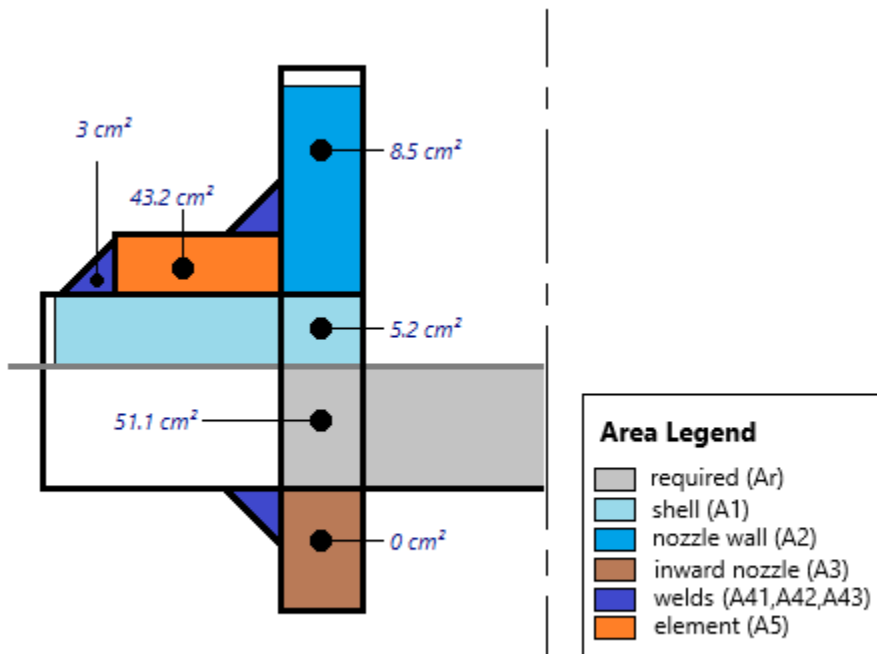


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	51.088	2.998	NA
Area in Shell	A1	5.233	50.324	NA
Area in Nozzle Wall	A2	8.548	15.163	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	2.960	2.960	NA
Area in Element	A5	43.200	43.200	NA
TOTAL AREA AVAILABLE	Atot	59.941	111.647	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 416.1650 24.0000 mm
 Based on given Pad Diameter: 453.0500 19.0820 mm
 Based on the Estimated Diameter Limit: 488.1621 15.9673 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (245 \cdot 20.9 \cdot 1 + 2 \cdot 14.1 \cdot 20.9 \cdot 1 \cdot (1 - 1))$
 $= 51.088 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 245(1 \cdot 23 - 1 \cdot 20.9) - 2 \cdot 14.1$
 $(1 \cdot 23 - 1 \cdot 20.9) \cdot (1 - 1)$



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$$= 5.233 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 57.5) \cdot (14.1 - 6.65) \cdot 1$$

$$= 8.548 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= Wo^2 \cdot fr3 + (Wi-can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1 + 14^2 \cdot 1$$

$$= 2.960 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$$

$$= (453 - 273) \cdot 24 \cdot 1$$

$$= 43.200 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 7.6547 mm
Wall Thickness per UG16(b),	tr16b = 2.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 21.8630 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 21.8630 mm
Wall Thickness, shell/head, external pressure	trb2 = 1.3386 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 2.5000 mm
Wall Thickness per table UG-45	tb3 = 9.1026 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[9.1, \max(21.9, 2.5)]$$

$$= 9.1026 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(7.65, 9.1)$$

$$= 9.1026 \text{ mm}$$

$$\text{Available Nozzle Neck Thickness} = 0.875 \cdot 15.088 = 13.202 \text{ mm} \rightarrow \text{OK}$$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 48.2,	Allowable	: 116.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 269.3 N./mm ²	Passed
Occasional	: 25.1,	Allowable	: 154.3 N./mm ²	Passed
Shear	: 15.1,	Allowable	: 81.2 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

SA-240 304, Min Metal Temp without impact per UHA-51: -196 °C

SA-312 TP304, Min Metal Temp without impact per UHA-51: -196 °C

Weld Size Calculations, Description: N4-Tube Outlet

Intermediate Calc. for nozzle/shell Welds	Tmin	14.0876 mm
Intermediate Calc. for pad/shell Welds	TminPad	19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	9.5000 = 0.5*TminPad	9.8980 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)



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Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (51.1 - 5.23 + 2 \cdot 14.1 \cdot 1 \cdot (1 \cdot 23 - 20.9)) \cdot 85.9)$
 $= 40693.33 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i - Can/.707)^2 \cdot f_r2) \cdot S_v$
 $= (8.55 + 43.2 + 2.96 - 0 \cdot 1) \cdot 85.9$
 $= 47919.98 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$
 $= (8.55 + 0 + 1 + (6.48)) \cdot 85.9$
 $= 14039.45 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$
 $= (8.55 + 0 + 2.96 + 43.2 + (6.48)) \cdot 85.9$
 $= 53596.23 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 273 \cdot 10 \cdot 0.49 \cdot 85.9$
 $= 18409. \text{ Kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.14/2.0) \cdot 453 \cdot 14 \cdot 0.49 \cdot 85.9$
 $= 42762. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 129) \cdot (15.1 - 1) \cdot 0.7 \cdot 116$
 $= 47449. \text{ Kgf}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.14/2) \cdot 273 \cdot 24 \cdot 0.74 \cdot 116$
 $= 90103. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 273 \cdot (14 - 0) \cdot 0.74 \cdot 116$
 $= 52560. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (S_{PEW} + S_{NW}) = (42762 + 47449) = 90211 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$
 $= (18409 + 90103 + 52560 + 0) = 161071 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (S_{pew} + T_{ngw} + S_{inw})$
 $= (42762 + 52560 + 0) = 95322 \text{ Kgf}$



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Summary of Failure Path Calculations:

Path 1-1 = 90210 Kgf, must exceed W = 40693 Kgf or W1 = 47919 Kgf
 Path 2-2 = 161071 Kgf, must exceed W = 40693 Kgf or W2 = 14039 Kgf
 Path 3-3 = 95321 Kgf, must exceed W = 40693 Kgf or W3 = 53596 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6570 KPa.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 35.5395 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 309.5395 mm

Input Echo, WRC297 Item 1, Description: N4-Tube Outlet:

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	1.0000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	560.000	mm	
Shell Thickness	24.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-240 304		
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²	
Vessel Hot S.I. Allowable	Smh 85.90	N./mm ²	
Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance	1.0000	mm	
Nozzle Diameter	242.875	mm	
Nozzle Thickness	15.0876	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-312 TP304		
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²	
Nozzle Hot S.I. Allowable	SNmh 116.00	N./mm ²	
Diameter of Reinforcing Pad	453.050	mm	
Thickness of Reinforcing Pad	24.0000	mm	

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	6105.39	KPa.
Radial Load	P	-2294.50	Kgf
Circumferential Shear	Vc	1657.15	Kgf
Longitudinal Shear	Vl	1657.15	Kgf
Circumferential Moment	Mc	1274.24	Kg-m.
Longitudinal Moment	Ml	1274.24	Kg-m.
Torsional Moment	Mt	1465.38	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$
 $= 273.05 + 2 \cdot 1.65 \cdot \sqrt{304.5(24.0 - 1.0)}$
 $= 549.217 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 609.000 mm
 Nozzle Outside Diameter (d) = 273.050 mm



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Vessel Thickness used (T) = 47.000 mm
 Nozzle Thickness used (t) = 14.088 mm
 T / t = 3.336
 d / t = 19.382

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.614$$

Nr/P = 0.086
 Mr/P = 0.079
 M0/P = 0.098
 N0/P = 0.220
 MrD/Mc = 0.157
 NrDL/Mc = 0.131
 M0d/Mc = 0.204
 N0DL/Mc = 0.140
 MrD/Ml = 0.121
 NrDL/Ml = 0.102
 M0D/Ml = 0.163
 N0DL/Ml = 0.278

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-5	-5	5	5
Outplane Bending (ML)	-20	20	20	-20
Normal Pressure Stress	33	39	33	39
Outplane Stress Summary	15	50	67	21

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	4	-4	4	-4
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-15	15	15	-15
Inplane Pressure Stress	16	16	16	16
Inplane Stress Summary	5	25	39	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Two * Max Shear Stress	16	50	67	21
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	C1	Du	D1
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	-2	-2	2	2
Outplane Bending (Mc)	-25	25	25	-25
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	16	16	16	16
Outplane Stress Summary	-3	35	53	-9

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	C1	Du	D1
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	4	-4	4	-4
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-19	19	19	-19
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	33	39	33	39
Inplane Stress Summary	17	52	61	19

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	C1	Du	D1
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2
Shear Stress Summary	1	1	3	3

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	C1	Du	D1
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	20	52	62	29

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	A1	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-5	-5	5	5
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	60	67	60	67
Hoop Stress Summary	56	64	68	75



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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	24	-24	24	-24
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-17	-17	17	17
Axial Bending (ML)	-96	96	96	-96
Axial Pressure Stress	30	30	30	30
Axial Stress Summary	-58	87	170	-71

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	8	8	8	8
Shear Stress Summary	11	11	6	6

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	117	91	171	147

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-2	-2	2	2
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	60	67	60	67
Hoop Stress Summary	59	66	65	72

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	24	-24	24	-24
Axial Membrane (Mc)	-17	-17	17	17
Axial Bending (Mc)	-126	126	126	-126
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	30	30	30	30
Axial Stress Summary	-87	116	200	-100

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
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 Nozzle Calcs.: N4-Tube Outlet Nozl: 10 10:04am Sep 9,2026

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	8	8	8	8

Shear Stress Summary	6	6	11	11
----------------------	---	---	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	147	117	201	174

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 273.05 + 2 \cdot 1.65 \cdot \sqrt{(292.5(24.0 - 1.0))}$
 $= 543.720 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	585.000 mm
Nozzle Outside Diameter	(d) =	453.050 mm
Vessel Thickness used	(T) =	23.000 mm
Nozzle Thickness used	(t) =	104.088 mm
T / t	=	0.221
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.906$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.050
Mr/P	=	0.084
M0/P	=	0.026
N0/P	=	0.056
MrD/Mc	=	0.256
NrDL/Mc	=	0.090
M0d/Mc	=	0.077
N0DL/Mc	=	0.085
MrD/Ml	=	0.069
NrDL/Ml	=	0.049
M0D/Ml	=	0.021
N0DL/Ml	=	0.059

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-3	-3	3	3
Outplane Bending (ML)	-6	6	6	-6
Normal Pressure Stress	71	77	71	77
Outplane Stress Summary	71	76	90	70

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to	Au Top	Al Top	Bu Bottom	Bl Bottom
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 Nozzle Calcs.: N4-Tube Outlet Nozl: 10 10:04am Sep 9,2026

longitudinal plane)	Outside	Inside	Outside	Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	21	-21	21	-21
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-21	21	21	-21
Inplane Pressure Stress	35	35	35	35
Inplane Stress Summary	35	35	83	-2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)		Top	Top	Bottom	Bottom
		Outside	Inside	Outside	Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		0	0	0	0
Outplane Shear (Mt)		1	1	1	1
Shear Stress Summary		2	2	0	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stress Intensities		Top	Top	Bottom	Bottom
		Outside	Inside	Outside	Inside
Two * Max Shear Stress		71	77	90	72

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)		Left	Left	Right	Right
		Outside	Inside	Outside	Inside
Outplane Membrane (P)		2	2	2	2
Outplane Bending (P)		6	-6	6	-6
Outplane Membrane (Mc)		-4	-4	4	4
Outplane Bending (Mc)		-23	23	23	-23
Outplane Membrane (ML)		0	0	0	0
Outplane Bending (ML)		0	0	0	0
Normal Pressure Stress		35	35	35	35
Outplane Stress Summary		16	51	73	11

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)		Left	Left	Right	Right
		Outside	Inside	Outside	Inside
Inplane Membrane (P)		2	2	2	2
Inplane Bending (P)		21	-21	21	-21
Inplane Membrane (Mc)		-4	-4	4	4
Inplane Bending (Mc)		-80	80	80	-80
Inplane Membrane (ML)		0	0	0	0
Inplane Bending (ML)		0	0	0	0
Inplane Pressure Stress		71	77	71	77
Inplane Stress Summary		10	133	179	-16

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)		Left	Left	Right	Right
		Outside	Inside	Outside	Inside



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Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary	0	0	2	2
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	16	133	179	29

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	33	39	33	39	33	39	33	39
Circ. Pl (SUS)	-3	-3	7	7	-2	-2	2	2
Circ. Q (SUS)	-15	15	25	-25	-15	15	23	-23
Long. Pm (SUS)	16	16	16	16	16	16	16	16
Long. Pl (SUS)	-2	-2	2	2	0	0	4	4
Long. Q (SUS)	-11	11	19	-19	-20	20	30	-30
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	33.0	39.0	33.0	39.0	33.0	39.0	33.0	39.0
Pm+Pl (SUS)	30.0	36.0	40.0	46.0	31.0	37.0	35.0	41.0
Pm+Pl+Q (Total)	15.3	51.2	65.1	22.4	20.4	52.2	58.5	28.3

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	39.00	85.90	Passed
Pm+Pl (SUS)	46.00	128.85	Passed
Pm+Pl+Q (TOTAL)	65.14	257.70	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	71	77	71	77	71	77	71	77
Circ. Pl (SUS)	-1	-1	5	5	-2	-2	6	6
Circ. Q (SUS)	0	0	12	-12	-59	59	101	-101
Long. Pm (SUS)	35	35	35	35	35	35	35	35
Long. Pl (SUS)	0	0	4	4	-2	-2	6	6



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Long. Q (SUS)	0	0	42	-42	-17	17	29	-29
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	71.0	77.0	71.0	77.0	71.0	77.0	71.0	77.0
Pm+Pl (SUS)	70.0	76.0	76.0	82.0	69.0	75.0	77.0	83.0
Pm+Pl+Q (Total)	70.0	76.0	88.1	73.0	16.2	134.0	178.0	30.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	77.00	85.90	Passed
Pm+Pl (SUS)	83.00	128.85	Passed
Pm+Pl+Q (TOTAL)	178.01	257.70	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	60	67	60	67	60	67	60	67
Circ. Pl (SUS)	-3	-3	7	7	0	0	4	4
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	30	30	30	30	30	30	30	30
Long. Pl (SUS)	-16	-16	18	18	-16	-16	18	18
Long. Q (SUS)	-72	72	120	-120	-102	102	150	-150
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	8	8	8	8	8	8	8	8
Pm (SUS)	60.0	67.0	60.0	67.0	60.0	67.0	60.0	67.0
Pm+Pl (SUS)	57.1	64.1	67.2	74.2	60.1	67.1	64.2	71.2
Pm+Pl+Q (Total)	116.7	89.9	168.4	146.5	148.5	116.7	198.7	174.2

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	67.00	116.00	Passed
Pm+Pl (SUS)	74.15	174.00	Passed
Pm+Pl+Q (TOTAL)	198.74	348.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



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 Nozzle Calcs.: N1-Shell Inlet Nozl: 11 10:04am Sep 9,2026

Input, Nozzle Desc: N1-Shell Inlet

From: 40

Pressure for Reinforcement Calculations	P	6100.000	KPa.
Temperature for Internal Pressure	Temp	300	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	300	°C
Parent Material		SA-240 304	
Parent Allowable Stress at Temperature	Sv	85.90	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	560.00	mm
Design Length of Section	L	9220.0000	mm
Shell Finished (Minimum) Thickness	t	24.0000	mm
Shell Internal Corrosion Allowance	c	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1960.98	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

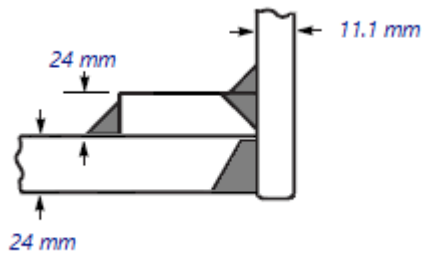
Type of Element Connected to the Parent : Nozzle

Material		SA-312 TP304	
Material UNS Number		S30400	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	116.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		10.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		60	
Flange Material		SA-182 F304	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	136.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	426.5400	mm
Thickness of Pad	te	24.0000	mm
Weld leg size between Pad and Shell	Wp	18.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	24.0000	mm
Reinforcing Pad Width		76.7450	mm
Flange Class		900	
Flange Grade		GR 2.1	

The Pressure Design option was Design Pressure + static head.



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Reinforcement CALCULATION, Description: N1-Shell Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	9.875 in.
Actual Thickness Used in Calculation	0.438 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6100 \cdot 280 / (85.9 \cdot 1 - 0.6 \cdot 6100)$
 $= 20.7696 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6100 \cdot 125 / (116 \cdot 1 - 0.6 \cdot 6100)$
 $= 6.8102 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.9026 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	501.6500 mm
Parallel to Vessel Wall, opening length	d	250.8250 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		51.7812 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 116 / 85.9)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 116 / 85.9)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 136 / 85.9)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$



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Nozzle Calcs.: N1-Shell Inlet

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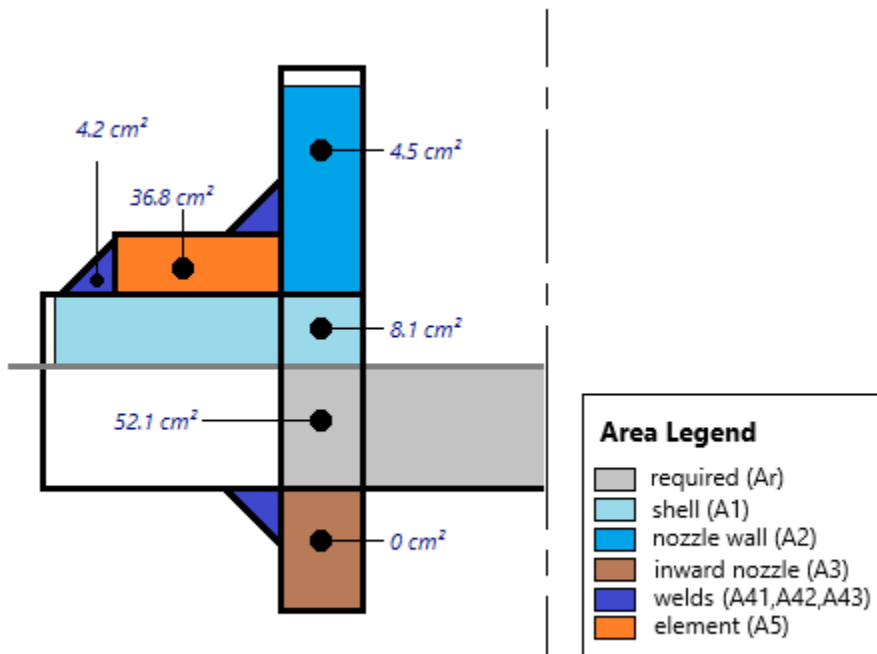


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	52.095	7.042	NA
Area in Shell	A1	8.103	46.114	NA
Area in Nozzle Wall	A2	4.456	10.574	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	4.240	4.240	NA
Area in Element	A5	36.838	36.838	NA
TOTAL AREA AVAILABLE	Atot	53.636	97.765	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Insufficient.

The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness

Based on given Pad Thickness: 420.1217 24.0000 mm

Based on given Pad Diameter: 426.5400 22.9964 mm

Based on the Estimated Diameter Limit: 500.0625 15.5486 mm

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (251 \cdot 20.8 \cdot 1 + 2 \cdot 11.1 \cdot 20.8 \cdot 1 \cdot (1 - 1))$$

$$= 52.095 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 251(1 \cdot 24 - 1 \cdot 20.8) - 2 \cdot 11.1$$

$$(1 \cdot 24 - 1 \cdot 20.8) \cdot (1 - 1)$$



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$$= 8.103 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 51.8) \cdot (11.1 - 6.81) \cdot 1$$

$$= 4.456 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= Wo^2 \cdot fr3 + (Wi-can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1 + 18^2 \cdot 1$$

$$= 4.240 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$$

$$= (427 - 273) \cdot 24 \cdot 1$$

$$= 36.838 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 6.8102 mm
Wall Thickness per UG16(b),	tr16b = 1.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 20.7696 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 20.7696 mm
Wall Thickness, shell/head, external pressure	trb2 = 0.3374 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 1.5000 mm
Wall Thickness per table UG-45	tb3 = 8.1026 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[8.1, \max(20.8, 1.5)]$$

$$= 8.1026 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(6.81, 8.1)$$

$$= 8.1026 \text{ mm}$$

Available Nozzle Neck Thickness = 11.1125 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 61.2,	Allowable	: 116.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 256.3 N./mm ²	Passed
Occasional	: 33.0,	Allowable	: 154.3 N./mm ²	Passed
Shear	: 18.3,	Allowable	: 81.2 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle to Flange Weld skipped as Nozzle is not a Carbon Steel material.

Nozzle-Pad Weld for Nozzle skipped, as Nozzle is not a Carbon Steel material.

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 11.1, tr = 5.7, c = 0 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.51, Temp. Reduction = 31 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B



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Govrn. thk, tg = 24, tr = 12.7, c = 0 mm, E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.53$, Temp. Reduction = 29 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-31 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Gov. MDMT of the Reinforcement Pad	: -31 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -31 °C

Weld Size Calculations, Description: N1-Shell Inlet

Intermediate Calc. for nozzle/shell Welds	Tmin	11.1125 mm
Intermediate Calc. for pad/shell Welds	TminPad	19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	9.5000 = 0.5*TminPad	12.7260 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (52.1 - 8.1 + 2 \cdot 11.1 \cdot 1 \cdot (1 \cdot 24 - 20.8)) \cdot 85.9)$
 $= 39163.27 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4 - (Wi - Can / .707)^2 \cdot fr2) \cdot Sv$
 $= (4.46 + 36.8 + 4.24 - 0 \cdot 1) \cdot 85.9$
 $= 39883.66 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (4.46 + 0 + 1 + (5.33)) \cdot 85.9$
 $= 9450.87 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (4.46 + 0 + 4.24 + 36.8 + (5.33)) \cdot 85.9$
 $= 44555.84 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.14/2.0) \cdot 273 \cdot 10 \cdot 0.49 \cdot 116$
 $= 24859. \text{ Kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.14/2.0) \cdot 427 \cdot 18 \cdot 0.49 \cdot 85.9$
 $= 51762. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.14 \cdot 131) \cdot (11.1 - 0) \cdot 0.7 \cdot 116$
 $= 37858. \text{ Kgf}$



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Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.14/2) \cdot 273 \cdot 24 \cdot 0.74 \cdot 136$
 $= 105636. \text{ Kg}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - c_{as}) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 273 \cdot (24 - 0) \cdot 0.74 \cdot 116$
 $= 90103. \text{ Kg}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (S_{PEW} + S_{NW}) = (51762 + 37858) = 89621 \text{ Kg}$

Failure Path [Path 2-2]:
 $= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$
 $= (24859 + 105636 + 90103 + 0) = 220598 \text{ Kg}$

Failure Path [Path 3-3]:
 $= (S_{pew} + T_{ngw} + S_{inw})$
 $= (51762 + 90103 + 0) = 141865 \text{ Kg}$

Summary of Failure Path Calculations:

Path 1-1 = 89620 Kg, must exceed W = 39163 Kg or W1 = 39883 Kg
 Path 2-2 = 220597 Kg, must exceed W = 39163 Kg or W2 = 9450 Kg
 Path 3-3 = 141865 Kg, must exceed W = 39163 Kg or W3 = 44555 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6181 KPa.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 35.5395 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 309.5395 mm

Input Echo, WRC297 Item 1, Description: N1-Shell Inlet:

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	0.0000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	560.000	mm	
Shell Thickness	24.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-240 304		
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²	
Vessel Hot S.I. Allowable	Smh 85.90	N./mm ²	
Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance	0.0000	mm	
Nozzle Diameter	250.825	mm	
Nozzle Thickness	11.1125	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-312 TP304		
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²	
Nozzle Hot S.I. Allowable	SNmh 116.00	N./mm ²	
Diameter of Reinforcing Pad	426.540	mm	
Thickness of Reinforcing Pad	24.0000	mm	

Note:
 External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.



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Design Internal Pressure	Dp	6100.00	KPa.
Radial Load	P	-2294.50	Kgf
Circumferential Shear	Vc	1657.15	Kgf
Longitudinal Shear	Vl	1657.15	Kgf
Circumferential Moment	Mc	1274.24	Kg-m.
Longitudinal Moment	Ml	1274.24	Kg-m.
Torsional Moment	Mt	1465.38	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 273.05 + 2 \cdot 1.65 \cdot \sqrt{(304.0(24.0 - 0.0))}$
 $= 554.925 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	608.000 mm
Nozzle Outside Diameter	(d) =	273.050 mm
Vessel Thickness used	(T) =	48.000 mm
Nozzle Thickness used	(t) =	11.113 mm
T / t	=	4.319
d / t	=	24.571

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.598$$

Nr/P	=	0.077
Mr/P	=	0.061
M0/P	=	0.110
N0/P	=	0.271
MrD/Mc	=	0.118
NrDL/Mc	=	0.114
M0d/Mc	=	0.242
N0DL/Mc	=	0.181
MrD/Ml	=	0.097
NrDL/Ml	=	0.097
M0D/Ml	=	0.192
N0DL/Ml	=	0.360

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-7	-7	7	7
Outplane Bending (Ml)	-22	22	22	-22
Normal Pressure Stress	32	38	32	38
Outplane Stress Summary	11	50	71	19

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	3	-3	3	-3
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-11	11	11	-11
Inplane Pressure Stress	16	16	16	16
Inplane Stress Summary	7	23	34	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	13	51	71	19

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	-3	-3	3	3
Outplane Bending (Mc)	-28	28	28	-28
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	16	16	16	16
Outplane Stress Summary	-6	37	57	-12

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	3	-3	3	-3
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-14	14	14	-14
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	32	38	32	38
Inplane Stress Summary	20	47	53	24

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0



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Torsional Shear (Mt)	2	2	2	2
Shear Stress Summary	1	1	3	3

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	28	48	59	37

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-7	-7	7	7
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	79	86	79	86
Hoop Stress Summary	74	81	88	96

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	25	-25	25	-25
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-21	-21	21	21
Axial Bending (ML)	-107	107	107	-107
Axial Pressure Stress	39	39	39	39
Axial Stress Summary	-61	102	196	-68

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	3	3	-3	-3
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	11	11	11	11
Shear Stress Summary	14	14	7	7

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	139	109	196	165

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-3	-3	3	3



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Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	79	86	79	86

Hoop Stress Summary	78	85	85	92
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	2	2	2	2
Axial Bending (P)	25	-25	25	-25
Axial Membrane (Mc)	-21	-21	21	21
Axial Bending (Mc)	-135	135	135	-135
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	39	39	39	39

Axial Stress Summary	-89	130	224	-96
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	11	11	11	11

Shear Stress Summary	7	7	14	14
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	168	131	225	191
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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 273.05 + 2 \cdot 1.65 \cdot \sqrt{292.0(24.0 - 0.0)}$$

$$= 549.306 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	584.000 mm
Nozzle Outside Diameter	(d) =	426.540 mm
Vessel Thickness used	(T) =	24.000 mm
Nozzle Thickness used	(t) =	87.857 mm
T / t	=	0.273
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.603$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.054
Mr/P	=	0.089
M0/P	=	0.029
N0/P	=	0.063
MrD/Mc	=	0.259
NrDL/Mc	=	0.092



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MØd/Mc = 0.078
 NØDL/Mc = 0.085
 MrD/Ml = 0.076
 NrDL/Ml = 0.052
 MØD/Ml = 0.022
 NØDL/Ml = 0.070

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-3	-3	3	3
Outplane Bending (ML)	-6	6	6	-6
Normal Pressure Stress	68	74	68	74
Outplane Stress Summary	67	73	87	66

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	20	-20	20	-20
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-23	23	23	-23
Inplane Pressure Stress	34	34	34	34
Inplane Stress Summary	31	35	83	-5

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	67	73	88	72

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	-4	-4	4	4
Outplane Bending (Mc)	-23	23	23	-23
Outplane Membrane (ML)	0	0	0	0



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Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	34	34	34	34

Outplane Stress Summary	15	49	71	10
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	20	-20	20	-20
Inplane Membrane (Mc)	-4	-4	4	4
Inplane Bending (Mc)	-79	79	79	-79
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	68	74	68	74

Inplane Stress Summary	7	129	175	-18
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	1	1	3	3
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	15	129	175	30
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WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	32	38	32	38	32	38	32	38
Circ. Pl (SUS)	-5	-5	9	9	-2	-2	2	2
Circ. Q (SUS)	-16	16	28	-28	-11	11	17	-17
Long. Pm (SUS)	16	16	16	16	16	16	16	16
Long. Pl (SUS)	-1	-1	1	1	-1	-1	5	5
Long. Q (SUS)	-8	8	14	-14	-22	22	34	-34
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	32.0	38.0	32.0	38.0	32.0	38.0	32.0	38.0
Pm+Pl (SUS)	27.0	33.0	41.0	47.0	30.0	36.0	34.0	40.0
Pm+Pl+Q (Total)	11.8	49.2	69.1	19.2	26.3	47.4	55.8	36.2



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Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	38.00	85.90	Passed
Pm+Pl (SUS)	47.00	128.85	Passed
Pm+Pl+Q (TOTAL)	69.10	257.70	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	68	74	68	74	68	74	68	74
Circ. Pl (SUS)	-1	-1	5	5	-2	-2	6	6
Circ. Q (SUS)	0	0	12	-12	-59	59	99	-99
Long. Pm (SUS)	34	34	34	34	34	34	34	34
Long. Pl (SUS)	0	0	4	4	-2	-2	6	6
Long. Q (SUS)	-3	3	43	-43	-17	17	29	-29
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	68.0	74.0	68.0	74.0	68.0	74.0	68.0	74.0
Pm+Pl (SUS)	67.0	73.0	73.0	79.0	66.0	72.0	74.0	80.0
Pm+Pl+Q (Total)	67.2	73.2	85.2	72.0	15.1	131.0	173.1	30.6

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	74.00	85.90	Passed
Pm+Pl (SUS)	80.02	128.85	Passed
Pm+Pl+Q (TOTAL)	173.09	257.70	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	79	86	79	86	79	86	79	86
Circ. Pl (SUS)	-5	-5	9	9	-1	-1	5	5
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	39	39	39	39	39	39	39	39
Long. Pl (SUS)	-19	-19	23	23	-19	-19	23	23
Long. Q (SUS)	-82	82	132	-132	-110	110	160	-160



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Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	11	11	11	11	11	11	11	11
Pm (SUS)	79.0	86.0	79.0	86.0	79.0	86.0	79.0	86.0
Pm+Pl (SUS)	74.2	81.1	88.3	95.3	78.2	85.1	84.4	91.3
Pm+Pl+Q (Total)	138.9	109.0	194.6	165.8	168.8	131.4	223.4	191.1

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	86.00	116.00	Passed
Pm+Pl (SUS)	95.27	174.00	Passed
Pm+Pl+Q (TOTAL)	223.41	348.00	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: N2-Shell outlet

From: 100

Pressure for Reinforcement Calculations	P	6105.420	KPa.
Temperature for Internal Pressure	Temp	300	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	300	°C
Parent Material		SA-240	304
Parent Allowable Stress at Temperature	Sv	85.90	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	560.00	mm
Design Length of Section	L	9220.0000	mm
Shell Finished (Minimum) Thickness	t	24.0000	mm
Shell Internal Corrosion Allowance	c	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		9680.97	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

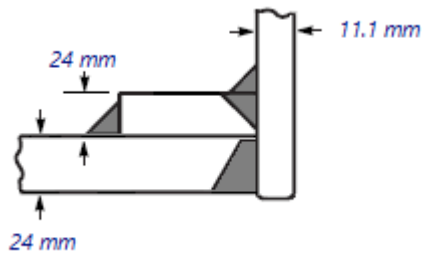
Type of Element Connected to the Parent : Nozzle

Material		SA-312	TP304
Material UNS Number		S30400	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	116.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		10.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		60	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516	70
Pad Allowable Stress at Temperature	Sp	136.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	426.5400	mm
Thickness of Pad	te	24.0000	mm
Weld leg size between Pad and Shell	Wp	18.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	24.0000	mm
Reinforcing Pad Width		76.7450	mm
Flange Class		900	
Flange Grade		GR 2.1	

The Pressure Design option was Design Pressure + static head.



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Reinforcement CALCULATION, Description: N2-Shell outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 9.875 in.
 Actual Thickness Used in Calculation 0.438 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6105 \cdot 280 / (85.9 \cdot 1 - 0.6 \cdot 6105)$
 $= 20.7889 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6105 \cdot 125 / (116 \cdot 1 - 0.6 \cdot 6105)$
 $= 6.8164 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.9026 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	501.6500	mm
Parallel to Vessel Wall, opening length	d	250.8250	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		51.7812	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 116 / 85.9)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 116 / 85.9)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 136 / 85.9)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$



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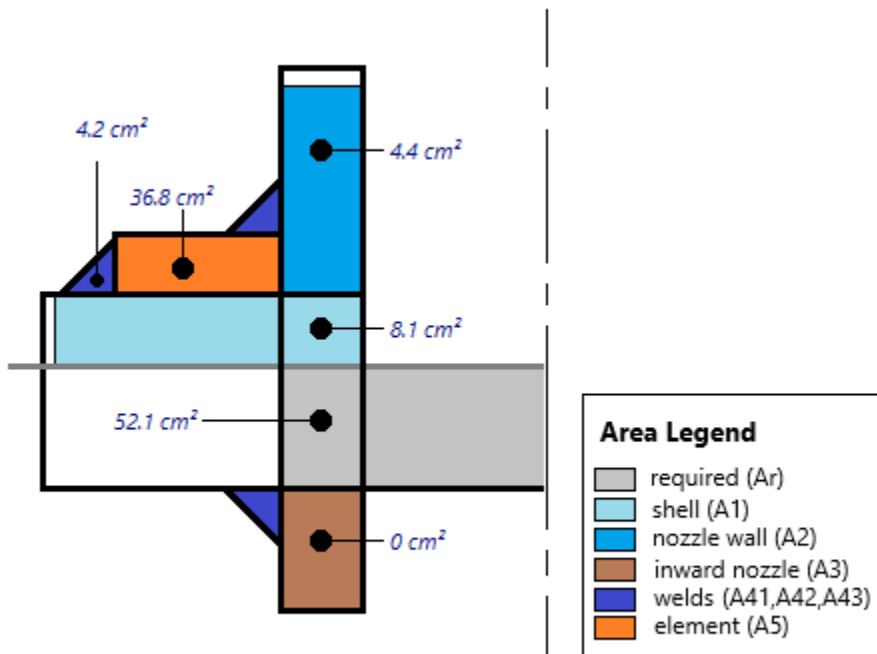


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	52.144	7.042	NA
Area in Shell	A1	8.054	46.114	NA
Area in Nozzle Wall	A2	4.449	10.574	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	4.240	4.240	NA
Area in Element	A5	36.838	36.838	NA
TOTAL AREA AVAILABLE	Atot	53.581	97.765	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Insufficient.

The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
Based on given Pad Thickness: 420.5516 24.0000 mm
Based on given Pad Diameter: 426.5400 23.0636 mm
Based on the Estimated Diameter Limit: 500.0625 15.5940 mm

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (251 \cdot 20.8 \cdot 1 + 2 \cdot 11.1 \cdot 20.8 \cdot 1 \cdot (1 - 1))$$

$$= 52.144 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 251(1 \cdot 24 - 1 \cdot 20.8) - 2 \cdot 11.1$$

$$(1 \cdot 24 - 1 \cdot 20.8) \cdot (1 - 1)$$



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= 8.054 cm²

Area Available in Nozzle Wall Projecting Outward [A2]:

= (2 • Tlwp) • (tn - trn) • fr2
 = (2 • 51.8) • (11.1 - 6.82) • 1
 = 4.449 cm²

Area Available in Welds [A41 + A42 + A43]:

= Wo² • fr3 + (Wi-can/0.707)² • fr2 + Wp² • fr4
 = 10² • 1 + (0)² • 1 + 18² • 1
 = 4.240 cm²

Area Available in Element [A5]:

= (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4
 = (427 - 273) • 24 • 1
 = 36.838 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 6.8164 mm
Wall Thickness per UG16(b),	tr16b = 1.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 20.7889 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 20.7889 mm
Wall Thickness, shell/head, external pressure	trb2 = 0.3374 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 1.5000 mm
Wall Thickness per table UG-45	tb3 = 8.1026 mm

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]
 = min[8.1, max(20.8, 1.5)]
 = 8.1026 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
 = max(6.82, 8.1)
 = 8.1026 mm

Available Nozzle Neck Thickness = 11.1125 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained :	61.2,	Allowable :	116.0 N./mm ²	Passed
Expansion :	0.0,	Allowable :	256.3 N./mm ²	Passed
Occasional :	33.0,	Allowable :	154.3 N./mm ²	Passed
Shear :	18.3,	Allowable :	81.2 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle to Flange Weld skipped as Nozzle is not a Carbon Steel material.

Nozzle-Pad Weld for Nozzle skipped, as Nozzle is not a Carbon Steel material.

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 11.1, tr = 5.7, c = 0 mm, E* = 1
 Thickness Ratio = tr • E*/(tg - c) = 0.51, Temp. Reduction = 31 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B



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Govrn. thk, tg = 24, tr = 12.7, c = 0 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.53$, Temp. Reduction = 29 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-31 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Gov. MDMT of the Reinforcement Pad	: -31 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -31 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $6105.42 / 14910.00 = 0.409$

Weld Size Calculations, Description: N2-Shell outlet

Intermediate Calc. for nozzle/shell Welds	Tmin	11.1125 mm
Intermediate Calc. for pad/shell Welds	TminPad	19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$
Pad Weld	$9.5000 = 0.5 \cdot T_{\text{minPad}}$	$12.7260 = 0.7 \cdot W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A_1 + 2 \cdot t_n \cdot f_r \cdot (E_1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (52.1 - 8.05 + 2 \cdot 11.1 \cdot 1 \cdot (1 \cdot 24 - 20.8)) \cdot 85.9)$
 $= 39244.22 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A_2 + A_5 + A_4 - (W_i - C_{an} / 707)^2 \cdot f_r^2) \cdot S_v$
 $= (4.45 + 36.8 + 4.24 - 0 \cdot 1) \cdot 85.9$
 $= 39877.99 \text{ Kg}$

Weld Load [W2]:
 $= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot f_r)) \cdot S_v$
 $= (4.45 + 0 + 1 + (5.33)) \cdot 85.9$
 $= 9445.20 \text{ Kg}$

Weld Load [W3]:
 $= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot f_r)) \cdot S$
 $= (4.45 + 0 + 4.24 + 36.8 + (5.33)) \cdot 85.9$
 $= 44550.18 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 273 \cdot 10 \cdot 0.49 \cdot 116$
 $= 24859. \text{ Kg}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot S_{EW}$



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$$= (3.14/2.0) \cdot 427 \cdot 18 \cdot 0.49 \cdot 85.9$$

$$= 51762. \text{ Kg}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 131) \cdot (11.1 - 0) \cdot 0.7 \cdot 116$$

$$= 37858. \text{ Kg}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.14/2) \cdot 273 \cdot 24 \cdot 0.74 \cdot 136$$

$$= 105636. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 273 \cdot (24 - 0) \cdot 0.74 \cdot 116$$

$$= 90103. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (51762 + 37858) = 89621 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (24859 + 105636 + 90103 + 0) = 220598 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (51762 + 90103 + 0) = 141865 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 89620 Kg, must exceed W = 39244 Kg	or W1 = 39877 Kg
Path 2-2 = 220597 Kg, must exceed W = 39244 Kg	or W2 = 9445 Kg
Path 3-3 = 141865 Kg, must exceed W = 39244 Kg	or W3 = 44550 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6175 KPa.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 35.5395 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 309.5395 mm

Input Echo, WRC297 Item 1, Description: N2-Shell outle:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	0.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	560.000	mm
Shell Thickness	24.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-240 304	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 85.90	N./mm ²

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	0.0000	mm
Nozzle Diameter	250.825	mm
Nozzle Thickness	11.1125	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-312 TP304	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²



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Nozzle Hot S.I. Allowable	SNmh	116.00	N./mm ²
Diameter of Reinforcing Pad		426.540	mm
Thickness of Reinforcing Pad		24.0000	mm

Note:
External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	6105.42	KPa.
Radial Load	P	-2294.50	Kgf
Circumferential Shear	Vc	1657.15	Kgf
Longitudinal Shear	Vl	1657.15	Kgf
Circumferential Moment	Mc	1274.24	Kg-m.
Longitudinal Moment	Ml	1274.24	Kg-m.
Torsional Moment	Mt	1465.38	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$
 $= 273.05 + 2 \cdot 1.65 \cdot \sqrt{304.0(24.0 - 0.0)}$
 $= 554.925 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	608.000 mm
Nozzle Outside Diameter	(d) =	273.050 mm
Vessel Thickness used	(T) =	48.000 mm
Nozzle Thickness used	(t) =	11.113 mm
T / t	=	4.319
d / t	=	24.571

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.598$$

Nr/P	=	0.077
Mr/P	=	0.061
M0/P	=	0.110
N0/P	=	0.271
MrD/Mc	=	0.118
NrDL/Mc	=	0.114
M0d/Mc	=	0.242
N0DL/Mc	=	0.181
MrD/Ml	=	0.097
NrDL/Ml	=	0.097
M0D/Ml	=	0.192
N0DL/Ml	=	0.360

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-7	-7	7	7
Outplane Bending (Ml)	-22	22	22	-22
Normal Pressure Stress	32	38	32	38



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Outplane Stress Summary	11	50	71	19
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	3	-3	3	-3
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-11	11	11	-11
Inplane Pressure Stress	16	16	16	16

Inplane Stress Summary	7	23	34	3
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2

Shear Stress Summary	3	3	1	1
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	13	51	71	19
------------------------	----	----	----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	-3	-3	3	3
Outplane Bending (Mc)	-28	28	28	-28
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	16	16	16	16

Outplane Stress Summary	-6	37	57	-12
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	3	-3	3	-3
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-14	14	14	-14
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	32	38	32	38

Inplane Stress Summary	20	47	53	24
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			0	0	0	0
Torsional Shear (Mt)			2	2	2	2
Shear Stress Summary			1	1	3	3

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress			28	48	59	37

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)			Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)			2	2	2	2
Hoop Bending (P)			0	0	0	0
Hoop Membrane (Mc)			0	0	0	0
Hoop Bending (Mc)			0	0	0	0
Hoop Membrane (ML)			-7	-7	7	7
Hoop Bending (ML)			0	0	0	0
Hoop Pressure Stress			79	86	79	86
Hoop Stress Summary			74	82	88	96

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)			Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)			2	2	2	2
Axial Bending (P)			25	-25	25	-25
Axial Membrane (Mc)			0	0	0	0
Axial Bending (Mc)			0	0	0	0
Axial Membrane (ML)			-21	-21	21	21
Axial Bending (ML)			-107	107	107	-107
Axial Pressure Stress			39	39	39	39
Axial Stress Summary			-61	102	196	-68

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)			Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)			3	3	-3	-3
Shear due to (Vl)			0	0	0	0
Shear due to Torsion			11	11	11	11
Shear Stress Summary			14	14	7	7

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)			Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress			139	109	196	165



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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-3	-3	3	3
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	79	86	79	86
Hoop Stress Summary	78	85	85	92

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	25	-25	25	-25
Axial Membrane (Mc)	-21	-21	21	21
Axial Bending (Mc)	-135	135	135	-135
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	39	39	39	39
Axial Stress Summary	-89	130	224	-96

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	11	11	11	11
Shear Stress Summary	7	7	14	14

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	168	131	225	191

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 273.05 + 2 \cdot 1.65 \cdot \sqrt{292.0(24.0 - 0.0)}$$

$$= 549.306 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	584.000 mm
Nozzle Outside Diameter	(d) =	426.540 mm
Vessel Thickness used	(T) =	24.000 mm
Nozzle Thickness used	(t) =	87.857 mm
T / t	=	0.273
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.603$$



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Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.054
 Mr/P = 0.089
 MØ/P = 0.029
 NØ/P = 0.063
 MrD/Mc = 0.259
 NrDL/Mc = 0.092
 MØd/Mc = 0.078
 NØDL/Mc = 0.085
 MrD/Ml = 0.076
 NrDL/Ml = 0.052
 MØD/Ml = 0.022
 NØDL/Ml = 0.070

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-3	-3	3	3
Outplane Bending (ML)	-6	6	6	-6
Normal Pressure Stress	68	74	68	74
Outplane Stress Summary	67	73	87	66

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	20	-20	20	-20
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-23	23	23	-23
Inplane Pressure Stress	34	34	34	34
Inplane Stress Summary	31	35	83	-5

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	67	73	88	72

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	-4	-4	4	4
Outplane Bending (Mc)	-23	23	23	-23
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	34	34	34	34
Outplane Stress Summary	15	49	71	10

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	20	-20	20	-20
Inplane Membrane (Mc)	-4	-4	4	4
Inplane Bending (Mc)	-79	79	79	-79
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	68	74	68	74
Inplane Stress Summary	7	129	175	-18

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	2	2	2	2
Shear Stress Summary	1	1	3	3

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	15	129	175	29

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	32	38	32	38	32	38	32	38
Circ. Pl (SUS)	-5	-5	9	9	-2	-2	2	2
Circ. Q (SUS)	-16	16	28	-28	-11	11	17	-17
Long. Pm (SUS)	16	16	16	16	16	16	16	16
Long. Pl (SUS)	-1	-1	1	1	-1	-1	5	5
Long. Q (SUS)	-8	8	14	-14	-22	22	34	-34
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0



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Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	32.0	38.0	32.0	38.0	32.0	38.0	32.0	38.0
Pm+Pl (SUS)	27.0	33.0	41.0	47.0	30.0	36.0	34.0	40.0
Pm+Pl+Q (Total)	11.8	49.2	69.1	19.2	26.3	47.4	55.8	36.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	38.00	85.90	Passed
Pm+Pl (SUS)	47.00	128.85	Passed
Pm+Pl+Q (TOTAL)	69.10	257.70	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	68	74	68	74	68	74	68	74
Circ. Pl (SUS)	-1	-1	5	5	-2	-2	6	6
Circ. Q (SUS)	0	0	12	-12	-59	59	99	-99
Long. Pm (SUS)	34	34	34	34	34	34	34	34
Long. Pl (SUS)	0	0	4	4	-2	-2	6	6
Long. Q (SUS)	-3	3	43	-43	-17	17	29	-29
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	68.0	74.0	68.0	74.0	68.0	74.0	68.0	74.0
Pm+Pl (SUS)	67.0	73.0	73.0	79.0	66.0	72.0	74.0	80.0
Pm+Pl+Q (Total)	67.2	73.2	85.2	72.0	15.1	131.0	173.1	30.6

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	74.00	85.90	Passed
Pm+Pl (SUS)	80.02	128.85	Passed
Pm+Pl+Q (TOTAL)	173.09	257.70	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl



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Circ. Pm (SUS)	79	86	79	86	79	86	79	86
Circ. Pl (SUS)	-5	-5	9	9	-1	-1	5	5
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	39	39	39	39	39	39	39	39
Long. Pl (SUS)	-19	-19	23	23	-19	-19	23	23
Long. Q (SUS)	-82	82	132	-132	-110	110	160	-160
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	11	11	11	11	11	11	11	11
Pm (SUS)	79.0	86.0	79.0	86.0	79.0	86.0	79.0	86.0
Pm+Pl (SUS)	74.2	81.1	88.3	95.3	78.2	85.1	84.4	91.3
Pm+Pl+Q (Total)	138.9	109.0	194.6	165.8	168.8	131.4	223.4	191.1

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	86.00	116.00	Passed
Pm+Pl (SUS)	95.27	174.00	Passed
Pm+Pl+Q (TOTAL)	223.41	348.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input Echo, Tubesheet Number 1, Description:

Shell Data:

Main Shell Description: Main Shell

Shell Maximum Design Pressure	Psd,max	6100.00 KPa.
Shell Maximum Operating Pressure	Psox,max	6100.00 KPa.
Shell Minimum Operating Pressure	Psox,min	0.00 KPa.
Shell Thickness	ts	24.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	560.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	300.00 °C
Shell Material		SA-240 304

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		S30400
Shell Allowable Stress at Temperature	Ss	85.90 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	6100.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	6100.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	24.0000 mm
Channel Internal Corrosion Allowance	cac	1.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	560.000 mm
Channel Design Temperature	TEMPC	300.00 °C
Channel Flange Matl		SA-240 304

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		S30400
Channel Allowable Stress at Temperature	Sc	85.90 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	900
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	12.7000 mm
Total Straight Tube Length	Lt	9600.00 mm
Straight Tube Length (bet. inner tubsht faces) L		9220.00 mm
Design Temperature of the Tubes		300.00 °C
Tube Material		SA-213 TP304L
Tube Material UNS Number		S30403
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		97.70 N./mm ²
Tube Allowable Stress At Ambient		115.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	108.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	15.8750 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	1.6500 mm
Groove Weld Leg	ag	1.6500 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	262.030 mm



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Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	8.0000 mm
Area of Tube Layout	Ap	3.6 cm ²

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf B

Tubesheet Design Metal Temperature	T	300.00 °C
Tubesheet Material		SA-965 F304

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		S30400
Tubesheet Allowable Stress at Temperature	S	85.90 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	190.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	13.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	13.0000 mm
Tubesheet Outside Diameter	A	824.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped?		No
Area of the Untubed Lanes	AL	0.0 cm ²

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	327.000 mm
Tube End condition corresponding to Span (l)	k	1.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Mean Shell Metal Temp. along Shell len.	Tsm	0.00 °C
Mean Tube Metal Temp. along Tube length	Ttm	0.00 °C
Junction Stress Reduction option		Perform Plastic Calculation

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Channel
Flange Outside Diameter	A	824.000 mm
Flange Inside Diameter	B	564.000 mm
Flange Face Outside Diameter	Fod	622.000 mm
Flange Face Inside Diameter	Fid	564.000 mm
Gasket Outside Diameter	Go	611.000 mm
Gasket Inside Diameter	Gi	585.000 mm
Small end Hub thk.	g0	24.0000 mm
Large end Hub thk.	g1	33.0000 mm
Gasket Factor,	m	3.00
Gasket Design Seating Stress	y	68.95 N./mm ²
Flange Facing Sketch		Code Sketch 1a
Gasket Thickness	tg	3.1750 mm
Full face Gasket Flange Option		Program Selects
Length of Partition Gasket	lp	550.000 mm
Width of Partition Gasket	wp	10.0000 mm
Partition Gasket Factor,	mPart	3.0000
Partition Gasket Design Seating Stress	yPart	68.95 N./mm ²
Partition Gasket Facing Sketch		Code Sketch 1a
Partition Gasket Column for Gasket Seating		Code Column II

Bolting Information:

Diameter of Bolt Circle	C	739.000 mm
Nominal Bolt Diameter	dB	38.1000 mm
Thread Series		TEMA Thread Series



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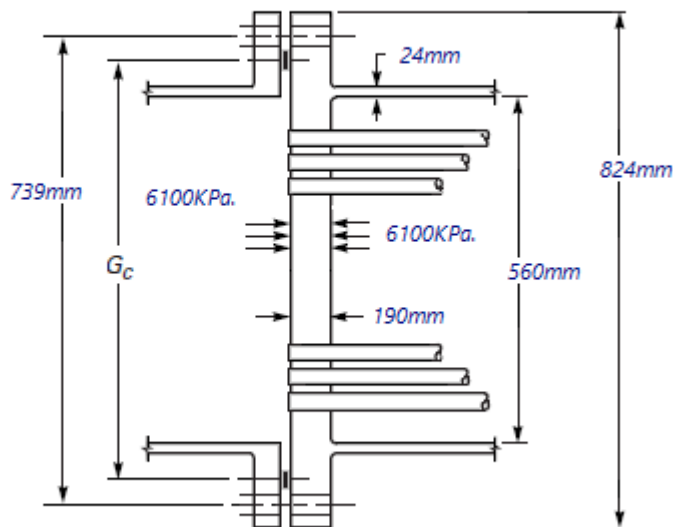
Number of Bolts	n	24
Bolt Material		SA-193 B7
Bolt Allowable Stress At Temperature	Sb	172.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00 N./mm ²
Weld between Flange and Shell/Channel		0.0000 mm

Note: The full bolt load Sa*Ab, was used for all load cases.

Tubesheet Integral with	Shell
Tubesheet Extended as Flange	Yes
Thickness of Extended Portion of Tubesheet	Tf 154.0000 mm
Is Bolt Load Transferred to the Tubesheet	Yes

Is Exchanger in Creep range (skip EP, Use 3S for Sps) No

Tubesheet Integral w/Shell, Gasketed w/Channel, Extended as a Flange



Configuration b:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Austenitic Group 3 Steels
 Shell - TM-1 Material Group G

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000152988 /°C
Elastic Mod. at Design Temperature	300.0 °C	0.17559E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000152988 /°C
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.19650E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.

Channel - TE-1 Austenitic Group 3 Steels
 Channel - TM-1 Material Group G

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000152988 /°C
Elastic Mod. at Design Temperature	300.0 °C	0.17559E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.

Tubes - TE-1 Austenitic Group 3 Steels
 Tubes - TM-1 Material Group G



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Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000152988 /°C
Elastic Mod. at Design Temperature	300.0 °C	0.17559E+09 KPa.
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.19650E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	300.0 °C	0.17559E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.

TubeSheet - TE-1 Austenitic Group 3 Steels
 TubeSheet - TM-1 Material Group G

Elastic Mod. at Design Temperature	300.0 °C	0.17559E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (6100 \cdot (12.7/2 - 0)) / (97.7 \cdot 1 + 0.4 \cdot 6100)$
 $= 0.3868 + 0.0000 = 0.3868 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart HA-3 at 300.00 °C
 Elastic Modulus for Material 170271248.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	12.70	9220.00	6.02	50.0000	0.0303634	60.23

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 16666 \text{ KPa.}$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.1766	12.70	9220.00	10.79	50.0000	0.0094423	49.39

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 49.4) / (3 \cdot 10.8) = 6101 \text{ KPa.}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.1766 mm
Allowable Internal Pressure at Corroded thickness	37438.61 KPa.
Required Internal Design Pressure	6100.00 KPa.
Allowable External Pressure at Corroded thickness	16666.02 KPa.
Required External Design Pressure	6100.00 KPa.
Required Thickness due to Shell Side pressure	1.1766 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (G_{oc} - G_{ic}) / 2$	13.000 mm
Basic Gasket Width,	$b_0 = N / 2$	6.500 mm
Effective Gasket Width,	$b = \text{sqrt}(b_0) \cdot 2.5$	6.422 mm
Gasket Reaction Diameter,	$G = G_o - 2.0 \cdot b$	598.156 mm

Results for ASME Fixed Tubesheet Calculations for Configuration b:

Results for Tubesheet Calculations Original Thickness:



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ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio $[\rho]$:

$$= l_{tx} / h \text{ (modified for corrosion if present)}$$

$$= 0/190$$

$$= 0.0000 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole diameter $[d^*]$:

$$= \max(dt - 2 \cdot tt(Et/E) \cdot (StT/S) \cdot (\rho); dt - 2 \cdot tt)$$

$$= \max(12.7 - 2 \cdot 2.11 (175593552/175593552) \cdot (0); dt - 2 \cdot 2.11)$$

$$= 12.7000 \text{ mm}$$

Equivalent outer tube limit circle diameter $[Do]$:

$$= 2 \cdot r_o + dt$$

$$= 2 \cdot 262 + 12.7$$

$$= 536.760 \text{ mm}$$

Determine the basic ligament efficiency for shear $[\mu]$:

$$= (p - dt) / p$$

$$= (15.9 - 12.7) / 15.9$$

$$= 0.200$$

Equivalent outer tube limit radius $[ao]$:

$$= Do / 2$$

$$= 537/2$$

$$= 268.3800 \text{ mm}$$

Effective tube pitch $[p^*]$:

$$= p / \sqrt{ (1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) }$$

$$= 15.9 / \sqrt{ (1 - 4 \cdot \min(0 \cdot 100 ; 4 \cdot 537 \cdot 15.9) / (\pi \cdot 537^2)) }$$

$$= 15.8750 \text{ mm}$$

Effective ligament efficiency for bending $[\mu^*]$:

$$= (p^* - d^*) / p^*$$

$$= (15.9 - 12.7) / 15.9$$

$$= 0.2000$$

Ratio $[ps]$:

$$= as / ao$$

$$= 280/268$$

$$= 1.0433$$

Ratio $[pc]$:

$$= ac / ao$$

$$= 299/268$$

$$= 1.1144$$

Compute parameter $[xt]$:

$$= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot ao))^2$$

$$= 1 - 900 ((12.7 - 2 \cdot 2.11) / (2 \cdot 268))^2$$

$$= 0.7754$$

Determine parameter $[xs]$:

$$= 1 - Nt(dt / (2 \cdot ao))^2$$

$$= 1 - 900 (12.7 / (2 \cdot 268))^2$$

$$= 0.4962$$

Determine the value $[h'g]$:

$$= \max((h_g - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((8 - 0); 0)$$

$$= 8.000 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:



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Axial Shell Stiffness [Ks]:

$$= \pi \cdot ts (Ds + ts) Es / L$$

$$= \pi \cdot 24 (560 + 24) 175593552 / 9220$$

$$= 85516.1562 \text{ Kg/mm}$$

Axial Tube Stiffness [Kt]:

$$= \pi \cdot tt (Dt - tt) Et / L$$

$$= \pi \cdot 2.11 (12.7 - 2.11) 175593552 / 9220$$

$$= 136.3335 \text{ Kg/mm}$$

Stiffness Factor [Ks,t]:

$$= Ks / (Nt \cdot Kt)$$

$$= 85516 / (900 \cdot 136)$$

$$= 0.69695$$

Rigidity Ratio [J]:

$$= 1 / (1 + Ks/Kt)$$

$$= 1 / (1 + 85516/0)$$

$$= 1.0000 (= 1 \text{ if No Exp. Jt.})$$

Shell coefficient [β_s]:

$$= ((12(1 - \nu_s^2))^{1/4}) / ((Ds + ts)ts)$$

$$= ((12(1 - 0.31^2))^{1/4}) / ((560 + 24)24)$$

$$= 0.0153 \text{ 1/mm}$$

Shell coefficient [ks]:

$$= \beta_s \cdot Es \cdot ts^3 / (6(1 - \nu_s^2))$$

$$= 0.015 \cdot 175593552 \cdot 24^3 / (6(1 - 0.31^2))$$

$$= 6860946944.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(Ds + ts) \cdot ks / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(560 + 24) \cdot 6.86E+09 / 190^3 \cdot (1 + 190 \cdot 0.015 + (190 \cdot 0.015)^2 / 2)$$

$$= 28579216.0000 \text{ KPa}$$

Shell coefficient [δ_s]:

$$= Ds^2 / (4 \cdot Es \cdot ts) \cdot (1 - Ds \cdot \nu_s / (2(Ds + ts)))$$

$$= 560^2 / (4 \cdot 175593552 \cdot 24) \cdot (1 - 560 \cdot 0.31 / (2(560 + 24)))$$

$$= 0.15838E-01 \text{ mm/N./mm}^2$$

Intermediate parameters for Tubesheet Gasketed on the Channel Side:

β_C , k_C , λ_C , δ_C are all = 0

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$h/p = 11.968504$; $\mu^* = 0.200000$; $E^* = 27278308 \text{ KPa}$.
 $E^*/E = 0.155349$; $\nu^* = 0.458572$

Note: As h/p (11.969) is > 2, data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor [X_a]:

$$= ((24(1 - \nu^{*2})Nt \cdot Et \cdot tt(dt - tt)ao^2) / (E^* \cdot L \cdot H^3))^{1/4}$$

$$= ((24(1 - 0.46^2)900 \cdot 175593552 \cdot 2.11$$

$$(12.7 - 2.11)268^2) / (27278308 \cdot 9220 \cdot 190^3))^{1/4}$$

$$= 1.2929$$

Values from Table 4.18.3

$$Z_d = 0.767414 ; Z_v = 0.166877 ; Z_m = 0.672338$$

$$Z_a = 0.487766E+00 ; Z_w = 0.166877$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$= A / D_o$$



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$$= 824/537$$

$$= 1.5351$$

Compute coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.46) / 27278308 \cdot (28579216 + 0 + 175593552 \cdot \ln(1.54))$$

$$= 2.0611$$

Compute parameter [ϕ]:

$$= (1 + \nu^*) F$$

$$= (1 + 0.46) 2.06$$

$$= 3.0062$$

Compute parameter [Q1]:

$$= (\psi_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$$

$$= (1.04 - 1 - 3.01 \cdot 0.17) / (1 + 3.01 \cdot 0.67)$$

$$= -0.151719$$

Compute Parameter [Qz1]:

$$= (Z_d + Q1 \cdot Z_w) / 2 \cdot X_a^4$$

$$= (0.77 + -0.15 \cdot 0.17) / 2 \cdot 1.29^4$$

$$= 1.0369$$

Compute parameter [Qz2]:

$$= (Z_v + Q1 \cdot Z_m) / 2 \cdot X_a^4$$

$$= (0.17 + -0.15 \cdot 0.67) / 2 \cdot 1.29^4$$

$$= 0.0906$$

Compute parameter [U]:

$$= (Z_w + (\psi_s - 1) Z_m) X_a^4 / (1 + \phi \cdot Z_m)$$

$$= (0.17 + (1.04 - 1) 0.67) 1.29^4 / (1 + 3.01 \cdot 0.67)$$

$$= 0.1813$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [γ_b]:

$$= (G_c - C) / D_o \text{ (config b)}$$

$$= (598 - 739) / 537 = -0.26$$

Compute Parameter [γ]:

$$= 0.000 \text{ mm (For Pressure only cases)}$$

Calculate parameter [ω_s]:

$$= \psi_s \cdot k_s \cdot \beta_s \cdot \delta S (1 + h \cdot \beta_s)$$

$$= 1.04 \cdot 6.86E+09 \cdot 0.015 \cdot 0.015837597 (1 + 190 \cdot 0.015)$$

$$= 6799.4673 \text{ mm}^2$$

Calculate parameter [ω^*s]:

$$= a o^2 (\psi_s^2 - 1) \cdot (\psi_s - 1) / 4 - \omega_s$$

$$= 268^2 (1.04^2 - 1) \cdot (1.04 - 1) / 4 - 6799$$

$$= -6730.4937 \text{ mm}^2$$

Parameter [ω_c]:

$$= \rho_c \cdot k_c [\delta C (C_m \theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \zeta c (C_m \theta + h \cdot C_q \theta \cdot h \cdot \beta C / 2)]$$

$$= 1.11 \cdot 0 [0.000 (0 + 0 \cdot 0 \cdot 190) - 0.000 (0/190 + 0 \cdot 0/2)]$$

$$= 0.0000 \text{ mm}^2$$

Parameter [ω^*c]:

$$= a o^2 [(\rho_c^2 + 1) \cdot (\rho_c - 1) / 4 - (\psi_s - 1) / 2] - \omega_c$$

$$= 268^2 [(1.11^2 + 1) \cdot (1.11 - 1) / 4 - (1.04 - 1) / 2] - 0$$

$$= 3058.1724 \text{ mm}^2$$



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Pressure [P*S]:
 = 0 For Pressure only cases or Configurations d,e,f,A,B,C,D

Pressure [P*C]:
 = 0 For Pressure only cases or Configurations b,c,d,B,C,D

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [P's]:

$$= P_s \{ x_s + 2(1 - x_s) v_t + [2/K_{st} \cdot (D_s/D_o)^2] v_s - [(p_s^2 - 1)/(J \cdot K_{st})] - [(1 - J)/(2J \cdot K_{st})][(D_j^2 - (D_s)^2)/D_o^2] \}$$

$$= 6100 \{ 0.5 + 2(1 - 0.5)0.31 + [2/0.7 \cdot (560/537)^2]0.31 - [(1.04^2 - 1)/(1 \cdot 0.7)] - [(1 - 1)/(2 \cdot 1 \cdot 0.7)][(0^2 - (560)^2)/537^2] \}$$

 = 10064.3506 KPa.

Pressure [P't]:

$$= [x_t + 2(1 - x_t) v_t + 1/(J \cdot K_{st})] P_t$$

$$= [0.78 + 2(1 - 0.78)0.31 + 1/(1 \cdot 0.7)] 6100$$

 = 14331.6885 KPa.

Pressure [P_γ]:

$$= N_t \cdot K_t \cdot \gamma / (\pi \cdot a_o^2)$$

$$= 900 \cdot 136 \cdot 0 / (3.14 \cdot 268^2)$$

 = 0.0000 KPa.

Pressure [P_w]:

$$= -\gamma_b \cdot U \cdot W^* / (2 \cdot \pi \cdot a_o^2)$$

$$= -0.26 \cdot 0.18 \cdot 381546 / (2 \cdot \pi \cdot 268^2)$$

 = 393.2560 KPa.

Calculate the pressure [Prim]:

$$= -(U/a_o^2) \cdot (\omega^* s \cdot P_s - \omega^* c \cdot P_t)$$

$$= -(0.18/268^2) \cdot (-6730 \cdot 6100 - 3058 \cdot 6100)$$

 = 150.2791 KPa.

Calculate the pressure [P_ω]:

$$= (U/a_o^2) \cdot (\omega_s \cdot P^* s - \omega_c \cdot P^* c)$$

$$= (0.18/268^2) \cdot (6799 \cdot 0 - 0 \cdot 0)$$

 = 0.0000 KPa.

Determine the effective pressure [Pe]:

$$= J \cdot K_{st} / (1 + J \cdot K_{st} (Q_{z1} + (p_s - 1) Q_{z2})) \cdot (P^* s - P^* t + P_\gamma + P_\omega + P_w + P_{prim})$$

$$= 0.1000E+01 \cdot 0.7 / (1 + 1 \cdot 0.7 (1.04 + (1.04 - 1)0.091)) \cdot (10064 - 14332 + 0 + 0 + 393 + 150)$$

 = -1504.1863 KPa.

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$= [((\omega^* s \cdot P_s - \omega^* c \cdot P_t) - (\omega_s \cdot P^* s - \omega_c \cdot P^* c)) + W^* \cdot \gamma_b / (2 \cdot \pi)] / (1 + \phi \cdot Z_m)$$

$$= [((-6730 \cdot 6100 - 3058 \cdot 6100) - (6799 \cdot 0 - 0 \cdot 0)) \cdot 0.0001 + 381546 \cdot -0.26 / (2 \cdot \pi)] / (1 + 3.01 \cdot 0.67)$$

 = -7289.4907 Kgf

Calculate factor [Q3]:

$$= Q_1 + 2 \cdot Q_2 / (P_e \cdot a_o^2)$$

$$= -0.15 + 2 \cdot -7289 / (-1504 \cdot 268^2)$$

 = 1.16784

Fm Value from Table 4.18.3 = 0.764021



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Tubesheet Bending Stress - original thickness [σ]:
 $= (1.5 \cdot F_m / \mu \cdot) \cdot (2 \cdot a_o / (h - h'g))^2 \cdot P_e$
 $= (1.5 \cdot 0.76 / 0.2) \cdot (2 \cdot 268 / (190 - 8))^2 \cdot -1504$
 $= -74.9740 \text{ N./mm}^2$

Allowable Tubesheet Bending Stress [σ allowed]:
 $= 1.5 \cdot S = 1.5 \cdot 85.9 = 129 \text{ N./mm}^2$

Tubesheet Bending Stress - final thickness [σ_f]:
 $= (1.5 \cdot F_m / \mu \cdot) \cdot (2 \cdot a_o / (h - h'g))^2 \cdot P_e$
 $= (1.5 \cdot 0.75 / 0.2) \cdot (2 \cdot 268 / (127 - 8))^2 \cdot -1118$
 $= -128.8502 \text{ N./mm}^2$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:
 $= h + C_{ats} + C_{atc}$
 $= 127 + 0 + 0$
 $= 126.8873 \text{ mm}$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ_{limit}]:
 $= (2 \cdot \mu \cdot h / a_o) \cdot \min(0.8S; 0.533S_y)$
 $= (2 \cdot 0.2 \cdot 190 / 268) \cdot 68.7$
 $= 19.4603 \text{ N./mm}^2$

Shear Stress is not required to be computed; [P_e] $\leq$ τ_{limit}

Tubesheet Shear Stress is probably low, use the following req. thk:

Tubesheet thickness (Incl. Corr.) = 14.6869 mm
 Tubesheet Shear Stress = 23.4422 N./mm²

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd]:
 $= \max(H_{reqB}; H_{reqS})$
 $= \max(127; 14.7)$
 $= 126.8873 \text{ mm}$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The F_{tmin} and F_{tmax} Coefficients from Table 4.18.4:
 $F_{tmin} = 0.64, F_{tmax} = 1.34$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:
 $= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmin}) / (X_t - X_s)$
 $= ((6100 \cdot 0.5 - 6100 \cdot 0.78) - (-1504) \cdot 0.64) / (0.78 - 0.5)$
 $= -2.6288 \text{ N./mm}^2$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:
 $= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmax}) / (X_t - X_s)$
 $= ((6100 \cdot 0.5 - 6100 \cdot 0.78) - (-1504) \cdot 1.34) / (0.78 - 0.5)$
 $= 1.1438 \text{ N./mm}^2$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:
 $= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 2.63 \text{ N./mm}^2$

Allowable Tube Stress [$\sigma_{t,A}$]:
 $= S_{ot} = 97.7009 \text{ N./mm}^2$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [F_s]:
 $= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w)X_{a^4}); 1.25)$
 $= \max((3.25 - 0.25(0.77 + 1.17 \cdot 0.17)1.29^4); 1.25)$



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$$= 2.0000 \text{ (Should be } \leq 2 \text{)}$$

Determine the factor [rt]:

$$= \left(\sqrt{\frac{dt^2}{4} + \left(\frac{dt}{2} - \frac{tt}{2} \right)^2} \right) / 4$$

$$= \left(\sqrt{\frac{12.7^2}{4} + \left(\frac{12.7}{2} - \frac{2.11}{2} \right)^2} \right) / 4$$

$$= 3.8177 \text{ mm}$$

Determine the factor [Ct]:

$$= \sqrt{\frac{2 \cdot \pi^2 \cdot Et / Syt}{2 \cdot \pi^2 \cdot 175593552 / 108}}$$

$$= 179.1504$$

Determine the factor [Ft]:

$$= k \cdot L / r$$

$$= 1 \cdot 327 / 3.82$$

$$= 85.6531$$

Buckling Allowable Stress [Stb]:

$$= (Sy, t / Fs) \cdot (1 - Ft / (2 \cdot Ct))$$

$$= (108 / 2) \cdot (1 - 85.7 / (2 \cdot 179))$$

$$= 41.092 \text{ N/mm}^2 \text{ (Never greater than } Sot)$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$= \sigma_{t, \max} \cdot \text{Tube Area}$$

$$= 2.63 \cdot 0.7$$

$$= 18.82 \text{ Kgf}$$

Tube Metal Area [At]:

$$= \pi (do - t) t$$

$$= \pi (12.7 - 2.11) 2.11$$

$$= 0.702 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (do - t) S$$

$$= \pi \cdot 2.11 (12.7 - 2.11) 97.7$$

$$= 699.356 \text{ Kgf}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot ag (do + 0.67 \cdot ag) Sw \text{ (but not } > Ft)$$

$$= 0.85 \cdot \pi \cdot 1.65 (12.7 + 0.67 \cdot 1.65) 85.9$$

$$= 532.8093 \text{ Kgf}$$

Weld Strength Factor [fw]:

$$= S / Sw \text{ where } Sw = \min(S \text{ and } St)$$

$$= 97.7 / 85.9$$

$$= 1.137$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw \text{ (but not } > Ft)$$

$$= 0.55 \cdot \pi \cdot 1.65 (12.7 + 0.67 \cdot 1.65) 85.9$$

$$= 344.7589 \text{ Kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{\frac{(0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) \cdot fw \cdot fd}{0.75 \cdot do}} \right) - 0.75 \cdot do$$

$$= 2 \left(\sqrt{\frac{(0.75 \cdot 12.7)^2 + 1.07 \cdot 2.11 \cdot (12.7 - 2.11) \cdot 1.14 \cdot 1}{0.75 \cdot 12.7}} \right) - 0.75 \cdot 12.7$$

$$= 2.668 \text{ mm}$$



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Minimum Required Fillet Weld Leg [afr]: 1.3341 mm
 Minimum Required Groove Weld Leg [agr]: 1.3341 mm

Tube-Tubesheet Jt allowable, 699 is >= tube strength 699 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:
 $= S_s \cdot E_{sw} = 85.9 \cdot 1 = 85.9 \text{ N./mm}^2$

Axial Membrane Stress in Shell [σ_s, m]:
 $= a_o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$
 $= 268^2 / ((560 + 24) 24) \cdot [-1504 + (1.04^2 - 1)(6100 - 6100)] + 280^2 \cdot 6100 / ((560 + 24) 24)$
 $= 26.3926 \text{ N./mm}^2$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:
 For a given Shell thickness of 24 mm, the minimum Shell length adjacent to the tubesheet should be 208.676 mm

Shell Membrane Stress due to Joint Interaction [σ_s, m]:
 $= a_o^2 / ((D_s + t_s) t_s) [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$
 $= 268^2 / ((560 + 24) 24) [-1504 + (1.04^2 - 1)(6100 - 6100)] + 280^2 \cdot 6100 / ((560 + 24) 24)$
 $= 26.3926 \text{ N./mm}^2$

Shell Bending Stress due to Joint Interaction [σ_s, b]:
 $= 6 \cdot k_s / t_s^2 \{ \beta_s [\delta_s \cdot P_s + a_s^2 \cdot P \cdot s / (E_s \cdot t_s)] + 6(1 - \nu \cdot s^2) / (E \cdot s) (a_o / h)^3 (1 + h \cdot \beta_s / 2) [P_e (Z_v + Z_m \cdot Q_1) + 2 / a_o^2 \cdot Z_m \cdot Q_2] \}$
 $= 6 \cdot 6.86E+09 / 24^2 \{ 0.015 [0.015837597 \cdot 6100 + 280^2 \cdot 0 / (4.21E+09)] + 6(1 - 0.46^2) / (27278308) (268 / 190)^3 (1 + 190 \cdot 0.015 / 2) [-1504 (0.17 + 0.67 \cdot -0.15) + 2 / 268^2 \cdot 0.67 \cdot -7289] \}$
 $= -17.2237 \text{ N./mm}^2$

Shell Stress Summation vs. Allowable
 $:: |\sigma_s, m| + |\sigma_s, b| \leq 1.5 \cdot S_s$
 $:: |26.4| + |-17.2| \leq 129 \text{ N./mm}^2$
 $:: 43.6, \text{ must be } \leq 129 \text{ N./mm}^2$

Computations completed for ASME tubesheet configuration: b

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	75.0 <=	128.9 N./mm ²	Pass
Tubesheet Shear Stress	5.3 <=	68.7 N./mm ²	Pass
Maximum Tube Stress	2.6 <=	97.7 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-2.6 <=	-41.1 N./mm ²	Pass
Maximum Force on any one Tube	18.8 <=	699.4 Kgf	Pass
Axial Membrane Stress in Shell	26.4 <=	85.9 N./mm ²	Pass
Shell Stress (jt. inter.)	43.6 <=	128.9 N./mm ²	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
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Tubesheet Thickness :	126.887	190.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.334	1.650	Ok
Tube-Tubesheet Groove Weld Leg :	1.334	1.650	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 14 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	163.012	77.880	100	129	20	69 Ps+Pt-Th D1 Ok
D2uc	90.750	...	33	129	15	69 Ps+Pt-Th D2 Ok
D3uc	126.887	...	75	129	5	69 Ps+Pt-Th D3 Ok
O1uc	53.617	...	91	258	20	69 Ps+Pt+Th 01 Ok
O2uc	91.739	...	74	258	15	69 Ps+Pt+Th 02 Ok
O3uc	67.174	...	69	258	5	69 Ps+Pt+Th 03 Ok
O4uc	66.218	...	44	258	1	69 Ps+Pt+Th 04 Ok
D1c	180.162	77.880	117	129	22	69 Ps+Pt-Th-c D1 Ok
D2c	107.100	...	38	129	17	69 Ps+Pt-Th-c D2 Ok
D3c	138.828	...	84	129	6	69 Ps+Pt-Th-c D3 Ok
O1c	79.608	...	115	258	22	69 Ps+Pt+Th-c 01 Ok
O2c	117.757	...	95	258	18	69 Ps+Pt+Th-c 02 Ok
O3c	93.264	...	84	258	6	69 Ps+Pt+Th-c 03 Ok
O4c	92.277	...	57	258	1	69 Ps+Pt+Th-c 04 Ok
Max: 180.1619 77.880 mm 0.911 0.327 (Str. Ratio)						

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+)-Th]:
 With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:
 Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Shell Axial Membrane Stress Summary:

	----	Shell Stresses			----	:	----	Shell Band Stress			----	:	Pass
Case#	Ten	Allwd	Cmp	Allwd	:	Ten	Allwd	Cmp	Allwd	:	Fail		
D1uc	3	85	...	...	:	...	...	...	...	:	Ok		
D2uc	24	85	...	...	:	...	...	...	...	:	Ok		
D3uc	26	85	...	...	:	...	...	...	...	:	Ok		
O1uc	3	258	...	...	:	...	...	...	...	:	Ok		
O2uc	25	258	...	...	:	...	...	...	...	:	Ok		
O3uc	26	258	...	...	:	...	...	...	...	:	Ok		
O4uc	1	258	...	...	:	...	...	...	...	:	Ok		
D1c	3	85	...	...	:	...	...	...	...	:	Ok		



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D2c	24	85	...	...	...	...	...	...	Ok
D3c	27	85	...	...	...	...	...	...	Ok
O1c	3	258	...	...	...	...	...	...	Ok
O2c	25	258	...	...	...	...	...	...	Ok
O3c	27	258	...	...	...	...	...	...	Ok
O4c	1	258	...	...	...	...	...	...	Ok
Max RATIO	0.314	...	...	...	...	...	...	...	...

Tube, Shell and Channel Stress Summary:

---- Tube Stresses Tube Loads Shell Stress Channel Stress Pass											
Case#	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	Fail
D1uc	5	97	...	...	38	699	127	128	...	...	Ok
D2uc	4	97	-4	-41	29	699	131	258	...	...	Ok
D3uc	3	97	-3	-41	19	699	44	128	...	...	Ok
O1uc	5	195	...	...	38	1398	127	258	...	...	Ok
O2uc	5	195	-5	-41	37	1398	50	258	...	...	Ok
O3uc	3	195	-3	-41	19	1398	44	258	...	...	Ok
O4uc	2	195	-2	-41	12	1398	83	258	...	...	Ok
D1c	6	97	-1	-41	41	699	170	258	...	...	Ok
D2c	4	97	-4	-41	30	699	134	258	...	...	Ok
D3c	4	97	-4	-41	27	699	84	128	...	...	Ok
O1c	6	195	-1	-41	41	1398	170	258	...	...	Ok
O2c	6	195	-6	-41	40	1398	26	258	...	...	Ok
O3c	4	195	-4	-41	27	1398	84	258	...	...	Ok
O4c	2	195	-2	-41	18	1398	112	258	...	...	Ok
Max RATIO	0.059	0.136	0.059	0.983	...	...	...	...	...	...	...

Summary of Thickness Comparisons for 14 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	180.162	190.000	Ok
Tubesheet thickness flanged extension :	77.880	154.000	Ok
Tube thickness :	1.177	2.110	Ok
Tube-tubesheet fillet weld leg :	1.334	1.650	Ok
Tube-tubesheet groove weld leg :	1.334	1.650	Ok

Min Shell length of thk, (24.000) adj. to tubesheet: 208.676 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm
 Due to Pressure Shell Compresses by : -0.290 mm
 Due to Pressure + Thermal Shell Compresses by : -0.290 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	8069.51	0.999	20418.71	1.000
Tubesheet Shear Stress	18173.99	1.000	24722.36	1.000
Tube Tensile Stress	110371.39	1.000	142217.50	1.000
Tube Compressive Stress	No Calc	No Calc	59814.61	1.000
Tube-Tubesheet Joint load	110371.39	1.000	142217.77	1.000
Shell Stress (Axial, Junction)	8069.51	0.999	11753.77	1.000
Tube Pressure Stress	37438.60	1.000	16666.01	1.000
Tubesheet Extension Stress	10114.50	...	No Calc	No Calc
Minimum MAWP	8069.51		11753.77	



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Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	15239.42	1.000	38691.76	1.000
Tubesheet Shear Stress	33259.53	1.000	45952.00	1.000
Tube Tensile Stress	140308.59	1.000	171190.95	1.000
Tube Compressive Stress	No Calc	No Calc	90531.76	1.000
Tube-Tubesheet Joint load	140308.59	1.000	171190.88	1.000
Shell Stress (Axial, Junction)	15239.42	1.000	19303.06	1.000
Tube Pressure Stress	44066.74	1.000	28106.38	1.000
Tubesheet Extension Stress	10114.50	...	No Calc	No Calc
Minimum MAPnc	10114.50		19303.06	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0166	0.0166	...	...	...	...
2	-0.2888	-0.2896	...	...	...	...
3	-0.2723	-0.2730	...	...	...	...
4	...	...	...	...	...	...
5	0.0166	0.0166	...	...	...	...
6	-0.2888	-0.2896	...	...	...	...
7	-0.2723	-0.2730	...	...	...	...
8	...	...	...	...	...	...

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PV Elite® Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: